



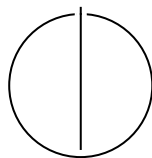
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNICAL UNIVERSITY OF MUNICH

Master's Thesis in Informatics

**Leveraging Grammar-Book Images in
Vision-Language Models for Low-Resource
Machine Translation**

Cagatay Iskenderoglu





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TECHNOLOGY — INFORMATICS

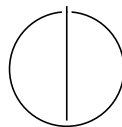
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for Low-Resource Machine Translation**

**Nutzung von Abbildungen aus Grammatikbüchern in
Vision-Language-Modellen für ressourcenarme maschinelle
Übersetzung**

Author:	Cagatay Iskenderoglu
Examiner:	Prof. Dr. Alexander Fraser
Supervisor:	Shu Okabe, Ph.D.
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Abstract

Glosses describe the meanings and grammatical roles of words or word parts and can support translation. This thesis asks whether grammar material improves translation when prompts already contain glossed examples. It translates from four low-resource languages, Gitksan, Lezgi, Natugu, and Tsez, into English with three vision-language models: Qwen3-VL-8B-Instruct, Qwen3.5-9B, and Gemini 2.5 Flash-Lite. Three methods use the same 21 annotated demonstrations per language, translating a new sentence without its gloss, generating its gloss first, or receiving an externally predicted gloss. The 351 primary configurations combine model, language, prompting method and grammar-context setting, including no book (no grammar material or its use instruction). I prepared summaries, tables, and cheat sheets from existing grammar documents and also supplied selected page images. Each grammar addition holds model, method, demonstrations, test items, and generation settings fixed while adding material and an instruction to use it.

The method receiving an external gloss has the highest observed no-book translation performance on the automatic BLEU and chrF++ metrics in every model-language group. Generating a gloss first sometimes leaves an incomplete response, especially for Gemini on Tsez. Grammar additions vary by model, with chrF++ changes mostly negative for Qwen3.5 and Gemini and more often positive for Qwen3: primary tests for Gitksan, Natugu, and Tsez, using valid reference translations, find one statistically significant increase and ten decreases. Nine significant paired text-image differences favour text, although the supplied packages can differ in content and instructions.

A separate Qwen supplement, including 37 new Qwen3 configurations, supports most external-gloss advantages and finds a Lezgi image-over-text BLEU advantage. Neural evaluation was explored, but its validity is unestablished for these source languages. The results support assessing each grammar package against the same gloss-supported prompt without grammar material; neither score changes nor the paired comparisons establish universal image benefits or reliable application of grammatical explanations.

Declaration of Independent Work and Resources

By signing below, I declare that I prepared this thesis independently and used only the sources and assistance identified in the thesis and in this declaration. I confirm that passages, ideas, data, and materials taken from other sources are identified as such, and that I have reviewed the final text and take responsibility for its contents.

I used GPT-5.6 Sol to help prepare derived grammar materials from source PDFs. I used Codex assistants with Sol and Astra models for drafting, editing, evidence checking, and preparation of the LaTeX source and PDF. I also used Claude to review the thesis and propose prose revisions, selected parts of which were incorporated into this version.

Place and date:

Munich, 30.09.2026

Signature:

A handwritten signature in blue ink, appearing to read 'Çağatay', is written over a horizontal line.

Contents

Abstract	ii
Declaration of Independent Work and Resources	iii
List of Tables	viii
List of Figures	x
Abbreviations	xi
1. Introduction	1
1.1. Motivation and Problem	1
1.2. Aim, Scope and Research Questions	2
1.3. Contributions	5
1.4. Thesis Organization	5
2. Background and Related Work	6
2.1. Low-Resource Translation and Linguistic Resources	6
2.1.1. Documentary Resources and Interlinear Glossed Text	6
2.1.2. Data Scarcity and Morphological Structure	7
2.2. Gloss Prediction and Gloss-Based Translation	8
2.2.1. Automatic Gloss Prediction	8
2.2.2. Trained and Prompted Gloss-Based Translation	9
2.3. Grammar Books and Linguistic Descriptions as Context	11
2.3.1. Resource-Based Translation Context	11
2.3.2. Examples, Explanations and Rule Use	14
2.4. Textual and Visual Grammar Context	15
2.4.1. Vision-Language Model Families	15
2.4.2. Visual Translation and Document Understanding	15
2.4.3. Grammar-Context Representations	16
3. Data and Grammar Resources	18
3.1. Languages and Source Documents	18
3.1.1. Gitksan	19

3.1.2.	Lezgi	19
3.1.3.	Natugu	19
3.1.4.	Tsez	20
3.2.	Translation and Gloss Data	20
3.2.1.	Dataset Provenance and Split Sizes	20
3.2.2.	Reference Quality and Predicted Glosses	21
3.3.	Prepared Grammar Representations	22
3.3.1.	Text Summaries, Tables, and Cheat Sheets	22
3.3.2.	Selected Original Pages	23
3.3.3.	Extracted PDF Text	24
4.	Methodology	25
4.1.	Translation Task and Gloss-Based Methods	25
4.1.1.	Translation Task and Matched Comparison	25
4.1.2.	Gloss-Based Prompting Methods	25
4.1.3.	Demonstrations and Query Inputs	26
4.2.	Models and Grammar-Context Experiments	28
4.2.1.	Models	28
4.2.2.	Grammar Context and Condition Matrix	28
4.2.3.	Prompt Assembly and Generation	30
4.3.	Evaluation and Statistical Analysis	31
4.3.1.	Evaluation Cohorts and Reference Policy	31
4.3.2.	Translation Metrics	32
4.3.3.	Comparison Sets and Research Questions	33
4.3.4.	Paired Bootstrap and Multiple Testing	33
4.3.5.	Supplementary Analyses and Diagnostics	35
4.4.	Reproducibility	36
5.	Results	37
5.1.	Evaluation Coverage and Baseline Performance	37
5.1.1.	Evaluation Coverage and Reporting	37
5.1.2.	No-Book Baselines and Method Trends	38
5.1.3.	Comparison on the Shared Tsez Cohort	40
5.2.	Effects of Grammar-Book Context	41
5.2.1.	Prepared Text: Content and Granularity	43
5.2.2.	Paired Text–Image Presentation	46
5.2.3.	Selected Original Pages	49
5.3.	Output Behaviour and Illustrative Examples	50
5.3.1.	Empty Outputs and Gloss-Format Compliance	50

5.3.2.	Examples from the Evaluated Predictions	52
5.3.3.	Supplementary Query-Gloss and Seed Diagnostics	53
6.	Discussion and Limitations	55
6.1.	Interpretation of the Research Questions	55
6.1.1.	No-Book Performance of the Adapted Prompting Methods (RQ1)	55
6.1.2.	What Adding Grammar Context Changes (RQ2)	56
6.1.3.	Text, Images, and Selected Original Pages (RQ3)	57
6.1.4.	Variation across Models, Languages, and Methods (RQ4)	59
6.2.	Relation to Previous Research	60
6.2.1.	Glosses and Grammar Resources in Translation	60
6.2.2.	What the Earlier MTOB Study Adds	62
6.3.	Validity, Uncertainty, and Scope	63
6.3.1.	Reference Quality and Evidence of Translation Quality	63
6.3.2.	Dependence and Statistical Uncertainty	64
6.3.3.	Generalizability and Reproducibility	65
7.	Conclusion and Future Work	66
7.1.	Research Summary and Contributions	66
7.2.	Conclusions by Research Question	67
7.2.1.	No-Book Performance of the Adapted Methods (RQ1)	67
7.2.2.	Incremental Effects of Grammar Context (RQ2)	68
7.2.3.	Grammar Representations and Paired Text-Image Packages (RQ3)	68
7.2.4.	Variation across Models, Languages, and Methods (RQ4)	69
7.3.	Future Work	69
A.	Prompt Templates	72
A.1.	Primary Translation Prompt Templates	72
A.1.1.	Grammar-Context Wrappers and Message Order	73
A.2.	Supplementary Gold-Query Diagnostic	74
A.3.	Earlier MTOB Grammar-Text Prompts	75
A.4.	Grammar-Material Preparation Template Excerpt	75
B.	Complete Experiment Matrix	77
B.1.	Primary matched conditions	78
B.2.	Separate Earlier MTOB Conditions	84
B.3.	Separate Retained Qwen3 Diagnostic Conditions	84
C.	Grammar-Context Material Examples	87
C.1.	Prepared Text Excerpts	87

C.2. Prepared and Original-Page Images	88
C.3. Embedded-PDF Text Extraction	89
D. Reproducibility and Evaluation Details	90
D.1. Frozen Inputs and Condition Identity	90
D.2. Models, Generation, and Response Parsing	91
D.3. Evaluation Cohorts, Metrics, and Saved Analyses	93
D.4. Retained Supplementary Provenance and Procedures	95
E. Earlier MTOB Methods and Results	97
E.1. Context Construction and Execution	97
E.1.1. Text Extraction and Context Construction	97
E.1.2. Prompting and Model Execution	98
E.2. Response Views, Metrics and Comparisons	99
E.2.1. Earlier MTOB Paired Tests	101
E.3. Earlier MTOB Results	101
E.3.1. Raw and Extracted Responses	101
E.3.2. Comparison of Grammar-Context Methods	104
E.3.3. Comparison with Earlier Matched Baselines	104
E.4. Provenance and Interpretation Limits	107
E.4.1. Retained MTOB Artifacts	107
E.4.2. A Response-Extraction Example	108
F. Primary and Supplementary Results	109
F.1. Complete Native chrF++ Matrices	109
F.2. Direction Counts and Saved Test Detail	113
F.3. Separate Supplementary Results	116
Bibliography	123

List of Tables

3.1. Five grammar source PDFs and selected pages	18
3.2. Accepted annotated and uncovered Track 1 records	21
3.3. Sizes of derived grammar materials	23
4.1. Primary grammar-context representations	29
4.2. Primary experiment matrix by model	29
4.3. Primary generation policies by backend	31
4.4. Research questions and corresponding analyses	34
4.5. Primary Holm-correction families	35
5.1. Primary generation coverage	37
5.2. Native no-book baselines across models	39
5.3. Shared first-99 Tsez no-book comparison	40
5.4. Native significant chrF++ context changes	42
5.5. Common99 significant chrF++ context changes	43
5.6. Native significant chrF++ TXT/JPG differences	47
5.7. Common99 significant chrF++ TXT/JPG differences	47
5.8. Exploratory xCOMET contrasts for Qwen3.5 Tsez	48
5.9. Native output diagnostics by model and method	50
5.10. Native parser and generation flags	52
5.11. Tsez Chain-gloss cheat-sheet example	53
5.12. Tsez ModelGloss table example	53
5.13. Natugu prepared-table and original-page example	54
B.1. Primary grammar-package availability	77
B.2. Gemini primary condition catalog	78
B.3. Qwen3.5 primary condition catalog	80
B.4. Qwen3 primary condition catalog	82
B.5. Earlier MTOB condition catalog	84
B.6. Retained supplementary Qwen3 conditions. Cap is the maximum number of new tokens; n and Empty count generation records.	85
D.1. Recorded software and primary evaluation identities	94

List of Tables

E.1. Primary and earlier evaluation conventions	100
E.2. Earlier MTOB results: Qwen3	102
E.3. Earlier MTOB results: Qwen3.5	103
E.4. Earlier significant within-MTOB chrF tests	105
E.5. Earlier cross-protocol significance counts	106
E.6. Earlier Qwen3 Natugu cross-protocol examples	107
F.1. Native chrF++ context changes: Gitksan	110
F.2. Native chrF++ context changes: Lezgi	111
F.3. Native chrF++ context changes: Natugu and Tsez	112
F.4. Native individual context-change directions	113
F.5. Saved lexical context p-values	114
F.6. Saved lexical pair p-values	115
F.7. Saved exploratory xCOMET test detail	115
F.8. Adjusted-significant supplementary no-book method contrasts. Parentheses give records/source clusters. Intervals are pointwise 95%; all Holm values retain the 72-test family.	116
F.9. Supplementary Qwen3.5 Lezgi Chain-gloss findings: main 83 and sensitivity 84 records. Intervals are pointwise 95%. Context uses the 264-test family; JPG–TXT uses the 96-test family.	119
F.10. Input-token ranges for existing Qwen conditions. Logged means range over conditions; logged extrema range over individual records. Reconstructed text and visual counts refer to first prompts.	119
F.11. Retained Qwen3 diagnostic scores. Lezgi 83 is the main quality view; 84 and first99 Tsez are overlapping sensitivities. xCOMET-XL and xCOMET-XXL are exploratory native-scale scores without supplementary tests.	120
F.12. Selected-pair seed diagnostics. Lexical columns are saved context-minus-baseline differences; neural columns show stored baseline-to-context scores. All outcomes are descriptive.	122

List of Figures

1.1. Aligned glossed example on a grammar page	3
2.1. Four-tier Natugu interlinear example	6
4.1. Primary prompting and evaluation flow	27
5.1. No-book chrF++ across four languages	41
5.2. Native chrF++ changes: Gitksan and original Lezgi	44
5.3. Native chrF++ changes: Cyrillic Lezgi, Natugu and Tsez	45
C.1. Display crop of frozen Brown summary-table JPG showing literal Mark- down syntax. Source: Brown et al. (2016); the excerpt refers to PDF page 9.	88
C.2. Display crop of frozen Brown cheat-sheet JPG showing literal Mark- down syntax. Source: Brown et al. (2016); the excerpt refers to PDF pages 9–10.	88
C.3. Display crop of frozen selected original Brown PDF page 9, preserving the source article’s layout. Source: Brown et al. (2016).	89

Abbreviations

IGT Interlinear Glossed Text

IPA International Phonetic Alphabet

LLM Large Language Model

MT Machine Translation

MTOB Machine Translation from One Book

OCR Optical Character Recognition

VLM Vision-Language Model

1. Introduction

Machine Translation (MT) is harder to develop and assess when parallel translation data are scarce. Descriptive grammars, dictionaries, and glossed examples may still document a language when aligned translation data are scarce (Tanzer et al., 2024; Ginn et al., 2023). This thesis examines one use of that documentation: supplying selected grammar material to a model translating sentences from four low-resource languages, Gitksan, Lezgi, Natugu, and Tsez, into English. It asks whether the added material helps when the prompt already includes source-language glosses and English translations as examples.

1.1. Motivation and Problem

For the present task, *low-resource* refers to the limited annotated and translated data available for developing and evaluating translation systems. It does not assign the four languages a common social status or imply that they are equally documented. A small parallel dataset can coexist with a grammar, a dictionary, and interlinear annotations, but these resources provide different types of information. Parallel sentences show particular translations; a descriptive grammar can state patterns and supply further examples. A benchmark on translating with one grammar book, for example, investigated a grammar together with a word list and a small parallel collection for Kalamang (Tanzer et al., 2024). The resources available here differ, but a document’s existence alone does not show how much it adds to an already informative translation prompt.

Morphological structure gives a reason to examine analyses alongside sentence pairs. Related forms can contain a stem and several grammatical elements, producing word forms that a small dataset may not contain as whole words. Work on gloss-based translation has used morphological analysis to address such sparsity (Zhou et al., 2020). An interlinear gloss assigns lexical meanings and grammatical labels to parts of a source sentence, whereas a free English translation expresses its meaning without requiring word-by-word alignment (Ginn et al., 2023). A gloss can thus expose information that the English sentence leaves implicit.

This linguistic information can also be supplied to a language model through its translation prompt. GrammarMT (Ramos et al., 2025) is a gloss-informed prompting approach that places source sentences, their glosses, and translations in example demonstrations. It distinguishes three ways to translate a new source sentence: translating without its gloss, asking the translating model to generate that gloss first, and supplying a gloss predicted outside the translator. This thesis adapts the three methods and asks what a *grammar package* adds when the demonstrations already provide gold glosses. Here, a package is selected grammar text or images supplied with the translation prompt.

A complementary study compared dictionaries, parallel examples, and selected grammar documents across 16 languages in both translation directions (Hus and Anastasopoulos, 2024). It used PanLex dictionaries, with MTOB resources for Kalamang, and reorganized the available evaluation partitions, including the SIGMORPHON data for Gitksan and Natugu. The present thesis instead keeps 21 gold-glossed demonstrations fixed for each language and uses its own evaluation partitions. The earlier study supports examining combinations of resources, but it does not provide an identical-data comparison for the grammar additions tested here.

A vision-language model can receive a grammar page as an image, making the page’s layout available alongside the translation prompt. A page can align source forms, interlinear glosses, labels, and translations in relationships that plain text conversion has to reconstruct. Figure 1.1 shows such alignment in the retained Brown Gitksan article. Document-parsing research separates text recognition, tables, and reading order (Ouyang et al., 2025); source-image translation likewise distinguishes recognizing text from using its visual context (Salesky et al., 2024). These distinctions motivate comparing supplied text and image packages within a specified translation configuration.

1.2. Aim, Scope and Research Questions

The primary study evaluates sentence-level translation from Gitksan, Lezgi, Natugu, and Tsez into English with Qwen3-VL-8B-Instruct (Qwen3), Qwen3.5-9B (Qwen3.5), and Gemini 2.5 Flash-Lite. Each model can receive text and images and is used here as a Vision-Language Model (VLM). The adapted GrammarMT conditions are Gloss-shot, explicit Chain-gloss, and ModelGloss. Each uses the same 21 *support examples* per language: source sentences from the training partition paired with gold glosses and English translations in the prompt. A *query* is the new source sentence to translate. Gloss-shot requests its translation without a query gloss; Chain-gloss asks the translating model to generate a query gloss before the translation; ModelGloss receives an

Orthographic transcription with interlinear English gloss

The top line of each entry is an orthographic representation. The second line indicates where the affix and clitic boundaries are within words.

Yukwt	laseexwhl	bahasxw	ganhl	hloxs
yukw=t	laseexw=hl	bax-asxw	gan=hl	hloxs
IPFV=3.I	discuss=CN	run-ANTIP	PH.CNJ=CN	sun
naahl	ky'aa	daxgyadit	as	'nidiit,
naa=hl	ky'aa	daxgyat-it	a=s	'nidiit
who=CN	most	strong-SX	PREP=PN	3PL.III
'The wind and the sun were discussing amongst themselves who was the strongest of them.'				

Figure 1.1.: Display crop from Brown et al.'s Gitksan article, PDF page 9 (printed page 375) (Brown et al., 2016). The heading introduces the orthographic transcription; the example aligns source and segmentation lines with lexical and grammatical glosses, followed by an English translation. Appendix C retains fuller examples of the original and prepared materials.

externally predicted query gloss from the dedicated GlossLM system. No primary method receives the test sentence's gold gloss (Figure 4.1).

The selected grammar material comes from five existing documents: Brown's Gitksan article, Rigsby's introductory Gitksan excerpt, Haspelmath's Lezgian grammar excerpt, Boerger's Natqgu sketch, and Comrie and Polinsky's Tsez manuscript. I prepared text summaries, summary tables, and cheat sheets from these PDFs with GPT-5.6 Sol assistance and also selected original-page images. Tables and sheets have paired text (TXT) and image (JPG) forms where supplied (Chapter 3). The 21 support examples remain fixed across these materials; further examples inside a grammar package are part of its varying content.

A *no-book baseline* is the local GrammarMT adaptation with the same glossed demonstrations and applicable query gloss, but without grammar-document material or its use instruction. Each grammar-addition comparison holds the model, method, supports, test items, and decoding policy fixed and compares a supplied package with its corresponding no-book baseline. Adding material also adds prompt wording that introduces the reference and asks the model to use it. A separate paired comparison evaluates the supplied text package against its corresponding image package.

These representations can differ in information as well as presentation. Prepared summaries select and reorganize content, whereas selected original pages retain surrounding explanations and examples. The original Lezgi cheat-sheet image also omits three mini-gloss rows and four examples present in its text version (Section 3.3). The paired comparisons therefore assess the supplied packages and their accompanying instructions, rather than isolating modality with content held identical.

These definitions lead to four research questions:

1. How do the adapted GrammarMT-based prompting methods perform without grammar-book context across Qwen3, Qwen3.5, and Gemini 2.5 Flash Lite?
2. How does adding grammar-book material affect translation quality compared with the corresponding no-book baseline?
3. How do text summaries, tables, cheat sheets, and selected grammar-book pages differ as context representations, including text-image comparisons where paired versions are available?
4. How do these comparisons vary across models, languages, and the Gloss-shot, explicit Chain-gloss, and ModelGloss conditions?

The first question concerns rankings of the adapted methods without grammar-book material. The second assesses the matched addition of grammar packages, while the third compares the supplied representations, including paired text and image versions. The fourth asks how these context and representation comparisons vary across models, languages, and methods. These are questions about observed performance under the evaluation defined in Chapter 4; they do not presume rule use or treat scores alone as proof of translation adequacy.

A separate post-hoc Qwen supplement adds lexical method tests, reference-valid Lezgi comparisons and input-size descriptions, with 37 saved new Qwen3 conditions that compare output caps, query-gloss inputs and selected random seeds. It extends the evidence within the same research questions while keeping the primary results and correction families distinct (Section 4.3.5).

A separate *earlier MTOB-inspired supplementary study* takes inspiration from Machine Translation from One Book (MTOB) (Tanzer et al., 2024) and examines grammar-text retrieval and longer contexts. It also compares saved responses with translations extracted from those same responses. Its cross-protocol comparison uses earlier matched baselines with different supports, prompts, and generation settings, not the current gold-support no-book baselines. This earlier MTOB study therefore provides a distinct view of grammar-text prompting and response evaluation, rather than another matched test of the primary study’s grammar addition (Appendix E).

1.3. Contributions

This thesis makes three contributions. First, it evaluates selected grammar packages against matched no-book prompts in 351 experimental configurations. The 36 baselines comprise three models, four languages, and three methods. The 315 grammar-context conditions comprise three models and three methods applied to 35 packages: six formats for each of five source documents, plus five Cyrillic-form formats for the Lezgi source. The Lezgi variant adds no second original-page condition (Chapter 4). Each context condition is paired with its relevant baseline. The saved configurations, responses, and evaluation inputs make these comparisons traceable.

Second, it compares paired supplied text and image packages, alongside summaries and selected original pages. Third, it provides empirical and diagnostic findings about when the packages help or hinder the evaluated prompts. ModelGloss has the highest observed no-book BLEU and chrF++ scores in each model-language group. Among the original primary tests with valid reference translations (Gitksan, Natugu, and Tsez), one grammar addition increases chrF++ and ten decrease it after multiple-testing correction; nine paired text-image differences favour text (Chapter 5). Separate supplementary Qwen analyses add method tests and valid-reference Lezgi comparisons, including an image-over-text BLEU advantage. Response diagnostics distinguish translation performance from completing the requested output format. The earlier MTOB study separately shows that scoring only the translation portion instead of the saved response with its explanations can change the ranking of grammar-text strategies, without generating a new translation. Chapter 6 develops the interpretation and measurement limits.

1.4. Thesis Organization

Chapter 2 relates gloss-supported translation to grammar resources and document-image processing. Chapter 3 describes the datasets, grammar documents, and prepared representations. Chapter 4 defines the gloss methods, demonstrations, matched comparisons, and evaluation. Chapter 5 presents baseline and grammar-package results, completion diagnostics, and supplementary Qwen analyses. Chapter 6 interprets these findings and their limits. Chapter 7 answers the research questions and identifies future work.

The appendices provide prompts, condition catalogs, material examples, reproducibility details, the earlier MTOB protocol (Appendix E), and detailed results (Appendix F).

2. Background and Related Work

Translation with limited parallel data can draw on dictionaries, glosses, and grammar books. This chapter explains what these resources provide and how previous studies have used them. It also examines the problems of reading grammar documents and applying their content to translation.

2.1. Low-Resource Translation and Linguistic Resources

2.1.1. Documentary Resources and Interlinear Glossed Text

MT systems have traditionally relied on parallel text. For endangered and other low-resource languages, such data may be scarce or unavailable (Tanzer et al., 2024). The available material may instead have been collected to document the language. This work commonly involves transcription, morphological segmentation, glossing, and free translation. Each step takes time and requires specialist knowledge (Ginn et al., 2023). A well-documented language may therefore still lack a large, consistent collection of sentence pairs for translation.

Interlinear Glossed Text (IGT) presents an utterance together with its analysis and translation. The SIGMORPHON shared-task records distinguish transcription, morphological segmentation, gloss, and free translation; access to segmentation depends on the track (Ginn et al., 2023). The gloss aligns linguistic analyses with source forms, while the free translation gives an ordinary target-language sentence. Figure 2.1 shows these four tiers in a training example.

Transcription	Buk	Ngr	Be	Rnge	:
Segmentation	buk	ngr	be	r-nge	:
Gloss	book	GEN1A	story	GEN1A-1MINII	:
Free translation	The Book of My Story				

Figure 2.1.: Four-tier Natugu example from the SIGMORPHON 2023 open-track (track 2) training set, record 10 (Ginn et al., 2023). Forms, capitalization, boundaries and punctuation are reproduced exactly.

In this example, *book* and *story* give lexical meanings. GEN1A labels a genitive relation, and 1MINII denotes first-person minimal in pronominal Set II (Boerger, 2022). The hyphen in *r-nge* marks the morpheme boundary and aligns with the boundary in GEN1A-1MINII. Lowercase words generally give lexical meanings, uppercase abbreviations indicate grammatical functions, and periods can join functions expressed by one morpheme. The free translation need not align word by word with the source or gloss (Ginn et al., 2023).

A gloss can give a lexical meaning or label grammatical information that has no direct English word equivalent. It therefore serves a different purpose from the free translation. Dictionaries, morphological analyzers and descriptive grammars provide further resources: a dictionary gives possible word meanings, an analyzer segments or labels forms, and a grammar describes patterns that a small example set may not contain. Their coverage and notation vary, and their analyses need not be compatible.

IGT collections differ in segmentation, spelling, abbreviations, and the treatment of fused or ambiguous morphemes (Ginn et al., 2023). Normalization can make records more consistent within a benchmark, but differences in the original linguistic analyses may remain. When using glosses for translation, it is necessary to specify which annotations are supplied and which are predicted, as well as their format.

2.1.2. Data Scarcity and Morphological Structure

The SIGMORPHON 2023 shared task on interlinear glossing (Ginn et al., 2023) shows how much these datasets can differ. Its training sets ranged from 261 tokens for Gitksan to 139,714 tokens for Arapaho. Most languages had roughly 2,000 to 15,000 training tokens. For Gitksan, the word- and morpheme-level out-of-vocabulary rates on the test set were 79.9% and 41.2%, respectively. The organizers linked these high rates to the exceptionally small training set (Ginn et al., 2023). These figures come from one glossing benchmark. They show why a dataset’s size, vocabulary coverage, annotations, and train–test split matter more than the broad label *low-resource* when interpreting results.

A word may combine a lexical stem with several grammatical morphemes. The same components can produce many word forms, some of which may be absent from a small training set. Analyzing these components can help a system handle forms it has not seen as whole words. The usefulness of this analysis depends on the language, its accuracy, and the translation model’s ability to use it.

The SIGMORPHON languages differ in how they express grammatical information. Gitksan ranges from analytic to synthetic and has productive derivation and compounding. Lezgi is highly agglutinative and suffixing, with features such as case stacking. Natugu is primarily agglutinative and has complex verbal morphology, while Tsez

has rich agglutinative suffixation (Ginn et al., 2023). A measure such as average morphemes per word can describe part of this variation. It cannot explain the particular grammatical patterns in each language.

Segmentation identifies components, glosses describe lexical and grammatical functions, and rules explain how those functions combine. The translation model must use this information to produce an English sentence. Making a word’s structure explicit does not ensure that it will be translated correctly.

2.2. Gloss Prediction and Gloss-Based Translation

2.2.1. Automatic Gloss Prediction

Task conditions. Automatic interlinear glossing predicts annotations that would otherwise require expert work. The SIGMORPHON 2023 shared task supplied transcriptions and, except for Nyangbo, translations. Systems could use the translations, although neither the baseline nor the winning Tü-CL system (Girrbach, 2023) did so. The open track supplied gold morphological segmentation. It also allowed external resources, except additional glossed text in the target language. The closed track supplied no segmentation and allowed only the shared-task data. Systems in this track had to handle morpheme boundaries while assigning glosses. Both tracks measured whole-word and morpheme-level accuracy (Ginn et al., 2023). Their scores thus reflect different combinations of segmentation and glossing ability.

Submitted systems included hard-attention models, conditional random fields combined with neural networks, recurrent models, and pretrained transformers. Some also used data augmentation or external data where permitted. In the open track, morpheme accuracy was generally higher than word accuracy. The closed track generally showed the reverse pattern (Ginn et al., 2023). These results depend on the task’s languages, splits, and normalization. They also show why a glossing score needs a clear evaluation unit and a statement of whether gold segmentation was available.

Multilingual gloss prediction. The GlossLM corpus (Ginn et al., 2024b) combines six IGT collections covering approximately 1,800 languages. Its roughly 451,000 instances include segmented and unsegmented versions of approximately 250,000 unique examples. The larger count therefore includes multiple versions of the same utterance. An additional normalized version maps common grammatical labels to a shared inventory and keeps labels that cannot be resolved reliably. This supports multilingual modeling while preserving some differences among the source annotations.

GlossLM continues pretraining ByT5-base, a byte-level pretrained model in the T5 family (Xue et al., 2022), on multilingual gloss data, then fine-tunes it for each language.

Evaluation on the seven SIGMORPHON languages shows improvements from shared training, with benefits that vary by language and data conditions (Ginn et al., 2024b). Its input can include a free translation when one is available. A known free translation can improve gloss prediction in some settings; it is unavailable as input when the new sentence’s translation is the task to be solved. The input given to the gloss predictor is therefore as relevant as the model itself.

Support selection. Support selection matters for a related but distinct task. Ginn et al. (2024a) prompt language models to produce interlinear glosses for Gitksan, Lezgi, Natugu, and Uspanteko. Their comparisons of demonstration counts and retrieval strategies usually favour retrieved over random examples, with variation by language and count. Gitksan, Lezgi and Natugu are shared with the present study; their fourth language is Uspanteko rather than Tsez, and shared language names do not imply identical records or splits. Their final test prompts omit query translations, and the reported outcome is gloss accuracy, not English translation quality. The 21 fixed supports in this thesis are not selected or evaluated by that retrieval procedure; the study instead suggests a separate question about support quality and predicted-query-gloss quality.

2.2.2. Trained and Prompted Gloss-Based Translation

Glosses can enter a translation system in two main ways. A trained pipeline can use them as an intermediate step between the source and target languages. An in-context system can instead include them in a pretrained model’s prompt. These approaches require different training resources and pass the analysis to different components.

Trained pipelines. The pipeline of Zhou et al. (2020) starts by analyzing a source sentence into lemmas and morphological tags. Dictionary-derived alignments map the lemmas to target-language interlinear glosses, which serve as the pivot between the analyzed source and target sentence. A trained gloss-to-target model then produces the translation. The experiments use resources beyond the small task datasets. For Turkish, 865 task lines were supplemented by 57,608 parallel lines for dictionary alignment. This training and resource use differs from prompting a fixed pretrained model with a few examples, so the results cannot be compared directly.

The intermediate analysis makes some errors visible. The authors report problems with normalization, segmentation, morphological disambiguation, and dictionary coverage. Generated sentences may also add unsupported content, repeat it, or express it incorrectly (Zhou et al., 2020). Errors from an earlier stage can pass to later stages, which must work with the analysis they receive.

Gloss-informed prompting. In-context learning supplies information through a Large Language Model (LLM)’s input without updating its parameters. The prompt may contain an instruction, example translations, or linguistic resources. In GrammarMT, zero-shot prompting supplies an instruction without demonstrations. Zero-shot chain-of-thought also requests intermediate reasoning, while few-shot prompting includes source–translation pairs. The main experiments use a fixed set of 21 demonstrations per language (Ramos et al., 2025). The translation model remains fixed, although a separate trained model may provide an input such as a predicted gloss.

GrammarMT represents each demonstration as a source–gloss–target triplet, whose gloss contains lexical lemmas and grammatical labels (Ramos et al., 2025). It defines three gloss-based prompting conditions, all with gold glosses in their support demonstrations. The query is the new sentence to be translated. Gloss-shot requests only its translation, with no supplied or requested query gloss at inference. Explicit Chain-gloss asks the translation model to generate an intermediate query gloss before translating. ModelGloss supplies a query gloss predicted by a separate model, GlossLM. The difference is therefore how the query is analyzed, rather than whether any gloss information appears in the prompt.

In ModelGloss, the translation model receives the predicted query gloss as input. This gloss provides an analysis of the source sentence, not a finished translation. An imperfect gloss may still provide useful cues. Even an accurate gloss leaves the translation model to handle word order and natural English phrasing. The result depends on both the gloss and the model’s use of it.

Published findings. In the Llama-3 70B experiments on Gitksan, Lezgi, Natugu, and Tsez, ModelGloss had the highest BLEU and chrF++ among GrammarMT’s gloss-shot, chain-gloss and model-gloss methods in each language. Gloss-shot and chain-gloss gave mixed results, and chain-generated glosses could be inaccurate. A broader oracle experiment with the same model supplied gold test glosses. Across the evaluated languages, this improved translation over few-shot prompting on average (Ramos et al., 2025). Gold glosses are unavailable for unannotated test sentences. The oracle therefore measures the benefit of having this annotation in that setting, not the performance of a method that can translate without it.

GrammarMT’s word-gloss-accuracy analysis compares the translation model’s generated glosses with GlossLM’s predictions and relates the more accurate external annotations to the stronger translation results (Ramos et al., 2025). This supports gloss quality as an explanation in that setting, while distinguishing linguistic annotation accuracy from English translation performance.

The BLEU and chrF++ translation metrics used in these comparisons are defined in Section 4.3. GrammarMT also tests the contribution of grammatical labels beyond the lexical meanings in a gloss. Its oracle-gloss versus oracle-empty ablation removes grammatical labels from gold glosses. This reduced validation BLEU for Gitksan, Natugu, and Tsez, but slightly improved it for Lezgi (Ramos et al., 2025). The labels helped in three of the four languages. Their value was not uniform, and the comparison does not show that all labels are equally useful.

The study mainly examines translation into English and provides limited direct evidence about how the model uses gloss-shot demonstrations. It also reports errors in generated glosses. ModelGloss was excluded from the additional low- to high-resource sets built from GlossLM training data, since prior exposure would have biased the comparison (Ramos et al., 2025). The reported ranking need not hold for other models, languages, or resources.

The experiments in this thesis adapt these methods using 21 fixed gold-glossed training examples per language. Gloss-shot supplies no test gloss, explicit Chain-gloss generates one, and ModelGloss receives an external prediction. No primary condition supplies a gold test gloss; the separate supplementary diagnostic later substitutes one in the same ModelGloss query field (Section 4.3.5). The demonstrations have gold training glosses, whereas primary queries have generated or externally predicted glosses only where the method requires them. The models, decoding, support selection, and prompt wording differ from the paper. Chapter 4 defines these adaptations and the added grammar-context conditions.

2.3. Grammar Books and Linguistic Descriptions as Context

2.3.1. Resource-Based Translation Context

Each type of context provides different information. A parallel example shows a translation, a dictionary entry gives possible meanings, and a gloss analyzes a particular form. A grammar passage can state a rule that applies to other sentences. These resources can supplement a small set of demonstrations, but they are not interchangeable. A useful comparison needs to state what each prompt supplies and hold the translation task constant.

Linguistic descriptions. Grammar descriptions can explain patterns beyond a single glossed sentence. LINGOLLM combines available analyzers, dictionaries, and grammatical descriptions across eight endangered or low-resource languages before calling the language model. Its results therefore concern this combined method, not grammar text alone. Scanned documents were converted to text through Optical Character

Recognition (OCR). GPT-4 summarized grammar texts that exceeded the context limit (Zhang et al., 2024). Its few-shot baseline uses three fixed validation examples; the full method adds linguistic descriptions and symbolic preprocessing. The translation model received the processed text rather than the original page images.

LINGOLLM evaluates Gitksan, Natugu and Tsez from the shared-task corpus used here, using supplied glosses and samples of 100 sentences. It does not evaluate Lezgi. Its results are not direct baselines for the present cohorts or evaluation settings (Zhang et al., 2024). Resource coverage also differs: the appendix lists no grammar for Tsez, while its Natugu entry cites Alfarano’s *Grammaire du Nalögo*, rather than the Boerger Natugu sketch used here (Zhang et al., 2024). The Tsez result therefore cannot be read as evidence for a grammar-book gain, and a shared language label does not imply use of the same grammar resource.

Morphological information can also help a model use a dictionary. In a Gitksan ablation, LINGOLLM found that adding morphological features to dictionary information improved translation (Zhang et al., 2024). A dictionary gives possible meanings for a stem, while segmentation and grammatical labels describe its role in a particular word form.

LINGOLLM’s chapter-level ablation found different effects for different grammar sections. Some morphological descriptions may repeat information already supplied by the analyzer. The eight languages also differed in dictionary quality and analyzer availability, and the experiments used romanized scripts. Its explicit script-normalization example concerns Manchu dictionary forms written with *q/x/v* versus input forms with *c/š/ü*; it is not evidence about Lezgi transliteration. The study reports incompatible analyses or spellings, as well as problems with rare scripts and complex page layouts (Zhang et al., 2024). The conversion and selection of grammar content can therefore affect what information reaches the model.

Translation from a grammar book. The MTOB benchmark (Tanzer et al., 2024) studied Kalamang using a 573-page grammar, a 2,531-entry word list, and a small parallel corpus. Kalamang was reported to have fewer than 200 speakers and virtually no web presence.

The grammar contains more than 1,000 examples. The separate parallel dataset contains 500 sentence pairs, divided into 400 training and 100 test items. In the experiments, grammar passages retrieved by lexical similarity were less useful than retrieved sentences and word-list entries. Shared words did not reliably identify the relevant grammatical explanation. Longer grammar contexts selected manually were more useful when combined with the other resources (Tanzer et al., 2024). This comparison

is specific to Kalamang and the tested settings; it does not establish that one resource type is always better.

MTOB combines a word list, retrieved parallel sentences, retrieved grammar passages, and manually selected long contexts. Its approximately 50,000-token context contains a grammar overview, a morpheme table, and the full word list. Its approximately 100,000-token context contains selected grammar-book chapters (Tanzer et al., 2024). The resource amounts and selection strategies differ from the fixed demonstrations and prepared grammar packages evaluated here.

The best MTOB model scored below the single human participant, although the model and human protocols differed. The human retrieved from 350 training pairs and translated two separate 50-sentence test halves. Model conditions could retrieve from 450 pairs (Tanzer et al., 2024). The authors also note that one grammar may not contain everything needed for translation and that prior web exposure cannot be ruled out completely. MTOB’s text-based evaluation differs from Kalamang’s mainly oral use (Tanzer et al., 2024). These limits affect how broadly the benchmark results can be interpreted.

Complementary resources. Adding a grammar does not always improve translation. Hus and Anastasopoulos (2024) compare dictionaries, parallel demonstrations, and grammar text using GPT-4-turbo. They evaluate 16 languages, including Gitksan and Natugu, in both translation directions. The best resource combination varies by language and direction. Words and sentences often give the best result without grammar, while grammar helps in some settings. This makes it useful to test what grammar adds to a prompt that already contains lexical and parallel resources. Comparing all resources together with a bare instruction cannot isolate that contribution.

For Gitksan-to-English, adding grammar to words and sentences produces only a small change without a significance marker. The corresponding Natugu-to-English configuration improves substantially and is marked as significantly better than the other tested resource combinations. In the reverse direction, the grammar-inclusive Gitksan and Natugu configurations also receive this marker (Hus and Anastasopoulos, 2024). Thus grammar’s contribution differs even between languages shared with this thesis and depends on translation direction. Their shared-task data were reorganized into development and evaluation partitions; their findings should not be treated as evaluations on the exact test cohorts used here.

2.3.2. Examples, Explanations and Rule Use

Examples and explanations. A grammar book itself contains several types of information. Aycock et al. (2025) separate parallel examples and word pairs from explanatory text in experiments on Kalamang, Nepali, and Guarani. Most of the observed translation benefit comes from the parallel material. The full books do not consistently outperform these subsets. For the tested languages and models, better translation after reading a grammar therefore need not mean that the model applied its explanations. An example sentence can help as a translation demonstration even if the accompanying rule is not used.

Marmonier et al. (2025) compare dictionary information with added grammatical excerpts and incidental bitexts for constructed encrypted languages. Their contrasts show that explanations can contribute information beyond incidental examples for some tested phenomena, while performance falls on more complex forms. The grammar excerpts themselves still contain examples. The result leaves a role for explanations, although it does not predict a gain for the natural-language documents or vision-language models evaluated here.

Explanatory material can still help with other tasks. Aycock et al. (2025) also evaluate typological questions, grammaticality judgments, and glossing. Combining typological information with parallel examples improves some linguistic-task results, including glossing compared with parallel examples alone or the full book. The useful content depends on the task and evaluation measure. Translation scores alone cannot show whether a model has learned a grammar’s explanations.

Finding and applying rules. Finding a relevant rule and applying it correctly are separate problems. Zhang et al. (2025) study both in *Read it in Two Steps*, using rule-annotated Chinese–Zhuang sentence pairs and supplied lexicons. Assessing rules individually improves retrieval in their setting. Representing rules as pseudocode can also help retrieval and application. Some conditions supply the annotated relevant rules, so their results depend on that assistance. The model receives the pseudocode as context; it is not executed as a translation program. The study separates three questions: whether grammar is supplied, whether a relevant rule is selected, and whether the model uses it correctly.

2.4. Textual and Visual Grammar Context

2.4.1. Vision-Language Model Families

A VLM can condition its text output on images as well as text. For a grammar document, the image provides visual features of the page, including letters, relative positions, table boundaries, and interlinear alignment. The language component must relate these features to the translation instruction and source sentence. The families below share multimodal input capabilities, but differ in architecture and interface; these features do not establish their accuracy on grammar documents.

Qwen3-VL. Qwen3-VL (Qwen Team, 2025a) is an image-text-to-text family with Instruct and Thinking editions. Its technical report describes a dynamic-resolution vision encoder, a feature merger, and visual features entering a Qwen3 language backbone at several depths (Qwen Team, 2025b). DeepStack combines features from multiple levels of the vision transformer (Qwen Team, 2025a).

Qwen3.5. Qwen3.5 (Qwen Team, 2026) combines a vision encoder with a causal language model to process image-text inputs and generate text. The 9B member has a 32-layer language component that combines Gated DeltaNet and gated attention.

Gemini 2.5 Flash-Lite. Gemini 2.5 Flash-Lite (Google, 2025) accepts text, images and other multimodal inputs and produces text. Its interface offers optional thinking and structured output. These documented capabilities describe how it can be used without identifying proprietary architectural details.

Chapter 4 identifies the configured model checkpoints, service route and decoding choices used in this thesis.

2.4.2. Visual Translation and Document Understanding

Translation from images. Visual text translation and document parsing expose different parts of this problem. Salesky et al. (2021) train a translation system on rendered images of the source sentences themselves, retaining visual character patterns through an image encoder. Their supervised experiments use TED and news parallel corpora, rather than the extremely limited documentary datasets considered here. In VISTRA, Salesky et al. (2024) compare translating English text in natural images into German, Spanish, Russian and Mandarin Chinese through OCR and machine translation with

direct image-based translation; error types and downstream systems affect the comparison. Here the source sentence remains text and a grammar page is supplemental, so neither study establishes robustness of the configured models.

Document structure. Page layout can help explain grammatical information. A paradigm table links forms to row and column labels, while an interlinear example aligns the source with gloss and translation lines. Converting a page to text requires a reading order and a way to preserve these links. Hus and Anastasopoulos (2024) report that OCR conversion can lose relative-position information in tables and multi-column documents. An image keeps the page layout available, but the model still has to read its symbols and understand their relationships.

For document structure, Lee et al. (2023) pretrain Pix2Struct to parse screenshots into simplified HTML while preserving layout through variable-resolution image inputs. Ouyang et al. (2025) evaluate document text, tables, layout and reading order separately in OmniDocBench. Both evaluate document understanding rather than low-resource translation. Together these works distinguish recognizing symbols from preserving their structural relations, without diagnosing any local translation error.

Linguistic symbols. Linguistic symbols can be difficult to recognize. Okabe et al. (2026) evaluate recognition of the International Phonetic Alphabet (IPA) using data from Japhug and Kagayanen grammars and synthetic material. They use rendered line images, including images degraded under controlled conditions. Off-the-shelf OCR systems and Qwen2.5-VL struggle with IPA symbols. Fine-tuning on synthetic data improves recognition in the tested settings. However, the study assumes that text has already been split into lines and does not evaluate real scanned pages (Okabe et al., 2026). Its results concern controlled symbol recognition. They do not establish how well a model reads a complete grammar page or uses it for translation.

2.4.3. Grammar-Context Representations

Direct image conditioning differs from a separate OCR pipeline that first converts a page into a text string. It leaves page structure available to the model, but visual pre-processing and recognition still determine what the model can use. A page image may retain an aligned paradigm while making small diacritics difficult to distinguish. Conversely, a text representation can make symbols explicit while losing their spatial relationships. In this thesis, the source sentence remains textual and the images provide additional grammar context. The model must read that context, identify relevant information, and use it for English translation; correct symbol recognition alone does not establish correct grammatical application.

Relevant visual context. Images may also carry useful or misleading context. Guerra and Khrapunova (2026) compare matching, random and, for one task, contradicting images in translation prompts; benefits vary by task and model, and the contradicting condition can harm translation. Their images are scene context rather than grammar pages, and their Gemini 2.5 Flash differs from the Flash-Lite service here. Their evaluation uses one generation, with greedy decoding for open models and provider defaults for proprietary ones, so it does not characterize generation variability. The comparison motivates checking image content rather than assuming that image presence alone helps. No irrelevant- or misleading-grammar-image control was run in this thesis.

Content and presentation. Preparing grammar context can change both its content and its presentation. A summary may focus on rules and omit examples. A selected page may keep examples, labels, and nearby explanations. Rendering the same summary as an image changes its modality. Replacing it with a different page also changes the information supplied. The examples, explanations and rule-use comparisons in Section 2.3.2 show why the information supplied matters alongside its presentation. If context formats differ in vocabulary, examples, or rules, a translation difference cannot be attributed to visual processing alone.

These studies address different parts of the translation problem: recognizing source text, reading document structure, judging image relevance, predicting glosses, and using grammatical explanations. MTOB also proposes rendered-PDF input as a multi-modal extension (Tanzer et al., 2024). The present study tests whether selected grammar packages add value to gloss-based translation prompts across four languages and three configured models. It compares the supplied text and image packages as complete inputs, since their information content is not established as equivalent. A separate earlier MTOB study examines retrieved and longer grammar text. Chapters 3 and 4 define both protocols.

3. Data and Grammar Resources

The study translates sentences from Gitksan, Lezgi, Natugu, and Tsez into English. It uses SIGMORPHON translation and gloss records alongside material prepared from five grammar documents. This chapter describes those sources and their coverage. Chapter 4 explains how they enter the prompts and evaluations.

3.1. Languages and Source Documents

The collection has two Gitksan documents and one for each other language. These five available, selected documents are not an exhaustive survey of the languages' grammars. A *source PDF* is one of the documents; a *derived material* is a summary, table, sheet, or image prepared from it. PDF page numbers refer to file pages, which sometimes show two printed pages. Selected original-page images reproduce those pages rather than rewritten summaries.

Language	Cited source	PDF pages	Selected pages	Document scope	Coverage in available PDF
Gitksan	(Brown et al., 2016)	12	9	Complete article	Phonetics, phonology, glossed narrative
	(Rigsby, n.d.)	9	6	Introductory excerpt	Dialects and structural overview
Lezgi	(Haspelmath, 1993)	59	9	Grammar excerpt	Introduction and opening phonology
Natugu	(Boerger, 2022)	71	9	Grammar sketch	Multiple grammatical domains
Tsez	(Comrie and Polinsky, n.d.)	40	9	Manuscript document	Phonology, morphology, syntax

Table 3.1.: Five source PDFs. Selected pages count original-page images used in the experiments. The Lezgi Cyrillic-form material is a second derived version of the same source and adds no original-page images.

3.1.1. Gitksan

Gitksan is a Tsimshianic language of British Columbia. The shared-task corpus contains three interlinearized stories, with a different story in each split (Ginn et al., 2023). The complete twelve-page *Gitksan* article by Brown et al. (Brown et al., 2016)¹ describes Eastern Gitksan phonetics, phonology, and orthography and includes an interlinear narrative with English translation. It gives limited coverage of noun and verb paradigms or general clause structure. The separate nine-page *Excerpts from the Introduction to Gitksan Grammar*, attributed to Rigsby (Rigsby, n.d.), covers language relationships, dialects, and introductory structural features, mainly for Eastern Gitksan. Its archive catalogue identifier is TS966R1986b.² The excerpt is a distinct source, not the remainder of the Brown article.

3.1.2. Lezgi

Lezgi belongs to the Nakh-Daghestanian family. The shared-task dataset represents the Qusar dialect of Azerbaijan (Ginn et al., 2023). Haspelmath’s *A Grammar of Lezgian* (Haspelmath, 1993)³ mainly describes the standard language based on the lowland Güne dialect. The available PDF is a 59-page excerpt of the longer book, including front matter and ending at printed page 39. It contains an introductory grammatical survey, notation, and opening phonological description. Later chapters listed in its contents are absent from this PDF. The dataset’s Lezgi sentences use Cyrillic while the source describes forms in Latin transcription. A second derived version presents selected Lezgi forms in Cyrillic while its explanations remain in English. The original strings *gun* and *gu-zwa*, for example, have Cyrillic-form counterparts in the retained variant; noisy negative forms such as *gu-zwa-6* and *ga-na-ö* also change. This is not established as strictly mechanical transliteration. Both versions use the same grammar document and test set.

3.1.3. Natugu

Natugu, written Natügu in the shared-task paper, is an Austronesian language of the Reefs–Santa Cruz group. Its data include narrative and written texts (Ginn et al., 2023). Boerger’s 71-page *A Grammar Sketch of Natqgu [ntu]: An Oceanic Language of*

¹<https://doi.org/10.1017/S0025100315000432>

²Alaska Native Language Archive catalogue search: <https://www.uaf.edu/anla/search/index.php?name=Rigsby%2C+Bruce+J..> This identifies the excerpt title; it does not establish that the local scan is byte-identical to an archive download.

³<https://doi.org/10.1515/9783110884210>

Santa Cruz, Solomon Islands (Boerger, 2022)⁴ identifies Natqgu and Natügu as spellings of the language. It covers noun phrases, verbal morphology, clause types, negation, and gloss abbreviations. This sketch has broader grammatical coverage than the short Gitksan documents or the available Lezgi excerpt.

3.1.4. Tsez

Tsez has extensive agglutinative suffixation. Its dataset comes from the Tsez Annotated Corpus Project (Abdulaev et al., 2022; Ginn et al., 2023). The available forty-page *Tsez* manuscript by Comrie and Polinsky (Comrie and Polinsky, n.d.)⁵ describes the Tsebari subdialect of Asakh unless noted otherwise. It covers transcription, phonology, nominal and verbal morphology, syntax, and abbreviations. That stated dialect does not establish a match to every corpus sentence.

3.2. Translation and Gloss Data

The SIGMORPHON 2023 shared task (Ginn et al., 2023) assembled interlinear data for automatic glossing. A record includes a source-language transcription, an interlinear *gloss* giving lexical meanings and grammatical labels, and an English *free translation* expressing the sentence meaning without morpheme-by-morpheme alignment. Depending on the track, a glossing system could receive gold morphological segmentation; the English translation could be an input to gloss prediction. Here the source transcription is the translation input and the English free translation is the evaluation *reference*. A dataset-provided linguistic annotation is a *gold gloss*; this identifies its origin, not an assertion that every analysis is unambiguous. GrammarMT (Ramos et al., 2025) previously used these resources for translation with gloss-bearing examples.

3.2.1. Dataset Provenance and Split Sizes

The shared task has training, development, and test partitions. Gitksan is split by story, while Lezgi, Natugu, and Tsez were split by sentence at approximately 80/10/10 (Ginn et al., 2023). Its Track 1 supplies the source transcription without a separate gold segmentation input. In the retained files, *uncovered* variants have the gloss annotations used here, whereas the present *covered* files have blank gloss fields. These names identify shared-task input variants, not whether a grammar book covers a sentence. The

⁴Source PDF: https://www.langlxmlanesia.com/_files/ugd/4bb486_7e4e85dfaba047998caf2a3c50be1c28.pdf.

⁵Author-hosted manuscript: https://www.mariapolinsky.com/wp-content/uploads/2013/12/Tsez_Final.pdf.

organizers standardized delimiters and gloss conventions and removed incompatible token alignments before release (Ginn et al., 2023). Table 3.2 counts records in the retained annotated, uncovered Track 1 inputs. The project’s own loader accepts a record when its source and first English free translation are nonempty, even when its gloss is blank. Gitksan’s annotated training count uses the pinned official data_v1 annotations; the other counts use locally retained annotated uncovered files. The counts reflect this project’s parsing, not a new shared-task corpus size. A literal nan passes the nonempty test; three such Lezgi references require a later evaluation mask.

Language	Training	Development	Test
Gitksan	31	—	37
Lezgi	701	88	87
Natugu	791	99	99
Tsez	3,558	445	445

Table 3.2.: Accepted records in annotated, uncovered Track 1 inputs, with Gitksan training from pinned official data_v1 annotations (annotation provenance in Appendix D). The Gitksan development file is absent from the local input snapshot; the dash does not mean the benchmark has no development partition. Lezgi has 87 parsed test records; 84 have non-invalid English references and 83 enter primary quality evaluation after the known support-reference overlap exclusion.

Tsez supplies 3,558 training and 445 test records here, compared with Gitksan’s 31 and 37. Dataset size and annotation differences matter when comparing languages. The project’s loader reads blank-line-separated source, gloss, and translation fields; it takes the first translation before `||` and does not require a nonempty gloss. The study draws its gold-glossed examples from annotated training records, including the pinned Gitksan annotations. All selected examples have nonempty gold glosses, including those in no-book conditions. Chapter 4 gives their selection, order, and test-overlap controls.

3.2.2. Reference Quality and Predicted Glosses

The 87 Lezgi test records include literal nan in the English-reference field at zero-based indices 37, 62, and 81. They remain generation items. Another test reference, at index 11, is embedded in a selected support translation. The uniform primary quality mask excludes all four items, leaving 83 records, as defined in Section 4.3.1. The 87 records

contain 85 distinct source strings and 86 distinct source–reference pairs. Exact-string comparisons with all training items find test-source reuse in Lezgi and Tsez; those full-training matches do not by themselves describe what the selected prompt examples contain. Primary ModelGloss uses test glosses predicted by the *GlossLM model*, rather than gold test annotations or entries taken directly from the GlossLM corpus (Ginn et al., 2024b). The saved vectors originate from the no-reference model-prediction resource distributed for GrammaMT (Ramos et al., 2025).⁶ The loader language identifiers are Gitksan git, Lezgi lez, Natugu ntu, and Tsez ddo. An explicit Chain-gloss prompt instead asks the translation model itself to produce an intermediate test gloss. The saved prediction vectors and limits of their source alignment are discussed in Chapter 4 and Appendix D.

3.3. Prepared Grammar Representations

I used GPT-5.6 Sol to help prepare summaries, tables and cheat sheets from the source PDFs, with a shared prompt adapted by language and file path. The prompt requested source-page links, faithful rendering and consistency checks. Logs of that model run are not retained, so the requested checks cannot be verified from them. Appendix A gives a short parameterized excerpt of the supplied template.

The prepared materials have four resource types: a text summary; structured summary tables in TXT or JPG; a cheat sheet in TXT or JPG; and selected original-page JPGs. Their six context formats are summary TXT, table TXT/JPG, sheet TXT/JPG, and original-page JPG. The TXT and corresponding Markdown files are byte-identical for all 18 prepared text assets. The 19 derived JPGs differ in rendering: Brown’s table and sheet display literal Markdown table syntax, while the other 17 images contain drawn tables. Their shared textual source does not guarantee complete visual content equivalence or legibility.

3.3.1. Text Summaries, Tables, and Cheat Sheets

The text summaries group grammatical information under headings and give PDF-page references for selected claims. The Brown article summary, for example, includes WORD_ORDER and PHONOLOGY_AND_ORTHOGRAPHY; an orthographic row points to PDF pages 2 and 4. Its scope note says the article lacks full case, pronoun, and tense–aspect–mood paradigms, general question formation, and a general word-order rule. The note describes source coverage; it does not verify the summary’s paraphrases.

⁶GlossLM repository, historical frozen/glosslm-v1 branch: <https://github.com/lecs-lab/polygloss>. The four original local prediction CSV caches are not retained.

Source/variant	Summary TXT	Tables TXT	Sheet TXT	Tables JPG	Sheet JPG
	approximate words			image pages	
Gitksan Brown	1,057	267	156	1	1
Gitksan Rigsby	792	162	143	1	1
Lezgi original	1,602	451	310	2	1
Lezgi Cyrillic forms	1,602	451	310	2	2
Natugu	1,720	537	277	2	2
Tsez	1,747	497	455	2	2

Table 3.3.: Sizes of the retained derived materials. Approximate word counts are whitespace-separated tokens containing at least one Unicode letter or digit, excluding punctuation-only Markdown; they are not model-token counts. Each derived JPG page is 1,900×2,600 pixels. The six rows yield 19 derived JPGs. Selected original-page images are counted separately in Table 3.1.

Summary lengths range from about 792 to 1,747 words, with no common numerical limit imposed by the retained builders.

Summary tables place forms, functions, English translation cues, and PDF-page references in labelled blocks. Cheat sheets make a still shorter selection of forms, clause cues, and gloss abbreviations. For example, a Brown table gives IPFV, PROSP, and CN rows, while its sheet lists compact *dim*, *nee*, and *dii* cues. Appendix C reproduces short material excerpts and image crops so that the distinction is visible. Each Gitksan source has its own materials; the Rigsby excerpt text for Gitksan is not a fuller version of the Brown article.

Longer prepared tables and sheets continue onto a second image page. Natugu and Tsez have two pages of each because one prepared material spans both pages. The original Lezgi table has two JPG pages, but its cheat-sheet JPG has one. Inspection of the derived images found that this sheet omits the last three mini-gloss rows and a four-example section present in its TXT. The two-page Cyrillic-form sheet includes those sections. The original Lezgi TXT/JPG comparison therefore changes coverage as well as modality; the original/Cyrillic JPG-sheet comparison changes coverage, script, and selected forms. Neither contrast isolates the cause of a score difference. No other inspected pair showed a missing substantive section or row, although character-level equivalence and model perception were not tested.

3.3.2. Selected Original Pages

The experiments use 42 selected page images: Brown PDF pages 1, 2, 5–11; Rigsby 1 and 3–7; Lezgi 19, 20, 24–26, 28, 30, 34, 43; Natugu 3, 8, 23, 25, 37–39, 50, 58; and Tsez

8–10, 12, 15, 16, 20, 21, 25. These are PDF file-page identifiers, and the second Lezgi derived variant does not add another set. The original images preserve the visible page layout while supplying only a selection of the document.

The author-supplied preparation prompt suggested ranking morphology tables, paradigms, glossed examples, case, agreement, and clause material while avoiding front matter and redundant pages. The retained source-specific implementations make fixed choices: Gitksan uses wanted lists with a fallback, and Lezgi, Natugu, and Tsez use specified page sets. Natugu’s list covers varied grammatical topics instead of repeated high-ranking valency pages; Tsez’s list has topic reasons for each page. These records explain the selected material but do not show tuning against translation scores. Chapter 4 describes the fixed image order in prompts.

An overlap screen compared all 668 test records with embedded text from the same-language PDFs and all 18 prepared TXT files. It used complete-string matching, both literal and after Unicode NFKC normalization, case folding, and removal of non-alphanumeric characters. No full test source or valid English reference matched text from the 42 selected pages or the prepared TXT files. Two Natugu test examples did occur on *unselected* PDF pages: test index 20 on page 26 and index 50 on page 47. The first has source-form and punctuation differences; the second reuses the source example with a differently worded English translation. Extraction can miss symbols, and the screen did not test partial, transliterated, fuzzy, semantic, or cross-page overlap. It also did not cover all training/development items or pretraining exposure. The negative result therefore does not establish that the selected images contain no test information. Appendix E describes the implications for the earlier MTOB study’s full-text contexts.

3.3.3. Extracted PDF Text

The earlier MTOB study also uses embedded text from the five source PDFs. Its extraction normalizes whitespace and removes repeated marginal lines without a new OCR pass. It retains source-page references but loses layout and sometimes characters. For example, the extracted Rigsby excerpt text for Gitksan contains transcription noise; this is an example of extraction limits, not a complete account of its defects. Appendix C shows a short extracted passage. Retrieval and long-context construction are methods of the earlier MTOB study and are defined in Appendix E.

4. Methodology

The primary study compares sentence-to-English translations with grammar context against matched no-book baselines. Each pair uses the same model, prompting method, training examples and test sentences. This chapter describes the prompts and conditions, then explains how the saved translations were evaluated.

4.1. Translation Task and Gloss-Based Methods

4.1.1. Translation Task and Matched Comparison

The source languages are Gitksan, Lezgi, Natugu, and Tsez. The data come from the SIGMORPHON 2023 glossing resources (Ginn et al., 2023), following their use for translation in GrammarMT (Ramos et al., 2025). Each query contains a source sentence whose English reference is reserved for evaluation. Chapter 3 describes the data and grammar resources. The translation models receive no task-specific parameter updates.

Every prompt contains 21 *support examples*: source–gold–gloss–English translation triplets from the training data. These few-shot examples are also called *demonstrations*. Even a *no-book* baseline contains them; no-book means no grammar-document package and no instruction to use one. A matched context comparison holds the model, prompting method, demonstrations, test sequence and decoding policy fixed. The context condition adds both grammar material and an instruction to use it. The comparison therefore measures their combined addition.

4.1.2. Gloss-Based Prompting Methods

The three methods adapt GrammarMT (Ramos et al., 2025), which also uses gloss-bearing support examples. They differ in how the *test* sentence is handled (Figure 4.1). The local models, decoding, support selection, wording and response delimiters differ from the paper.

Gloss-shot. The prompt supplies the test source without a test gloss and requests its English translation. Its gold-glossed training examples remain present.

Explicit Chain-gloss. The translation model first generates a test gloss, then the English translation. The instruction asks for English lexical meanings and grammatical abbreviations in source-word order, preserves identifiable morpheme boundaries, uses a question mark for uncertainty and forbids copying the source as its gloss. The method uses a distinct system message and requires `Gloss:` and `FINAL_TRANSLATION:` lines; `Gloss-shot` and `ModelGloss` instead request one translation line. A formatted gloss is not assumed to be linguistically correct.

ModelGloss. The prompt supplies a test gloss predicted by a separate `GlossLM` model and requests the English translation. The local instruction omits `GrammarMT`'s warning that this gloss may be inaccurate; the effect of this adaptation was not isolated in a test. The primary templates are excerpted in Appendix A.

4.1.3. Demonstrations and Query Inputs

The 21 supports have nonempty gold glosses and remain in the same training order across models, methods and context conditions. As in `GrammarMT`, the selection begins with 21 training examples (Ramos et al., 2025). `Gitksan` uses the pinned official `data_v1` revision in Table 3.2. `Gitksan`, `Natugu` and `Tsez` use original training indices 0–20. `Lezgi` uses 0–4 and 6–21: item 5 was replaced by item 21 because its source exactly matches a test source.

In the primary study, the query excludes its English reference and gold gloss. `ModelGloss` instead uses predictions from the `GlossLM model`, distributed in the no-reference test-prediction collection used by `GrammarMT` (Ginn et al., 2024b; Ramos et al., 2025). These are model outputs, not annotations from the `GlossLM` corpus. Within each model and language, an ordered test item receives the same saved predicted gloss in baseline and context conditions. Appendix D describes the positional loader and the missing original CSV caches that prevent an independent recheck of source-sentence alignment.

No selected support source exactly matches a test source. One `Lezgi` reference does occur within a support translation, as described in Section 3.2.2. The primary 83-reference evaluation excludes that item, though other semantic overlap with prompts or grammar materials remains possible.

Figure 4.1 shows the three query routes and the optional grammar package. The English reference enters evaluation; `XCOMET-XXL` also uses the source sentence, as explained in Section 4.3.2.

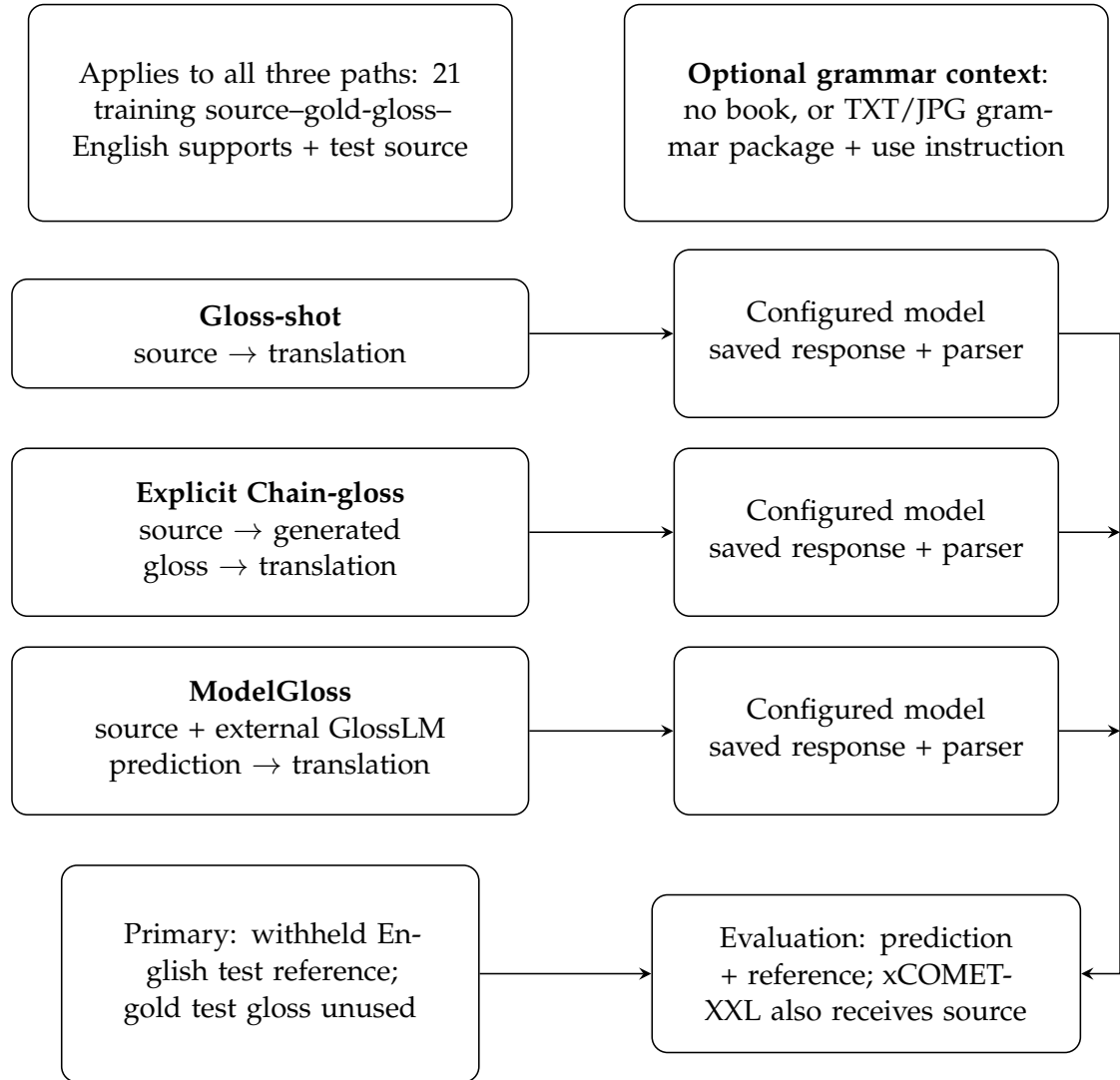


Figure 4.1.: Shared supports and the optional context choice apply to each separate method path. Only ModelGloss receives an external predicted test gloss.

4.2. Models and Grammar-Context Experiments

4.2.1. Models

The configured Qwen models are Qwen/Qwen3-VL-8B-Instruct (Qwen3) and Qwen/Qwen3.5-9B (Qwen3.5). These are the released checkpoint names recorded in the local run configurations. Both accept image and text input and produce text (Qwen Team, 2025a, 2026). The Qwen3-VL technical report describes its family architecture (Qwen Team, 2025b); it does not establish the behavior of the specific saved run. Chapter 2 describes the models separately.

The configured Gemini identifier is google/gemini-2.5-flash-lite (Gemini), which accepts image and text input and produces text (Google, 2025). It names the OpenRouter route to the Google provider rather than an immutable provider-weight revision. This common input capability permits a text/image comparison across the three systems, while their implementations and decoding differ. Appendix D records the limits of the available design history.

4.2.2. Grammar Context and Condition Matrix

Context conditions use five original grammar sources: two for Gitksan and one for each other language. Each source has six formats: summary text, table text, table images, cheat-sheet text, cheat-sheet images, and original-page images. An additional Cyrillic-form Lezgi bundle has five derived formats without a second original-page condition. Section 3.3 describes their preparation and source-specific sizes. Table 4.1 summarizes what enters a prompt.

TXT/JPG table and sheet pairs differ in visual layout and tokenization. The original Lezgi sheet JPG also omits three mini-gloss rows and four examples present in TXT; the two-page Cyrillic-form sheet includes them. That pair changes coverage as well as modality. The original/Cyrillic sheet-image contrast changes script, forms, and coverage. Summaries and selected pages differ in wording, examples, and length. There is no common token budget or demonstrated exact content equivalence. The comparisons therefore concern complete context packages.

The supplementary input-size analysis records the number of token IDs in the actual processor input for existing Qwen runs. It also reconstructs first-prompt text, grammar-only text and visual token counts with pinned tokenizers and processors; these counts cannot be added to obtain the logged total. Appendix D.4 defines the quantities, and Table F.10 summarizes their ranges without treating length as a demonstrated cause of score changes.

A *condition* fixes a model, language, prompting method, grammar-source variant and representation for one ordered test sequence. A *sentence-condition record* is one saved

4. Methodology

Representation	Modality	Content	Recorded size	Pair relationship
Summary TXT	Text	Prepared prose on selected grammar topics	792–1,747 words	No image pair
Tables TXT	Text	Prepared forms, functions and examples	162–537 words	Source for table JPG
Tables JPG	Image	Rendered prepared tables	1–2 pages	Layout differs from TXT
Sheet TXT	Text	Compact forms and grammar cues	143–455 words	Source for sheet JPG
Sheet JPG	Image	Rendered prepared sheet	1–2 pages	Original Lezgi JPG omits TXT content
Original pages JPG	Image	Selected source-document pages	6–9 pages	Different coverage from summaries

Table 4.1.: Six representation types per original grammar source. Word counts are approximate, not model-token counts; image sizes are page counts per source. The five derived Cyrillic-form Lezgi representations share the original selected-page condition. Table 3.3 gives source-specific sizes.

test-item response under a condition, including a failure. Gitksan’s two grammar sources share the same language test set and method-specific baseline.

For each method and model, the sources and formats give $5 \times 6 + 5 = 35$ grammar contexts. The three methods are Gloss-shot, explicit Chain-gloss and ModelGloss, so $35 \times 3 = 105$ grammar conditions per model. Four languages with three methods give $4 \times 3 = 12$ no-book baselines per model. Thus $12 + 105 = 117$ conditions per model and $3 \times 117 = 351$ primary conditions: 36 baselines and 315 context conditions. Table 4.2 gives the model totals.

Model	Baselines	Contexts	Total	Sentence-condition records
Qwen3	12	105	117	15,999
Qwen3.5	12	105	117	15,999
Gemini 2.5 Flash Lite	12	105	117	8,733
Total	36	315	351	40,731

Table 4.2.: Completed primary experiment matrix. Record totals count repeated evaluations of the test sentences across conditions, not distinct sentences. The separate MTOB-inspired suite is excluded.

All models have 37 Gitksan, 87 Lezgi and 99 Natugu items per condition. The Qwen

models use all 445 Tsez items, while Gemini uses the first 99. Their native Tsez scores therefore cover different test sets; the common99 view aligns them on the shared prefix. Every primary condition contains its expected records, including failures. The individual primary conditions and the separate earlier MTOB matrix are listed in Appendix B.

4.2.3. Prompt Assembly and Generation

I used GPT-5.6 Sol to help prepare the summaries, tables and sheets from the source PDFs with the shared template excerpted in Appendix A. Logs of that model run are not retained. Section 3.3 describes the materials. Corresponding Markdown and plain-text files are byte-identical; text conditions use the plain-text version. Appendix D records the renderer’s dimensions and settings.

The textual prompt places demonstrations before the grammar context and test sentence. ModelGloss also inserts the predicted query gloss. Image conditions replace the grammar text with an instruction referring to attached pages. Image order follows each fixed configuration. The baseline and context prompts retain their method-specific output requirements. Appendix D specifies the backend message-block order. The Qwen models use FP32 inference on one H100 per job, with one example processed at a time and thinking disabled. Their recorded decoding samples at temperature 0.7 rather than using the greedy decoding reported for GrammarMT’s main setup (Ramos et al., 2025). Earlier development diagnostics crossed support-gloss availability with greedy or sampled decoding, giving four configurations on the same eight Tsez examples for each Qwen model. Sampled configurations reduced invalid or truncated responses, and the sampled settings were carried into the matched runs. The diagnostic varied the whole decoding policy; it did not isolate sampling or the presence penalty, or validate the full matrix. Table 4.3 gives the fixed settings. Seeds depend on the original test index, so processing order does not change the seed assigned to a sentence.

The Qwen presence penalty subtracts 1.5 from the logit of each token type already generated at each decoding step, without accumulating the penalty for repeated occurrences; prompt tokens are excluded. It can therefore affect the second line of a Chain-gloss answer after its gloss has been generated. Its differential effect across methods was not tested. Gemini has no such penalty.

The protocol retains one persisted semantic attempt per item and does not add a corrective instruction after a malformed answer. Resuming an interrupted condition processes unsaved test indices and preserves previously saved failures. Following an API transport failure, the identical request may be resent; this is distinct from seeking a revised answer. A response lost before persistence cannot be recovered from the

Setting	Qwen3 and Qwen3.5	Gemini 2.5 Flash Lite
Temperature	0.7	0
Sampling controls	top- p 0.8; top- k 20; min- p 0	No additional controls specified
Penalties	Presence 1.5; repetition 1	No additional penalties specified
Maximum new tokens	512	512
Seed	20260910 + $2i$, test index i	42
Thinking / reasoning	Disabled in chat template	Requested effort: none
Semantic attempts	One	One

Table 4.3.: Generation policies for the primary matched experiments. The decoding settings differ between the local and API backends but remain fixed within each matched comparison.

retained records. Appendix D gives the transport-attempt limit and resume procedure. The parser extracts the final translation. Known truncated, filtered, empty, reasoning-only or unextractable responses become empty predictions at their original test indices. A missing or multiline Chain-gloss is flagged but does not erase an extractable translation. Appendix D specifies the edge rules for labelled and unmarked answers.

4.3. Evaluation and Statistical Analysis

4.3.1. Evaluation Cohorts and Reference Policy

An *evaluation cohort* is the specified set of test records used to calculate a condition’s score or paired comparison. The cohorts below reuse saved predictions with different index masks; they are not independent replications.

Native cohort. Each model uses its complete configured test sequence: 37 Gitksan, 87 Lezgi and 99 Natugu items, plus 445 Tsez items for Qwen or 99 for Gemini. This describes each model’s recorded run, including failed responses.

Common99 cohort. Qwen’s Tsez records are restricted to the same first 99 used by Gemini, while other languages are unchanged. This gives a common Tsez index set for cross-model description. The first-99 subset is a fixed prefix of the native cohort, neither a random sample nor a second independent test set. Native and common99 findings therefore cannot be counted as independent confirmations.

General response retention. Known failed outputs remain empty predictions at their test indices when those indices have valid references. Comparisons are not restricted to sentences for which both systems produced an answer. This rule applies to all four languages.

Lezgi reference exclusions. Of 87 generated test items, three have literal nan English references at zero-based indices 37, 62 and 81 (Section 3.2.2). Test index 11 has a reference embedded in a selected support translation. The primary descriptive evaluation excludes these four indices, leaving $87 - 3 - 1 = 83$ records. This same mask applies to every model, method, context and metric; failed predictions at retained indices remain included. The original primary significance tests use all 87 Lezgi records and are not attached to the 83-record scores. Separate post-hoc Qwen comparisons use the 83-record mask, as described in Section 4.3.5; Gemini remains descriptive under this mask. Appendix D retains the intermediate 84-record provenance.

4.3.2. Translation Metrics

BLEU measures word overlap with an English reference and allows comparison with GrammarMT; chrF++ complements it with character and word overlap. Both evaluate the English translation. XCOMET-XXL supplies a learned quality score, subject to the source-language coverage limit below. The Lezgi scores use the reference exclusions defined in Section 4.3.1; retained failures remain empty predictions on included items.

Lexical metrics. BLEU measures modified word n-gram precision with a brevity penalty (Papineni et al., 2002). Tokenization and preprocessing affect comparability (Post, 2018). The saved SacreBLEU 2.6.0 analysis uses one reference, mixed case, 13a tokenization, exponential smoothing, and no effective-order adjustment.¹ chrF++ combines character and word n-gram F-scores (Popović, 2017); here it uses character order six, word order two, and $\beta = 2$.² Both scores use a 0–100 scale. Primary predictions and references receive no additional punctuation removal.

Neural metric. The selected neural evaluator is Unbabel/XCOMET-XXL, a learned translation-quality model from the XCOMET family (Guerreiro et al., 2024).³ Its actual

¹Saved analysis signature: nrefs:1|bs:100000|seed:20260915|case:mixed|eff:no|tok:13a|smooth:exp|version:2.6.0. This signature includes bootstrap metadata and does not describe the filtered Lezgi scores.

²Saved analysis signature: nrefs:1|bs:100000|seed:20260915|case:mixed|eff:yes|nc:6|nw:2|space:no|version:2.6.0.

³Official model card and usage example: <https://huggingface.co/Unbabel/XCOMET-XXL>.

inputs are the source sentence, retained English prediction, and English reference. XCOMET also supports reference-only scoring from the prediction and reference; that mode was not used for the retained scores. The reported system score is the arithmetic mean of item scores on the metric’s native scale, without multiplication by 100. All 351 conditions have xCOMET-XXL scores on both cohorts. XCOMET-XL remains supplementary and is not substituted for xCOMET-XXL.

The official model card does not list Gitksan, Lezgi, Natugu, or Tsez among its covered languages and warns that scores for pairs containing uncovered languages are unreliable (Unbabel, n.d.). An English reference does not remove that limitation because the scorer also conditions on the source sentence. The retained xCOMET-XXL scores and tests describe its outputs, but their interpretation as translation adequacy for these four source languages is exploratory; statistical significance cannot establish metric validity.

4.3.3. Comparison Sets and Research Questions

A paired comparison requires identical record indices, source sentences, and references. Primary baseline–context comparisons also verify that demonstrations, decoding policy, and ModelGloss inputs match. Text–image comparisons verify the same controls within each material pair. No failed sentence is dropped to obtain a common successful subset.

Scores remain identified by language, grammar source, model, method, representation, and cohort. Gitksan’s two resources use the same 37 test sentences and share a baseline, so they do not provide independent language samples. Pooling all language records into one score would give a 445-item Tsez set more weight than a 37-item Gitksan set simply because it contains more records. The 315 context–baseline contrasts and 108 TXT/JPG contrasts are each available per cohort. Primary rankings across models and methods are descriptive. The supplementary Qwen no-book tests compare methods directly, but neither analysis tests between-model or between-method interactions. Table 4.4 places the four research questions from Section 1.2 in these defined comparison sets.

4.3.4. Paired Bootstrap and Multiple Testing

Lexical paired bootstrap. BLEU and chrF++ contrasts resample the same sentence indices for both systems, with replacement and the original cohort size (Koehn, 2004; Dror et al., 2018). Corpus scores are recomputed for each of 100,000 samples; sentence BLEU values are not averaged. The stored confidence half-width describes an individual system’s bootstrap score, not an interval for the paired score difference. Be-

RQ	Comparison	Interpretation
1	36 no-book baselines on native and common99 cohorts	Observed primary performance; supplementary Qwen method tests in Section 4.3.5
2	315 matched context-baseline contrasts per cohort	Changes associated with a package plus use instruction; applicable paired tests
3	Material scores and 108 paired TXT/JPG contrasts per cohort	Representation differences; no isolated modality effect where content differs
4	Language, model and method patterns within RQs 1–3	Descriptive variation; no model-by-material or method interaction test

Table 4.4.: Primary research questions and analyses. Original primary tests do not match the Lezgi 83-reference mask; the supplementary Qwen tests are separate. Native and common99 cohorts overlap.

cause the resampling unit is the record, duplicate Lezgi strings are not grouped and document dependence is not modelled. Appendix D specifies the SacreBLEU p-value calculation and seed.

Holm correction. Holm’s sequential multiple-testing correction controls family-wise error within each specified family without requiring independent tests (Holm, 1979). The saved adjusted p-value is compared with 0.05. A signed score difference shows direction and size separately from this decision. Significance alone does not establish practical importance; non-significance does not establish equivalence.

Correction families. A *family* is the set of tests whose p-values are corrected together. Table 4.5 distinguishes the lexical and xCOMET-XXL pooling rules. Native and common99 lexical families are separate, so family-wise error is not controlled across their union. xCOMET-XL has its own families of the same sizes as xCOMET-XXL and is not pooled with it. Every saved family includes the 87-record Lezgi comparisons; omitting those from valid-reference inference does not redefine the family or recalculate adjusted values.

Neural paired bootstrap. xCOMET-XXL tests resample paired item-score differences 100,000 times. Their 95% percentile interval describes the *mean paired difference* between systems, unlike the lexical half-width for one system score. The interval is unadjusted for multiple testing. Appendix D gives the null-centred p-value rule and deterministic comparison seeds. The original primary tests do not compare models or prompting

Metric analysis	Context-baseline family	TXT/JPG family	What each family pools
BLEU + chrF++	630	216	Both lexical metrics within one cohort; native and common99 corrected separately
XCOMET-XXL	630	216	One neural metric across both cohorts, within each comparison type

Table 4.5.: Numbers of tests corrected together within each comparison type. The lexical counts apply separately to each cohort; the xCOMET-XXL counts already pool both cohorts. All stored families retain the original 87-record Lezgi comparisons.

methods against one another, or test model-by-material interactions. The supplement adds Qwen lexical method tests and a limited descriptive seed check; no human adequacy evaluation is available, and different significance decisions in two models do not test an interaction.

4.3.5. Supplementary Analyses and Diagnostics

A post-hoc supplement analyses saved Qwen3 and Qwen3.5 predictions and adds 37 Qwen3 conditions with 6,049 recorded responses, including 192 empty predictions with truncation flags. Operational failure left no saved new Qwen3.5 predictions. These conditions are separate from the 351-condition primary study and the earlier MTOB study. Appendix B lists the retained conditions once, including controls shared between diagnostics.

Analysis A compares no-book methods. Analysis B rechecks Lezgi context and paired presentation contrasts on the main 83-record mask and the overlapping 84-record sensitivity mask. Their lexical bootstrap samples distinct source texts with replacement, retaining every record of each sampled source on both sides. This groups repeated source strings into one resampling unit. The saved Holm families contain 72 method tests, 264 context tests and 96 presentation tests, pooling both Qwen models and both lexical metrics within each family. These decisions remain separate from the original primary families and finding counts.

The new Qwen3 diagnostics compare 512- and 1,024-token caps for no-book Chain-gloss (F), predicted and gold query glosses for no-book ModelGloss (G), and four selected context-baseline pairs across three random seeds (I). G changes only the final query gloss field in the existing prompt; its instructions and supports remain

unchanged. F and G have incomplete planned test families without adjusted p-values, and I is descriptive. The supplementary xCOMET-XL and xCOMET-XXL scores also have no significance tests.

Analysis C describes logged and reconstructed input sizes for the existing Qwen conditions. Appendix D.4 records runtime identities, source-cluster resampling and input-size definitions. Appendix F.3 gives the significant A/B contrasts and descriptive F/G/I outcomes, with the 84-record Lezgi and common99 Tsez views identified as overlapping sensitivities.

4.4. Reproducibility

The saved configurations fix each run’s supports, ordered test inputs, model settings, predicted glosses and context/code hashes. Saved records link source and reference to a configuration fingerprint; metric provenance links scores to prediction hashes and record counts. System and user text can be reconstructed from saved inputs and templates and checked against persisted messages. The primary query builder receives the source and, for ModelGloss, its predicted gloss, but neither test-reference nor gold-test-gloss fields. This separation does not exclude incidental support overlap, including the Lezgi case in Section 4.3.1.

Qwen generation. Primary Qwen identifiers do not pin immutable weight revisions; the retained supplementary Qwen3 runs do pin a revision. The processor’s complete image-and-text token sequence has not been independently reconstructed, and the preserved files do not constitute a complete portable runtime. Image hashes identify the files supplied, not their legibility or how the model read them.

Gemini generation. Requested and returned API identifiers identify the OpenRouter/Google service route, not immutable provider weights or a fixed service revision. API responses lost before persistence cannot be recovered from the retained records. These limits distinguish checking stored outputs from exact regeneration.

The ModelGloss vectors are consistent across matched conditions, but the four original prediction CSV caches are absent from this transfer. Their source-sentence alignment cannot be independently rechecked from those files. Appendix D details the positional loader, renderer, parser, retry and analysis procedures.

The earlier MTOB-inspired supplementary study (Tanzer et al., 2024) uses retrieved or longer grammar-text excerpts without the primary demonstrations. It compares against earlier `matched_v1` baselines rather than gold-support runs. Appendix E describes its separate protocol; Table E.1 lists metric conventions, evaluated records and exclusions.

5. Results

This chapter reports no-book performance, the effects of three types of grammar-book context, and output behaviour in the primary matched study. Selected predictions illustrate how changes in automatic scores relate to individual English outputs. Separate post-hoc Qwen analyses and diagnostics are reported beside the comparisons they extend. The complete context matrices are in Appendix F; the earlier MTOB-inspired study remains separate in Appendix E.

5.1. Evaluation Coverage and Baseline Performance

5.1.1. Evaluation Coverage and Reporting

All 351 conditions contain a saved response record for every planned test item: 36 no-book baselines and 315 grammar-context conditions. Table 5.1 counts sentence-condition records, so the same sentence contributes again in each condition. These counts include empty retained predictions and therefore measure recording coverage rather than successful translation.

Model	Conditions	Expected	Recorded	Empty	Empty (%)
Qwen3	117	15,999	15,999	197	1.23
Qwen3.5	117	15,999	15,999	20	0.13
Gemini	117	8,733	8,733	372	4.26
Total	351	40,731	40,731	589	1.45

Table 5.1.: Primary generation coverage before reference filtering. Counts include failures and repeated use of the same sentences. Gemini denotes Gemini 2.5 Flash Lite.

Generation uses 37 Gitksan, 87 Lezgi and 99 Natugu items per condition for every model. Tsez uses 445 items for each Qwen model and the first 99 for Gemini. The common99 reporting view uses the already generated Qwen responses at these same first 99 Tsez indices and all 99 Gemini Tsez responses. Other languages keep the same records. No new responses are generated for this view. The Qwen subset is

contained in the native 445, and Gemini’s two views are identical, so common99 is not an independent sample.

All primary Lezgi quality scores use the same 83-record mask, retaining failures at included indices; generation diagnostics count all 87 items. Section 4.3.1 defines the reference and inference policy, including the separate supplementary Qwen tests and descriptive Gemini comparisons.

Tables round lexical scores to two decimals and xCOMET scores to three; differences and rankings use unrounded values. Metric definitions and the original Holm correction families are in Section 4.3. The reported xCOMET scores use the XXL checkpoint with source, retained prediction and English reference as inputs. Its source-language coverage does not include the four study languages, so these outputs are exploratory rather than established measures of adequacy for these pairs (Section 4.3.2).

5.1.2. No-Book Baselines and Method Trends

The no-book conditions are locally adapted GrammarMT methods: Gloss-shot, explicit Chain-gloss and ModelGloss (Ramos et al., 2025). All receive 21 gold-glossed demonstration triplets. Gloss-shot receives no query gloss; Chain-gloss generates one before its translation; ModelGloss receives an external predicted query gloss. Thus the methods differ in query information and response requirements, with no control for query-gloss accuracy. Chapter 4 describes their prompts.

Table 5.2 combines all 36 native baselines. The three model groups report the same metrics, but Gemini’s native Tsez cohort is smaller. ModelGloss ranks highest on every metric within each of the twelve model-language groups. Its chrF++ advantage over Gloss-shot ranges from 9.58 points for Gemini Lezgi to 26.07 for Gemini Tsez. This recurring pattern concerns complete prompting methods, rather than an isolated effect of gloss quality.

The Qwen comparison depends on the method. Qwen3.5 ModelGloss exceeds Qwen3 ModelGloss on all three native metrics in every language; its Chain-gloss chrF++ is also higher in every language. Gloss-shot is more mixed: Qwen3.5 has higher chrF++ in Gitksan, Lezgi and Tsez but lower chrF++ in Natugu, while its Gitksan and Tsez BLEU is lower. Both Qwen models use the same decoding policy (Chapter 4). These comparisons concern complete checkpoints and prompting methods; they do not isolate architecture, size or query-gloss quality.

Chain-gloss falls below Gloss-shot in chrF++ for every Qwen3 and Gemini language. Qwen3.5 instead improves in Lezgi and Natugu and declines in Gitksan and Tsez. Gemini has higher native ModelGloss chrF++ than either Qwen model in Gitksan, Lezgi and Natugu, and higher BLEU in Lezgi and Natugu. Qwen3.5 has the highest Gitksan ModelGloss BLEU. Gemini’s Tsez Chain-gloss is the clearest exception to

5. Results

Language	Method	Qwen3			Qwen3.5			Gemini		
		BLEU	chrF++	xC	BLEU	chrF++	xC	BLEU	chrF++	xC
Gitksan	S	4.86	22.93	0.220	4.01	23.61	0.197	5.74	26.75	0.234
	C	3.80	18.11	0.184	4.14	23.46	0.189	6.71	24.71	0.240
	MG	14.98	46.10	0.432	21.06	49.14	0.452	19.52	50.53	0.487
Lezgi	S	3.29	19.40	0.153	6.35	25.00	0.201	12.41	33.55	0.318
	C	3.63	18.78	0.151	5.80	25.82	0.210	13.45	32.93	0.314
	MG	8.46	34.51	0.333	13.57	39.52	0.353	18.83	43.13	0.390
Natugu	S	2.34	19.54	0.166	3.54	18.82	0.170	5.07	22.99	0.201
	C	3.19	18.70	0.176	3.60	19.64	0.166	3.33	22.09	0.196
	MG	14.09	39.14	0.397	16.55	41.59	0.409	20.68	44.19	0.423
Tsez	S	2.17	18.14	0.145	1.82	18.44	0.142	2.74	19.24	0.151
	C	1.22	15.52	0.131	1.00	17.20	0.130	0.16	6.95	0.153
	MG	12.08	38.95	0.362	13.72	39.95	0.366	20.06	45.31	0.398

Table 5.2.: All 36 native no-book baselines. S: Gloss-shot; C: explicit Chain-gloss; MG: ModelGloss. xC abbreviates exploratory xCOMET with the XXL checkpoint. Evaluated records: Gitksan 37, Lezgi 83, Natugu 99; Tsez 445 for Qwen and 99 for Gemini. Bold marks the highest unrounded score within each model–language–metric block.

otherwise stronger lexical baselines: chrF++ is 6.95 against Gloss-shot’s 19.24, with 45 empty predictions among 99 outputs. The slightly higher xCOMET score for Chain-gloss does not resolve this disagreement because the evaluator remains exploratory. Section 5.3.1 examines the completion behaviour.

The strongest parallel with the original GrammaMT results is the ModelGloss ranking. Among its gloss-shot, chain-gloss and model-gloss methods, model-gloss has the highest BLEU and chrF++ in each displayed language in Table 2 (Llama-3 70B) and Table 6 (across models); Table 2 also gives it the highest mean xCOMET (Ramos et al., 2025). Chain-gloss trends are mixed there as well: in Table 2, chain-gloss has higher average BLEU but lower average chrF++ than gloss-shot, while Table 6 varies by model. ModelGloss thus shows a recurring descriptive advantage across the two studies, whereas a Chain-gloss benefit is not consistent. Differences in models, decoding, support handling, query-gloss provenance, prompt adaptation and the local Lezgi exclusions prevent treating this comparison as a controlled replication or comparing numerical improvements across studies.

The supplementary no-book tests support ModelGloss over both Gloss-shot and Chain-gloss in chrF++ for both Qwen models in every tested cohort, including the 83 retained Lezgi records and the shared first 99 Tsez items. Its BLEU advantage also passes the 72-test Holm correction except for the two Qwen3 Lezgi comparisons, whose positive differences remain descriptive. Chain-gloss falls significantly below Gloss-shot in chrF++ only on native Tsez for both Qwen models; this contrast does not pass correction on the shared 99-item cohort, and no BLEU contrast between Chain-gloss and Gloss-shot passes. Appendix F.3 gives the exact differences and pointwise intervals. These tests do not cover Gemini, between-model comparisons or neural method scores.

5.1.3. Comparison on the Shared Tsez Cohort

Table 5.3 aligns the models on the same 99 Tsez items. ModelGloss remains highest within each model, as in the other three languages. Gemini ModelGloss is descriptively highest across models on all three metrics in this aligned view. The Qwen ordering varies by method: Qwen3 Gloss-shot has slightly higher BLEU and chrF++ than Qwen3.5, whereas Qwen3.5 has higher Chain-gloss chrF++ and higher ModelGloss scores on all three metrics. Between-model rankings remain descriptive; the supplementary Qwen lexical method tests cover this shared cohort.

Model	Method	BLEU	chrF++	xCOMET
Qwen3	S	1.25	17.33	0.137
	C	1.26	15.64	0.130
	MG	11.84	38.18	0.352
Qwen3.5	S	1.24	17.28	0.135
	C	0.85	16.64	0.136
	MG	12.96	39.72	0.359
Gemini	S	2.74	19.24	0.151
	C	0.16	6.95	0.153
	MG	20.06	45.31	0.398

Table 5.3.: No-book scores on exactly the same first 99 Tsez items ($n = 99$ for every model and metric), called common99. These are a subset of the 445 saved Qwen responses and the full Gemini Tsez cohort in Table 5.2. Method codes follow that table; xCOMET uses XXL.

Figure 5.1 shows this recurring ModelGloss advantage across the four languages and

the unusually low Gemini Tsez Chain-gloss score. It uses the shared Tsez items, allowing the baseline pattern to be read alongside the aligned Gitksan, Lezgi and Natugu results. The Qwen native Tsez results in Table 5.2 concern the larger 445-item sequence.

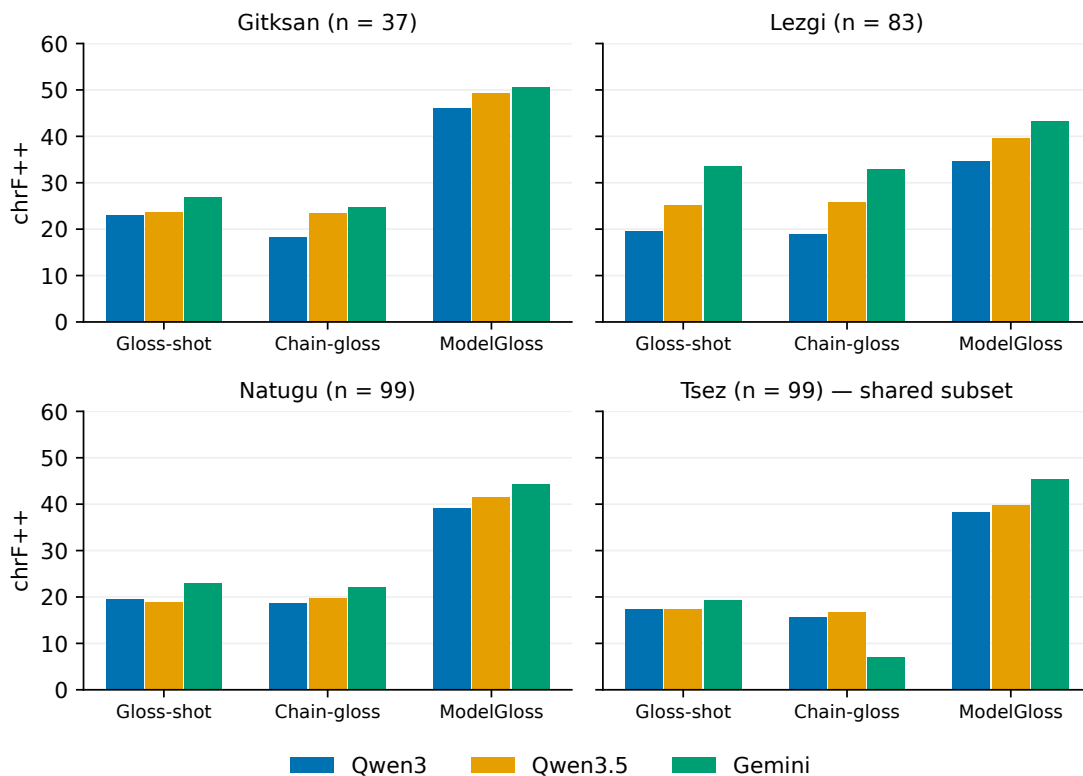


Figure 5.1.: No-book chrF++ with Gitksan $n = 37$, Lezgi $n = 83$, Natugu $n = 99$, and the same first 99 Tsez items for all models. The bars describe primary scores; supplementary Qwen method tests are reported separately.

5.2. Effects of Grammar-Book Context

The comparisons address three questions: how prepared text packages with different content and granularity change a no-book score; how paired TXT/JPG presentations of tables or cheat sheets differ; and how selected original pages change the baseline. Context changes are context minus the matched no-book score. Paired presentation changes are JPG minus TXT. The saved tests compare context with baseline or the

prepared TXT/JPG pair; they do not test summary against table, sheet against table, or prepared text against original pages.

Tables 5.4 and 5.5 list every eligible contrast passing the original Holm correction for BLEU and chrF++. All original primary lexical findings are in chrF++: the native view has one increase and ten decreases, whereas common99 has three decreases. No eligible BLEU context contrast passes correction. The lone native increase is Qwen3.5 Tsez Chain-gloss with cheat-sheet TXT; every corrected decrease involves an image package. Positive descriptive changes elsewhere in the matrix are distinct from corrected findings. The direction counts in Appendix F, Table F.4 show predominantly negative chrF++ changes for Qwen3.5 and Gemini and more often positive changes for Qwen3. These descriptive contrasts share test sentences and baselines.

The four lexical finding tables below use original materials and the saved Holm-adjusted $p < 0.05$ threshold. Native Tsez has 445 Qwen and 99 Gemini records; common99 has 99 for each. Lezgi is excluded from the original primary finding summaries; supplementary 83-record Qwen tests are separate. Method codes follow Table 5.2; Sheet means cheat sheet and Pages means selected original pages. Raw p-values are in Appendix F.

Model	Language	Method	Context	Baseline	Context	Δ	p_{Holm}
Qwen3	Tsez	C	Sheet JPG	15.52	12.96	-2.57	0.00630
Qwen3	Tsez	C	Table JPG	15.52	13.92	-1.60	0.03726
Qwen3.5	Tsez	S	Sheet JPG	18.44	17.41	-1.02	0.04340
Qwen3.5	Tsez	C	Sheet TXT	17.20	18.57	+1.38	0.00630
Qwen3.5	Tsez	C	Table JPG	17.20	16.01	-1.18	0.01869
Qwen3.5	Tsez	MG	Sheet JPG	39.95	37.88	-2.07	0.00630
Qwen3.5	Tsez	MG	Pages JPG	39.95	37.79	-2.17	0.00630
Qwen3.5	Tsez	MG	Table JPG	39.95	37.99	-1.97	0.00630
Gemini	Natugu	S	Pages JPG	22.99	19.00	-3.99	0.00630
Gemini	Natugu	S	Table JPG	22.99	20.01	-2.98	0.01869
Gemini	Tsez	S	Pages JPG	19.24	16.27	-2.97	0.00630

Table 5.4.: All 11 native chrF++ context contrasts with saved Holm-adjusted $p < 0.05$ in the 630-test lexical context family. Δ : context minus its own no-book baseline. Natugu: 99 records; Tsez: 445 for Qwen, 99 for Gemini. S: Gloss-shot; C: explicit Chain-gloss; MG: ModelGloss; Sheet: cheat sheet; Pages: selected original pages. Displayed findings exclude primary Lezgi inference.

The common99 findings reuse the same Gemini Natugu and Tsez responses as the native view. Their adjusted p-values can differ because the two lexical correction families contain Qwen comparisons on different sentence cohorts. The raw Gemini paired

Model	Language	Method	Context	Baseline	Context	Δ	p_{Holm}
Gemini	Natugu	S	Pages JPG	22.99	19.00	-3.99	0.00630
Gemini	Natugu	S	Table JPG	22.99	20.01	-2.98	0.01884
Gemini	Tsez	S	Pages JPG	19.24	16.27	-2.97	0.00630

Table 5.5.: All 3 common99 chrF++ context contrasts with saved Holm-adjusted $p < 0.05$ in the 630-test lexical context family for this cohort. Δ : context minus its own no-book baseline. Natugu and Tsez: 99 records; displayed systems: Gemini. S: Gloss-shot; Pages: selected original pages. Displayed findings exclude primary Lezgi inference.

evidence is unchanged; only the other p -values in its correction family differ. The original families, including the 87-record Lezgi tests, are preserved without recalculation when Lezgi inference is excluded from the displayed findings. As explained above, the overlapping reporting views supply no independent confirmation.

5.2.1. Prepared Text: Content and Granularity

Summary TXT supplies prose explanations, table TXT organizes selected forms and functions, and cheat-sheet TXT supplies compact cues. These are different prepared packages with their own content, instructions and length. Their results describe the packages as supplied; they do not isolate granularity or length as a single cause. Figures 5.2 and 5.3 provide a reading guide for all three comparisons in this section. Each column is a model–method combination, and each row identifies a resource, variant and package. The value is a change from that column’s own baseline, so a large gain from a weak baseline need not yield the highest absolute score. Black outlines identify the eleven eligible native corrected contrasts. Appendix F retains all 315 numeric cells.

Language and resource patterns. Gitksan shows how results vary by resource and method. Qwen3 gains with all three prepared TXT packages from both Brown and Rigsby across all methods. For Qwen3.5 ModelGloss, Brown’s summary, tables and sheet instead all decline; tables fall from 49.14 to 46.85 and the sheet to 46.65. Rigsby’s summary gives a small positive change, while its tables and sheet decline. Even the two grammars for this language therefore yield different package orderings. Gemini Gitksan Chain-gloss gains with Brown’s summary and Rigsby’s tables, but declines with Rigsby’s summary and Brown’s tables.

The separate three-seed check illustrates variation within one configuration: Qwen3

5. Results



Figure 5.2.: Native context-minus-baseline chrF++: Gitksan $n = 37$ and original Lezgi $n = 83$. S: Gloss-shot; C: Chain-gloss; MG: ModelGloss. Black outlines mark saved native Holm-adjusted $p < 0.05$ for eligible non-Lezgi contrasts. Figure 5.3 continues the matrix.

5. Results

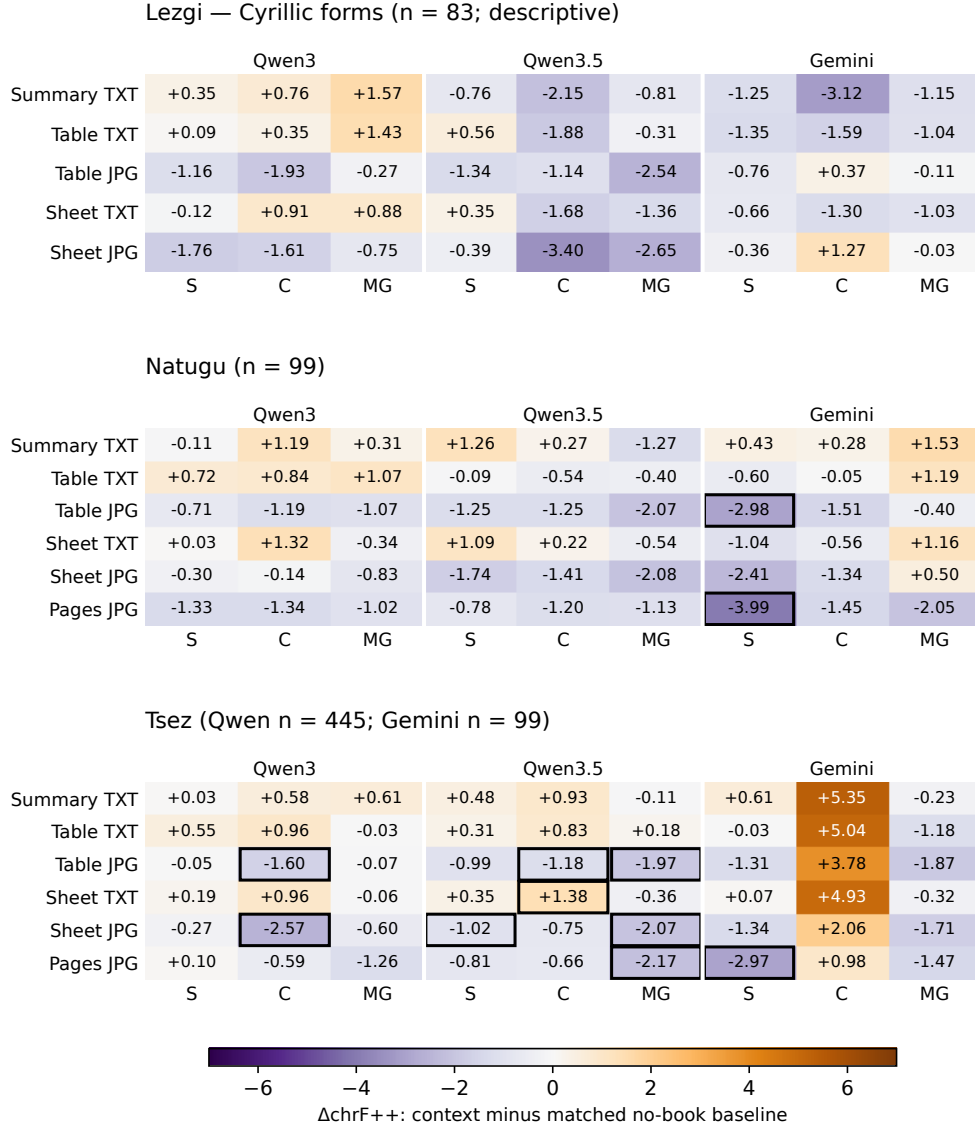


Figure 5.3.: Native context-minus-baseline chrF++ continued: Cyrillic-form Lezgi $n = 83$, Natugu $n = 99$, and Tsez $n = 445$ for Qwen and $n = 99$ for Gemini. Method codes and outlines follow Figure 5.2. Outlines show original primary tests; supplementary Lezgi tests are separate.

Gitksan Gloss-shot baselines are 22.93, 25.90 and 23.83 chrF++ for base seeds 20260910, 20260925 and 20261001, while Brown sheet TXT adds approximately +2.30, +0.22 and +0.45, respectively. All three gains are positive but differ in magnitude; this selected check is descriptive (Appendix F.3).

Lezgi’s original and Cyrillic-form packages also show different patterns. Qwen3 Chain-gloss and ModelGloss gain with all three TXT packages in both variants; Gloss-shot has small changes in both directions. Qwen3.5 Chain-gloss and ModelGloss decline with all six TXT packages, while Gloss-shot gains with tables and sheets but declines with both summaries. Gemini declines with every Lezgi TXT package. The largest prepared-text loss is its original Chain-gloss summary, from 32.93 to 29.50 (−3.43). These 83-record changes are descriptive; the variant bundles do not establish a controlled script effect.

Natugu’s text effects are generally smaller and method dependent. Qwen3 Chain-gloss gains with each TXT package, while its ModelGloss sheet declines despite gains with summary and tables. Gemini ModelGloss gains with all three, while its Gloss-shot and Chain-gloss sheets decline. For Natugu ModelGloss, table TXT is Qwen3’s highest-scoring context package (+1.07 chrF++), while summary TXT is Gemini’s (+1.53); these package orderings are descriptive. Tsez shows especially large changes for Gemini Chain-gloss: summary, tables and sheet move its low 6.95 baseline to 12.30, 12.00 and 11.88. None passes the saved context correction ($p_{\text{Holm}} = 0.15886, 0.08008$ and 0.19520 , respectively), and all remain below its 19.24 Gloss-shot baseline.

Model and method patterns. The Qwen3 columns show more broadly positive text changes than the Qwen3.5 columns, especially in Gitksan and Lezgi. Yet the native corrected TXT increase occurs for Qwen3.5 Tsez Chain-gloss: 17.20 to 18.57, $\Delta = +1.38$ and $p_{\text{Holm}} = 0.00630$. Its stronger ModelGloss baseline changes little with Tsez TXT: summary and sheet decline slightly, while tables increase slightly. The contrast between Gemini Tsez Chain-gloss gains and its ModelGloss declines likewise shows why the best delta and the best absolute method can differ. These patterns describe model-method combinations; no model-by-material interaction test was performed.

5.2.2. Paired Text–Image Presentation

Tables 5.6 and 5.7 list all eligible native and common99 lexical pairs that pass correction. All nine native and four common99 findings favour TXT in chrF++; no eligible BLEU pair passes correction. The pairs use the original Holm threshold within families separate from the context–baseline tests. Eight native findings concern Tsez and one Natugu; all use drawn-table images. Brown’s literal Markdown rendering is therefore distinct from the packages involved in these nine findings.

5. Results

Model	Language	Method	Context	TXT	JPG	Δ	p_{Holm}
Qwen3	Tsez	C	Sheet	16.49	12.96	-3.53	0.00216
Qwen3	Tsez	C	Table	16.48	13.92	-2.56	0.00216
Qwen3.5	Natugu	S	Sheet	19.91	17.08	-2.83	0.00418
Qwen3.5	Tsez	S	Sheet	18.79	17.41	-1.38	0.00216
Qwen3.5	Tsez	S	Table	18.75	17.44	-1.31	0.00216
Qwen3.5	Tsez	C	Sheet	18.57	16.45	-2.13	0.00216
Qwen3.5	Tsez	C	Table	18.02	16.01	-2.01	0.00216
Qwen3.5	Tsez	MG	Sheet	39.59	37.88	-1.71	0.00624
Qwen3.5	Tsez	MG	Table	40.13	37.99	-2.15	0.00216

Table 5.6.: All 9 native chrF++ pairs with saved Holm-adjusted $p < 0.05$ in the separate 216-test lexical presentation family. Δ : JPG minus TXT; all favour TXT. Natugu: 99 records; Tsez: 445 Qwen records. S: Gloss-shot; C: explicit Chain-gloss; MG: ModelGloss; Sheet: cheat sheet. Displayed findings exclude primary Lezgi inference.

Model	Language	Method	Context	TXT	JPG	Δ	p_{Holm}
Qwen3	Tsez	C	Sheet	16.02	12.87	-3.15	0.00216
Qwen3.5	Natugu	S	Sheet	19.91	17.08	-2.83	0.00430
Qwen3.5	Tsez	C	Sheet	18.08	15.95	-2.13	0.03604
Qwen3.5	Tsez	C	Table	17.54	15.41	-2.13	0.00430

Table 5.7.: All 4 common99 chrF++ pairs with saved Holm-adjusted $p < 0.05$ in the separate 216-test lexical presentation family for this cohort. Δ : JPG minus TXT; all favour TXT. Natugu and Tsez: 99 records. S: Gloss-shot; C: explicit Chain-gloss; Sheet: cheat sheet. Displayed findings exclude primary Lezgi inference.

Cheat sheets. Native corrected decreases cover Qwen3 Tsez Chain-gloss, all three Qwen3.5 Tsez methods, and Qwen3.5 Natugu Gloss-shot. A negative pair does not mean both packages reduce the baseline. Qwen3.5 Tsez Chain-gloss sheet TXT rises from 17.20 to 18.57, whereas JPG falls to 16.45; JPG minus TXT is -2.13 . Qwen3.5 Natugu Gloss-shot likewise has TXT 19.91 and JPG 17.08 around baseline 18.82, giving a -2.83 pair. By contrast, Qwen3.5 Tsez ModelGloss has both sheet formats below baseline, with the larger loss for JPG.

The original Lezgi sheet JPG omits three mini-gloss rows and four examples present in TXT, whereas its Cyrillic-form sheet includes them (Section 3.3). That original TX-T/JPG pair changes coverage as well as presentation. Across packages, layout and

use instructions can also differ, and character-level identity was not established. The paired findings therefore concern the supplied representations without isolating visual encoding alone.

Summary tables. Native corrected table pairs cover Qwen3 Tsez Chain-gloss and every Qwen3.5 Tsez method. Qwen3.5 Chain-gloss table TXT improves on baseline while JPG declines; ModelGloss table TXT is slightly above baseline and JPG below it. The common99 table finding is Qwen3.5 Tsez Chain-gloss. Large descriptive JPG losses also occur outside these corrected pairs: Qwen3.5 Brown Gitksan Chain-gloss tables fall from 23.46 to 19.42 relative to no book (-4.04), and its Brown ModelGloss sheet JPG falls from 49.14 to 45.08 (-4.06). These are different conditions from the corrected Tsez set and are not additional significant findings.

The separate supplementary presentation analysis finds a Lezgi exception on the 83-record mask: Qwen3.5 Chain-gloss with Cyrillic-form summary tables has BLEU 4.15 for TXT and 6.77 for JPG ($+2.62$; $p_{\text{Holm}} = 0.01045$ in the 96-test family). The no-book BLEU is 5.80, and the JPG-baseline contrast does not pass correction. On chrF++, JPG and TXT score 24.69 and 23.94 against a 25.82 baseline; neither the paired nor the JPG-baseline difference passes correction. The corrected finding is a paired BLEU advantage on the 83-record cohort.

Exploratory xCOMET scores. Table 5.8 contains the only eligible corrected context and presentation differences for xCOMET with XXL. Both concern Qwen3.5 Tsez ModelGloss table JPG on the native 445 items. The baseline comparison falls from 0.366 to 0.352; the TXT/JPG comparison falls from 0.367 to 0.352. Each row uses a different correction family, as specified in the caption. The corresponding chrF++ pair is 40.13 to 37.99 (-2.15). Agreement in direction does not establish the evaluator’s adequacy validity for Tsez. No eligible common99 xCOMET contrast passes correction.

Comparison	A	B (JPG)	ΔxCOMET	p_{Holm}
Context – baseline	0.366	0.352	-0.01425	0.03780
JPG – TXT	0.367	0.352	-0.01476	0.02808

Table 5.8.: All eligible corrected exploratory xCOMET contrasts, using XXL. Both rows are Qwen3.5 Tsez ModelGloss with original table JPG, native $n = 445$. A is the no-book baseline in row 1 and table TXT in row 2. Row 1 uses the 630-test context family; row 2 uses the separate 216-test TXT/JPG family. Each family pools native and common99. Saved unadjusted paired intervals and raw p-values are in Appendix F.

The supplementary context analysis on the 83 retained Lezgi records finds one corrected decrease: Qwen3.5 Chain-gloss with the Cyrillic-form cheat-sheet JPG falls from 25.82 to 22.42 chrF++ (-3.40 ; $p_{\text{Holm}} = 0.01584$ in the separate 264-test family). This compares the prepared sheet with the no-book baseline, distinct from the original primary findings and from selected-page conditions. Its paired interval and the overlapping 84-record sensitivity are in Appendix F.3.

5.2.3. Selected Original Pages

Selected original-page JPGs preserve source-page surroundings and coverage that differ from the prepared packages. Their scores are compared with the same no-book baseline; they have no paired TXT original-page counterpart. Comparisons with summaries, tables or sheets are therefore descriptive package comparisons.

Every Qwen3.5 original-page condition declines in chrF++ across the five resource/language blocks and three methods. The largest page loss overall is Gemini Gitksan Rigsby ModelGloss, 50.53 to 44.04 (-6.49); its Brown pages also decline in every method. Neither the Rigsby loss nor the largest positive page change, Qwen3 Brown Chain-gloss 18.11 to 20.01 ($+1.90$), passes correction: both have adjusted $p = 1$. Qwen3's Brown and Rigsby page changes have four increases and two decreases, demonstrating resource and method variation within Gitksan.

For Lezgi, the largest page loss is Qwen3.5 Chain-gloss, 25.82 to 22.71 (-3.11). Gemini Gloss-shot and Chain-gloss instead increase descriptively; Qwen3 ModelGloss has a near-zero positive change ($+0.01$). Natugu pages decline for every model and method. Its largest loss, Gemini Gloss-shot 22.99 to 19.00 (-3.99), passes native correction. Tsez pages have small positive exceptions for Qwen3 Gloss-shot and Gemini Chain-gloss, but the largest loss is Gemini Gloss-shot 19.24 to 16.27 (-2.97), which passes correction. Qwen3.5 Tsez ModelGloss also has a corrected decline, 39.95 to 37.79 (-2.17). These original-page findings appear in Table 5.4, separately identified from prepared table and sheet JPG.

The page conditions show predominantly negative baseline changes, with exceptions tied to particular methods and resources. Their distinct coverage prevents assigning the pattern solely to images or to less compact grammatical explanations. Chapter 3 documents how original pages and derived materials were selected and checked.

5.3. Output Behaviour and Illustrative Examples

5.3.1. Empty Outputs and Gloss-Format Compliance

Table 5.9 separates generation diagnostics by method. Its denominators include every generated item, including all 87 Lezgi items, rather than the 83 scored references or Qwen common99 subsets. There are 589 empty retained predictions: 197 for Qwen3, 20 for Qwen3.5 and 372 for Gemini, corresponding to the rates in Table 5.1.

Model	Method	Records	Empty	Truncated	Missing gloss	Multiline
Qwen3	S	5,333	29	18	—	—
	C	5,333	166	159	166	16
	MG	5,333	2	2	—	—
Qwen3.5	S	5,333	1	1	—	—
	C	5,333	19	19	14	3
	MG	5,333	0	0	—	—
Gemini	S	2,911	27	20	—	—
	C	2,911	339	339	338	0
	MG	2,911	6	0	—	—

Table 5.9.: Native output diagnostics over 39 conditions per model and method. S: Gloss-shot; C: Chain-gloss; MG: ModelGloss. Missing gloss and Multiline concern the extracted Chain-gloss string only. Flags can overlap and are not accuracy labels.

Completion and extracted glosses. An empty prediction is the retained string after fixed response handling; it does not imply that the raw response contains no text. A Qwen truncation flag means at least 512 generated tokens with a final token other than EOS; Gemini uses the API finish reason length. Known truncation forces an empty prediction even if some translation text could otherwise be extracted. A missing final marker alone is not this truncation predicate.

For Chain-gloss, a missing-gloss flag means that the parser extracted an empty gloss string. It searches for Gloss: before the last qualifying FINAL_TRANSLATION label. A raw response can therefore contain visible gloss text but have no extracted gloss when the final label is absent. A multiline-gloss flag means that the stripped extracted gloss contains a newline. These are extraction and formatting diagnostics, rather than

measures of linguistic accuracy. Neither flag alone rejects a nonempty recoverable translation.

Chain-gloss accounts for 524 empty predictions, of which 517 have known truncation: Qwen3 has 159 among 166, Qwen3.5 19 among 19, and Gemini 339 among 339. Its gloss and translation share the 512-token limit, so a long gloss can exhaust the space for the final answer. This completion mechanism is consistent with the saved records, while the primary study does not quantify how much a larger budget would change scores.

The supplementary Qwen3 no-book Chain-gloss cap check changes little. Raising the limit from 512 to 1,024 tokens leaves lexical scores unchanged in Gitksan, the 83-record Lezgi view and Natugu; full-generation empty counts remain 1/37, 1/87 and 3/99. Native Tsez empties fall from 35/445 to 34/445, with chrF++ changing from 15.52 to 15.55 and BLEU from 1.219 to 1.221. Lexical scores on the shared 99-item Tsez cohort are unchanged. These four paired checks are descriptive and do not test Gemini’s completion problem; Appendix F.3 includes their lexical and neural scores.

All 45 empty predictions in Gemini’s 99-item Tsez no-book Chain-gloss condition were inspected in their raw form. Every response has a visible `Gloss:` line, 512 completion tokens and a length stop, but no final-translation marker. The glosses contain repeated affixes/pronoun tags or expanding enumerations. No separate English translation sentence or partial translation clause was found; English lexical items inside a gloss are not a translated sentence. Thus the missing-extracted-gloss count does not mean that Gemini generated no gloss, and these records provide no partial translation to assess or salvage.

Additional response flags. Table 5.10 covers all methods. The unmarked single-line fallback allows Gloss-shot and ModelGloss to retain a single-line translation without the required final-translation marker, subject to format checks. It does not apply to explicit Chain-gloss and is not itself a failure; an empty result remains an empty prediction. Appendix D.2 gives the exact prefix and extraction rules.

The active Qwen repetition warning detects repeated spans in the raw response and adds a warning without automatically rejecting the translation. The Gemini API path has no corresponding detector, so zero saved Gemini warnings does not establish absence of repetition. A reasoning violation is triggered by API reasoning fields or reasoning-token metadata, generated special-token think tags, or raw think tags or line-initial Analysis, Reasoning or Thinking labels; it rejects the response. None was recorded, which establishes the saved flag count rather than the absence of unobservable reasoning. The exact repeated-span rule and active parsing path are in Appendix D.2.

Additional flag	Qwen3	Qwen3.5	Gemini
Unmarked single-line fallback	10,074	2	3,226
Repetition warning	62	10	0
Reasoning violation	0	0	0

Table 5.10.: Additional native parser or generation flags. Counts cover 15,999 records per Qwen model and 8,733 Gemini records. The permitted unmarked single-line fallback can retain a translation and is not itself a failure. Gemini does not record the Qwen repetition detector; its zero is a saved-warning count.

5.3.2. Examples from the Evaluated Predictions

The following three groups were selected before sentence scoring for three comparison types: Chain-gloss cheat-sheet TXT/JPG, ModelGloss table TXT/JPG, and Gloss-shot prepared tables versus original pages. They comprise ten saved predictions at fixed original zero-based indices. Strings, including punctuation, are copied verbatim from the evaluated records. Each output is scored against its displayed English reference using sentence-level chrF++ from SacreBLEU 2.6.0, with character order 6, word order 2 and $\beta = 2$. These scores measure lexical overlap for individual outputs, not corpus performance or human judgments of adequacy. The fixed selection is illustrative and not representative.

Table 5.11 shows Qwen3.5 Tsez Chain-gloss index 0. All three outputs differ from the reference’s many houses and loss of fortune. TXT introduces a fox, mats, a woman and sons; JPG introduces a fox, a wedding and a wealthy man. Both context outputs have lower sentence chrF++ than the no-book prediction. This sentence does not overturn the positive corpus sheet-TXT change: other retained items contribute to the corpus result.

Table 5.12 shows Qwen3.5 Tsez ModelGloss index 1. The no-book and table-TXT predictions are exactly identical. Both change the reference’s hiding of one person with his face outside the leaves into people hiding themselves. JPG further changes participants and actions, introducing *we*, one person and hair, with lower sentence chrF++. These observations compare English strings with the reference; they do not assess the Tsez source through expert knowledge.

Table 5.13 shows Gemini Natugu Gloss-shot index 2 and adds selected original pages. The reference describes a father hearing Brother John’s speech and thanking him. The baseline instead calls the talk bad; TXT adds inability to speak; JPG describes talking about the past; original pages introduce an upcoming meeting. TXT’s slightly higher

Item	Verbatim text	chrF++
Source	Raq'ʿu ɣʿutkubin rodin, žedałxortow kinaw idu yałru maʿiʂatno bik'in iyn.	—
Reference	He built many houses and the entire fortune that was in his home went away through them.	—
No book	They built a house for the request, and they went to their father's home with wealth.	28.12
Sheet TXT	After the fox made mats in the house, a woman and her two sons came from there and gathered them.	22.32
Sheet JPG	The fox gathered the house, and while they were like this, a wealthy man brought something to the wedding and said, "Come!"	22.33

Table 5.11.: Qwen3.5 Tsez Chain-gloss, zero-based test index 0, with no book and original cheat-sheet TXT/JPG. The last column is sentence-level chrF++ against the displayed reference.

Item	Verbatim text	chrF++
Source	Žeda ža sidaɣ ʔ'ebozał ʔ'iriɣor hut'-mʿalin roʔik'no uqʿno.	—
Reference	They hid him somewhere with only his face out of the leaves.	—
No book	They pulled out the leaves from their faces and hid themselves.	32.19
Table TXT	They pulled out the leaves from their faces and hid themselves.	32.19
Table JPG	We, the one person, pulled out his hair and hid him.	23.07

Table 5.12.: Qwen3.5 Tsez ModelGloss, zero-based test index 1, with no book and original table TXT/JPG. Scores are sentence-level chrF++; the baseline and TXT strings are identical.

sentence chrF++ therefore does not establish semantic correctness. Original pages have the lowest displayed score for this item, illustrating a different output from the prepared table images without proving a general page effect.

5.3.3. Supplementary Query-Gloss and Seed Diagnostics

Replacing the predicted query gloss with its gold annotation in the otherwise unchanged Qwen3 no-book ModelGloss prompt decreases both lexical scores in Gitksan, the 83-record Lezgi view and Natugu, but increases them on native Tsez. Gold-minus-predicted chrF++ changes are respectively -7.43 , -0.87 , -2.93 and $+0.40$; the shared 99-item Tsez cohort rises by $+0.92$. Supplying a gold query gloss therefore does not

Item	Verbatim text	chrF++
Source	Kx xlrbz trtenge nzycmnekrbz Tzsiu Jon , sc tqrpipebzle kx , “ Awitxpe Jon , mz nzycmne-krm la .	—
Reference	When my father heard the speech of Brother John, he said to him, "Thank you very much John, for that speech of yours.	—
No book	When they were talking to Tasiu John, he told them, "John, your talk is not good.	20.93
Table TXT	When the two of them were not yet able to speak, Tzsiu John said to them, "John, we are not yet able to speak.	22.63
Table JPG	When they were talking to Tasiu Jon, they told him, "Jon, we are talking about the past."	16.89
Pages JPG	And he said to me, "Jon, the meeting is coming."	14.38

Table 5.13.: Gemini Natugu Gloss-shot, zero-based test index 2, with no book, original table TXT/JPG and selected original-page JPG. Scores are sentence-level chrF++. Original punctuation, including unmatched quotation marks, is retained.

guarantee a higher translation score in these prompts. The neural scores do not all follow the lexical direction: native Tsez xCOMET-XXL declines. This diagnostic assesses the substitution, without measuring gloss accuracy or isolating which cues account for a translation change.

The three-seed check also has a limited scope. For the selected Qwen3 pairs, context-minus-baseline chrF++ stays positive for Gitksan Gloss-shot with Brown cheat-sheet TXT, Lezgi Chain-gloss with cheat-sheet TXT and Tsez Chain-gloss with summary TXT, and stays negative for Natugu ModelGloss with cheat-sheet JPG. Other metric directions vary; the stable chrF++ signs apply only to these four selected pairs. Appendix F.3 reports each seed’s scores, including the overlapping common99 Tsez view. The cap, query-gloss and seed outcomes are descriptive.

6. Discussion and Limitations

The results in Chapter 5 show that supplying linguistic information does not have a uniform effect on translation. An externally predicted query gloss accompanies the strongest observed no-book performance, whereas grammar-context changes vary by configuration: chrF++ changes are predominantly negative for Qwen3.5 and Gemini and more often positive for Qwen3. This chapter interprets these findings through the four research questions, relates them to previous research, and considers what the evaluation can establish. The central distinction is between the performance of a supplied grammar package and evidence that a model has learned or applied its grammatical explanations.

6.1. Interpretation of the Research Questions

6.1.1. No-Book Performance of the Adapted Prompting Methods (RQ1)

ModelGloss has the strongest observed no-book performance across the three models and four languages, while explicit Chain-gloss does not consistently improve on Gloss-shot. The baseline tables show the ModelGloss ranking on both lexical metrics in every model–language combination (Table 5.2). It also holds on the shared Tsez cohort in Table 5.3. Supplementary Qwen lexical tests support the ModelGloss advantage, except for the Qwen3 Lezgi BLEU comparisons reported in Chapter 5; Gemini and neural method rankings remain descriptive. XCOMET-XXL has the same ModelGloss ranking in the saved source–prediction–reference evaluation; interpretation of these local scores remains exploratory (Section 6.3.1).

The information available at the query is central to interpreting this pattern. All three methods already receive the same gold-glossed supports. ModelGloss adds an external analysis of the particular sentence to be translated, potentially supplying lexical meanings and grammatical labels that the translator would otherwise have to infer. Chain-gloss instead asks the translator to produce that analysis itself. Thus, the baseline comparison concerns different ways of obtaining and using query information, rather than the presence or absence of glosses in general. For use within these configurations, the results favour supplying the external analysis over relying on a request to generate one.

This distinction also affects completion. Chain-gloss must produce a gloss and a labelled translation within the same 512-token output budget. Across baseline and context conditions, 517 of its 524 empty native predictions have a truncation flag (Table 5.9). In the Gemini Tsez no-book condition, all 45 empty predictions among 99 records stopped for length without a final-translation marker. The method comparison therefore includes whether the response completes and can be extracted, as well as translation content. The supplementary Qwen3 cap check changes little in four no-book Chain-gloss pairs; it does not establish how Gemini would respond to a larger budget or whether generated glosses are linguistically worse than supplied ones.

More informative external glosses are a plausible explanation for the ModelGloss ranking, consistent with GrammarMT’s word-gloss-accuracy analysis (Ramos et al., 2025). Linguistic gloss accuracy was not assessed at the word or morpheme level for the primary Chain-gloss conditions, with or without grammar books. Earlier eight-example development diagnostics retain gloss_chrf similarity summaries, but those decoding checks do not answer whether grammar books improve gloss accuracy. Annotation correctness, cues useful for translation and compatibility with the prompt and demonstrations are distinct properties. The supplementary gold-query substitution improves lexical scores only in Tsez and lowers them in the other three languages. These translation results weigh against explaining ModelGloss’s advantage by annotation accuracy alone; they do not measure the linguistic quality of gold glosses. The local translation scores do not separate lexical and grammatical contributions.

The exact query-gloss strings are retained in the matched configurations and are identical across models and materials for shared records within each language, including the shared prefix of the two Tsez cohorts. The original prediction-cache CSV files are unavailable, so their source/record-ID alignment cannot be independently rechecked. The results concern the complete method using these retained external predictions.

6.1.2. What Adding Grammar Context Changes (RQ2)

Adding grammar-book context does not yield a broadly consistent improvement over the matched no-book conditions. The directions in Appendix F, Table F.4 vary by model and metric, and the original primary reference-valid native lexical tests identify one corrected increase and ten decreases, all in chrF++ (Table 5.4). The increase occurs for Qwen3.5 Tsez Chain-gloss with cheat-sheet text: +1.38 chrF++ on the 445-record cohort, with Holm-adjusted $p = 0.00630$. All ten decreases concern image packages. No reference-valid BLEU context contrast passes correction. This is evidence of a selective benefit alongside losses in particular configurations, rather than a general gain from adding a grammar resource.

The baseline determines what such a gain means. A no-book prompt already includes translation examples and their gold glosses; ModelGloss also includes the predicted query gloss. The added grammar must therefore contribute beyond information that is already available. A package could repeat cues in the supports, supply useful information for only some sentences, or contain material that the model cannot relate to the current query. The experiment does not distinguish these possibilities, but they explain why the amount of linguistic information supplied and its incremental value are separate questions. A grammar need not be intrinsically uninformative for its addition to leave a corpus score unchanged.

Incremental benefit must also be separated from absolute performance. The Qwen3.5 Tsez Chain-gloss gain does not make it the highest-scoring configuration. Conversely, a ModelGloss context condition can retain a higher score than another prompting method while falling below its own no-book baseline, as the context matrices show (Tables F.1–F.3). Selecting a translation configuration and deciding whether to add a grammar package are therefore distinct decisions. The latter requires comparison with that configuration’s own baseline.

The matched design supports this incremental comparison because the model, supports, test inputs, decoding policy, and applicable query gloss remain fixed. The addition includes both grammar material and an instruction to use it. A score increase cannot identify whether the model used an explanation, copied a useful lexical mapping, followed an example, or changed its response to the instruction. Likewise, the predominance of nonsignificant comparisons does not establish that grammar is useless. Relevant linguistic content alone is therefore insufficient reason to expect an improvement from adding one of these packages to a functioning gloss-based translation prompt.

6.1.3. Text, Images, and Selected Original Pages (RQ3)

The clearest paired representation pattern favours the supplied text versions. All nine original primary reference-valid native text–image differences that pass the lexical correction favour TXT, and all concern chrF++ (Table 5.6). Eight occur in Qwen Tsez conditions and one in Qwen3.5 Natugu; all use drawn-table images. Brown’s two derived images instead display literal Markdown syntax (Section 3.3), so that rendering difference does not explain these nine findings. This supports preferring those text packages over their paired images for the tested configurations.

Resource choice is nevertheless informative within a stated configuration. For native Qwen3.5 Tsez Chain-gloss, cheat-sheet TXT has the highest observed chrF++ among the evaluated packages: 18.57, compared with 18.12 for summary TXT, 18.02 for table TXT and 16.53 for selected-page JPG, against a 17.20 no-book baseline (Table F.3). It

also has the largest observed gain over that baseline, and its gain is the corrected increase discussed under RQ2. These comparisons do not test sheet-versus-summary or sheet-versus-table differences. Gemini Tsez Chain-gloss instead scores highest with summary TXT (12.30), followed by table TXT (12.00) and sheet TXT (11.88), against a 6.95 baseline; none of these baseline gains passes correction. The preferred package thus depends on the model and prompting method.

The separate supplementary Lezgi analysis adds a JPG-over-TXT BLEU advantage on the 83-record cohort for Qwen3.5 Chain-gloss Cyrillic-form summary tables. That table pair has no corrected chrF++ difference, and its JPG does not significantly improve on no book. A different package, the Cyrillic-form sheet JPG, shows a corrected chrF++ decrease against no book. Taken together, these comparisons do not establish a general ordering of summaries, tables, cheat sheets and original pages, or a text advantage across all models and languages.

Preserving page layout makes visual context worth examining: row and column structure, interlinear alignment, and neighbouring examples remain visible in an image. Yet preserving those features does not ensure that they improve translation. The model must still recognize the forms and connect the information to the query. Text and images can differ in the effort required to access the same cue, while prepared summaries and selected pages can differ in whether that cue is supplied at all. The observed comparisons combine these questions. Neither character recognition nor the selection and application of relevant rules was measured separately, so the text-favouring results cannot be diagnosed as an OCR or visual-reasoning failure.

The verified Lezgi cheat sheet illustrates a more direct problem with treating format labels as controls. Its original JPG omits three mini-gloss rows and four examples present in the TXT, while the two-page Cyrillic-form JPG includes them (Section 3.3). The retained history does not establish whether this omission was intentional or an oversight (Appendix D.1). The original TXT/JPG comparison therefore changes information coverage along with modality. Comparing the original and Cyrillic-form JPGs changes coverage, script, and selected forms. A difference between these versions cannot identify which change mattered. This omission concerns Lezgi and does not explain the significant non-Lezgi text-image findings. Conversely, the absence of another detected missing section does not establish character-level or perceptual equivalence for the other pairs.

Comparisons with selected original pages involve a further choice about content. A summary reorganizes and compresses the source; selected pages preserve the local wording and layout but supply different examples and surrounding discussion. Table 3.3 shows unequal lengths and image-page counts. Table F.10 adds model-specific logged processor-input lengths and separately reconstructed text and visual counts; these quantities cannot be added to recover the logged totals. There is no common

model-token budget or matched irrelevant-context control, and the representation instructions differ. As a result, these conditions assess alternative ways of packaging the available resource. The stored analyses provide paired TXT/JPG tests, but no general significance test that ranks summaries against original-page selections.

Preparing grammar context requires attention to both fidelity and usefulness. A faithful image can contain information that does not help the current translation task, while a concise text selection can be useful without preserving the whole source. The text-favouring pairs show no measured advantage from access to the rendered layout in those comparisons. Whether recognition, cue presentation, or content selection accounts for the difference remains unresolved. A choice based on these results therefore applies to the evaluated packages, not to text and vision as abstract capabilities.

6.1.4. Variation across Models, Languages, and Methods (RQ4)

ModelGloss ranks highest in every model–language no-book group on the lexical metrics, whereas added grammar packages produce both positive and negative changes across configurations (Table 5.2 and Appendix F). A package that helps one configuration cannot be assumed to help another, even when its language and source document remain the same. This variation describes the evaluated combinations; the analyses do not test model-by-material or method interactions.

Shared evaluation sentences make cross-model descriptions more informative, but they do not equalize the complete systems. On the common Tsez cohort, Gemini ModelGloss has the highest observed baseline scores, whereas the smaller differences between the Qwen baselines depend on the method and metric (Table 5.3). These comparisons include distinct model families and decoding policies. They therefore cannot identify a superior architecture or attribute a difference to model size. Response completion means producing a final English translation that can be extracted within the output cap. The saved Chain-gloss truncation and missing-marker examples in Sections 5.3.1 and 5.3.2 make this relevant to the cross-model comparison alongside the linguistic information supplied; an extractable answer need not be linguistically correct.

Language differences also coincide with resource differences. The two Gitksan sources are a phonological article and a separate introductory excerpt; the available Lezgi document covers an introduction and opening phonology, with additional script/form variants in the derived materials. The Natugu sketch and Tsez manuscript cover broader grammatical domains (Table 3.1). These are unequal opportunities to supply useful context, not equivalent grammars applied to four interchangeable datasets. A weak effect for one language cannot establish that its morphology is harder for the

model, that its dialect mismatch caused errors, or that a longer grammar would have performed better.

The Tsez cohort comparison demonstrates why apparent variation also depends on which sentences are evaluated. Restricting the Qwen results to the same first 99 records used by Gemini changes some small rankings and leaves no Qwen context contrast significant after the common99 correction (Table 5.5). This does not refute the native-cohort gain: the subset changes the evaluated sentences, sample size, and family of lexical tests. The common99 view supports a comparison on shared items, while the native view retains the larger Qwen sample. Together they show the scope of the observed patterns, not their replication in a second test set.

6.2. Relation to Previous Research

6.2.1. Glosses and Grammar Resources in Translation

Gloss-informed prompting. ModelGloss has the highest observed no-book BLEU and chrF++ in every local model–language group. This ranking has a qualitative parallel in GrammaMT: among its Gloss-shot, Chain-gloss and ModelGloss methods, ModelGloss leads on both metrics in the four shared source languages in Table 2’s Llama-3 70B comparison (Ramos et al., 2025). GrammaMT’s gloss-accuracy analysis and gold-query oracle experiments also show that the intermediate analysis matters in its setting. The local gold-query diagnostic is narrower and includes no gloss-accuracy or label-ablation analysis, so the translation scores cannot separate lexical and grammatical contributions. Prompts, support selection, decoding and output requirements differ from the paper, and the local ModelGloss prompt omits its warning about possible gloss errors. No rationale for omitting the warning is recorded (Appendix D.1). Its effect was not isolated. The shared ranking supports the relevance of query-gloss information across these settings, without establishing a controlled reproduction or a numerical improvement over the published systems.

Gloss prediction and support selection. For gloss prediction rather than English translation, Ginn et al. (2024a) find that retrieved supports often improve morpheme accuracy over random selection, with language- and count-dependent exceptions. Their study shares Gitksan, Lezgi and Natugu with this study, but their records and splits need not be identical, and their outcome measures gloss accuracy. This makes support selection relevant to the fixed 21 demonstrations here; it does not establish that changing them would improve the stored translations.

Grammar content and rule use. The uneven local grammar gains raise the question of what information a package adds to an existing prompt. Hus and Anastasopoulos (2024) find that this value depends on language and translation direction: adding grammar to words and sentences gives a substantial Natugu-to-English gain, but a small Gitksan change without a significance marker. Their evaluation partitions and Gitksan resource differ from those used here, so the published scores cannot explain the local language pattern. Aycock et al. (2025) separate parallel examples and word pairs from explanations in Kalamang, Nepali and Guarani. Parallel material accounts for much of their translation benefit, and full books do not consistently exceed those subsets; some grammaticality and glossing tasks instead benefit from combining typological information with examples. These distinctions matter because the present packages were not ablated into examples and explanations.

Marmonier et al. (2025) show that explanations can add information beyond incidental bitexts for some constructed-language phenomena, although their excerpted grammar also contains examples. Their controlled resources and accuracy task differ from these natural-language translation packages. Together, these studies make incremental information the relevant comparison, while leaving the contribution of explanations to the local gains unisolated.

Zhang et al. (2025) make a related distinction between finding a relevant rule and applying it. Their rule-annotated Chinese–Zhuang experiments use supplied lexicons, and some conditions supply the relevant rules themselves. Individual rule assessment improves retrieval in their setting, while pseudocode representations can assist retrieval and application. Those controls expose stages that an end-to-end translation score alone leaves unresolved. The present packages may make information available without establishing that a needed rule was found or correctly used. Together, these studies suggest that the useful unit of comparison is the information made accessible for a particular task. More grammar content, preserved layout, or an explicit request for analysis is not by itself evidence of successful grammatical reasoning.

Visual and document context. The local TXT/JPG differences leave recognition, document relations and relevance unresolved. Visual source-text translation evaluates images containing the query sentence (Salesky et al., 2021, 2024); here images supply grammar context beside a textual query. Pix2Struct studies screenshot parsing (Lee et al., 2023), while OmniDocBench distinguishes text, table and reading-order performance (Ouyang et al., 2025). Scene-image relevance is a separate control in Guerra and Khrapunova (2026). These task distinctions identify possible requirements for useful visual information, without diagnosing a local score difference as a recognition, layout or relevance failure.

6.2.2. What the Earlier MTOB Study Adds

The earlier MTOB study contributes evidence about grammar-text strategies and the evaluation of unconstrained responses under a separate protocol. Original MTOB combines grammar retrieval or long contexts with word lists and parallel examples in selected conditions, and prepares plaintext from grammar LaTeX with manual correction (Tanzer et al., 2024). The local supplement instead uses extracted PDF text without those additional resources. This distinction matters when relating these local results to the original MTOB benchmark: the shared retrieval labels do not imply equivalent information or document quality. Original MTOB also proposes rendered-PDF input as an extension; the primary image comparisons here address supplied visual packages rather than reproducing its Kalamang benchmark.

The earlier MTOB results show that a method ranking can depend on where the evaluated translation ends. Their plain chrF is character-only, with word n-gram order zero, unlike the primary chrF++ setting with word n-gram order two. For Qwen3 Tsez, raw chrF favours embedding retrieval over substring retrieval, while extracted chrF favours substring retrieval; both within-view differences pass the saved correction (Table E.4). The underlying responses are unchanged. Removing labelled commentary can therefore change a comparison without generating or correcting a translation. The Gitksan example in Appendix E illustrates this limit: extracting the short answer removes a breakdown but leaves the omission of the action expressed in the reference. Response boundaries are thus part of the evaluation definition, with consequences for both scores and method rankings.

The earlier MTOB comparison with earlier matched baselines favours those baselines under the common evaluator (Table E.5). It cannot establish that grammar caused the disadvantage. The comparator has empty support-gloss fields and belongs to the earlier study, while the MTOB prompts have no demonstrations or predicted query gloss. Their instructions, decoding, output limits, and retry policies also differ. Applying one evaluator makes the resulting texts comparable under that scoring procedure; it does not remove the differences in how they were generated. There is no matched no-book MTOB condition, and no current gold-support result enters this comparison. Consequently, the supplement does not confirm the primary study’s image-package pattern or measure the same incremental grammar effect.

Its language-specific rankings also need their own qualifications. The saved Natugu prompts contain a known test reference in substring and long-context conditions, and another test source example in all three strategies. The effect of those inclusions on scores is unknown, so higher earlier MTOB Natugu scores cannot be attributed solely to more useful grammar retrieval. Earlier MTOB Lezgi still includes the three invalid

references and has no corresponding valid-reference analysis. In addition, absent finish metadata and overwritten earlier attempts limit reconstruction of the response process. These constraints leave the supplement useful for examining the retained configurations and response views, while preventing it from serving as an independent validation of the primary findings.

6.3. Validity, Uncertainty, and Scope

6.3.1. Reference Quality and Evidence of Translation Quality

The automatic scores establish differences under specified evaluation procedures, but do not establish that a translation is adequate for a reader. BLEU and chrF++ measure overlap with the retained English reference. A change may reflect lexical choice or phrasing as well as a correction in meaning; a fluent, well-formatted response can still mistranslate the source. The fixed examples in Section 5.3.2 illustrate this distinction without supplying a representative human assessment. The study has no human adequacy evaluation or predefined threshold connecting a score change to practical usefulness.

The stored XCOMET-XXL evaluation supplies the source sentence, predicted English translation and English reference. The xCOMET framework also supports a reference-only mode using the prediction and reference (Guerreiro et al., 2024), but that was not the local run. Its official xCOMET-XXL model card omits all four source languages from the listed coverage and warns about unreliable scores for uncovered-language pairs (Unbabel, n.d.). For the retained combined-input evaluation, the English target and reference do not establish that the evaluator can assess the source meaning (Section 4.3.2). No local source-language or human evaluation validates these scores. A significant difference describes consistency in the evaluator’s scoring under the analysis, rather than validated adequacy judgments. Agreement with chrF++ likewise cannot supply the missing validity, and disagreement should not be resolved by choosing whichever metric supports a preferred interpretation.

Lezgi illustrates why inference must match the reported scores: ModelGloss remains the highest-scoring baseline on the uniform 83-record quality mask, but the original 87-record p-values do not apply. Separate supplementary Qwen source-cluster tests use 83 records; Gemini remains descriptive (Section 4.3.1; Appendix D.3). Removing the known overlap does not establish independence from all supplied materials.

6.3.2. Dependence and Statistical Uncertainty

The large number of sentence-condition records does not create a correspondingly large sample of independent sentences. Conditions repeatedly evaluate the same items, and the two Gitsan sources share both a test set and a method-specific baseline. The common99 Tsez cohort is a fixed prefix of the native Qwen cohort. Lezgi also contains repeated source strings, while the original record bootstrap does not model their clustering. The supplementary A/B bootstrap groups identical source strings, but does not model document dependence. These dependencies mean that agreement across packages or cohorts can describe a persistent pattern within the evaluated material without providing new independent evidence for generalization.

The multiple-testing corrections address defined families of comparisons rather than every possible conclusion drawn from the experiment. As specified in Chapter 4, the lexical analysis pools BLEU and chrF++ within separate context and material families for each cohort; the xCOMET-XXL analysis instead pools the two cohorts within its own families. The primary material family contains paired TXT-versus-JPG comparisons for cheat sheets and summary tables within each model, language and prompting method. These corrections neither cover the union of all metrics and cohorts nor test primary prompting-method or model differences; supplementary A tests compare Qwen methods within a separate family. Supplementary B context and presentation tests likewise retain their own source-cluster families. Different significance decisions across configurations therefore do not establish a difference between their effects.

Magnitude and uncertainty must be interpreted together. A nonsignificant contrast leaves uncertainty about its effect and is not evidence of equivalence. A significant contrast can still be too small, or too weakly related to adequacy, to establish useful translation improvement. The lexical system-score intervals also cannot be treated as intervals for paired differences; the neural paired-difference intervals have a different definition and are unadjusted. The clearest supported claims are the observed deltas together with their applicable tests and cohorts, rather than a count of significant cells treated as the strength of one general hypothesis.

Finally, the stored tests concern variation over evaluation items conditional on the retained outputs. Fixed seed policies and matched decoding settings within comparisons improve control, but do not measure variability across repeated generations, provider operations, or model revisions. The supplementary three-seed check describes only four selected Qwen3 pairs: their chrF++ signs persist, while other metric signs vary. It does not measure full-matrix or cross-model generation uncertainty. This matters especially when small descriptive differences are used to choose between closely scoring configurations.

6.3.3. Generalizability and Reproducibility

The study covers sentence-level translation into English for four languages, using selected resources, fixed supports, and three configured models. It does not establish performance for reverse-direction or document translation, other languages, or community use. The available grammars differ in coverage, and their prepared packages represent particular selections rather than the full range of linguistic documentation. The findings apply first to the evaluated configurations. Extending them to other settings requires evidence beyond a shared low-resource-language label.

Independence from prior linguistic material remains only partly known. The bounded text-matching screen described in Chapter 3 found no complete test-source or valid-reference match in extracted text from the selected original pages or prepared TXT files. It did not cover all partial, semantic, transliterated, or image-only overlap, and it cannot establish absence of pretraining exposure. The known support overlap and earlier MTOB Natugu examples have narrower, identifiable scopes. Addressing them does not turn the remaining evaluation into a demonstrated test of learning an entirely unseen language from grammar materials alone.

The retained evidence makes the reported evaluation traceable. Configurations bind supports, query inputs, context files, and settings to saved responses, and prediction hashes link those responses to the metric inputs. This allows inspection of what was compared without regenerating it. Exact regeneration is a stronger requirement: primary model weight revisions are not fully pinned, the portable runtime is incomplete, provider operations are not recoverable, and the complete Qwen image-and-text token stream has not been independently reconstructed. Missing gloss-cache alignment and earlier MTOB attempt information impose further limits, as discussed in Sections 6.1.1 and 6.2.2.

The new Qwen3 runs pin a weight revision and runtime versions. In 13 corresponding conditions using the original base seed and 512-token cap, all 2,041 aligned saved predictions and raw responses match the primary records exactly. This supports same-seed agreement for those Qwen3 runs, within the regeneration limits above. The input-size analysis records actual input lengths and reconstructs selected text and visual quantities, but neither reconstructs the complete original visual stream nor establishes length as a cause of score changes.

The study documents how supplied grammar packages perform in gloss-based translation configurations. Its findings support checking a package's actual content and incremental value before adopting it. They establish neither a general benefit from grammar context nor a general disadvantage of visual presentation, and they leave the model's use of grammatical explanations unresolved.

7. Conclusion and Future Work

This thesis examined whether supplied grammar resources improve sentence-level translation into English when prompts already contain glossed examples. It compares methods with different query-gloss inputs, measures grammar packages against matched baselines without grammar material or its use instruction (no-book baselines), and examines paired text and image packages.

7.1. Research Summary and Contributions

The primary study translated sentences from four low-resource languages, Gitksan, Lezgi, Natugu, and Tsez, into English with three vision-language models: Qwen3, Qwen3.5, and Gemini 2.5 Flash-Lite. It adapted the Gloss-shot, explicit Chain-gloss, and ModelGloss methods from GrammarMT (Ramos et al., 2025). Each method used the same 21 gold-glossed training demonstrations per language; ModelGloss also received an externally predicted gloss for the test sentence, while Chain-gloss asked the translating model to generate one first. Matched no-book baselines omitted grammar material and its use instruction.

I prepared summaries, tables, and cheat sheets from Brown’s Gitksan article, Rigsby’s introductory Gitksan excerpt, Haspelmath’s Lezgian excerpt, Boerger’s Natqgu sketch, and Comrie and Polinsky’s Tsez manuscript, with GPT-5.6 Sol assistance. The packages included text, corresponding images where supplied, and selected original pages. The main design contribution is the matched comparison of each grammar package with a no-book condition using the same model, method, supports, test inputs, and generation policy. The 21 glossed demonstrations are fixed across models, methods, and materials for a language. Additional examples inside the grammar packages vary with their coverage and preparation. Adding a package also adds an instruction to use it, while packages differ in length and layout; the design therefore does not isolate grammatical explanations. Linked configurations, responses, and evaluation inputs make the comparisons traceable to saved outputs, subject to the reproducibility limits in Chapter 6.

Empirically, ModelGloss ranks highest among the no-book methods, with observed chrF++ differences of roughly 9–26 points over Gloss-shot across the model-language groups. Completed supplementary Qwen method tests support its chrF++ advantages;

two Qwen3 Lezgi BLEU comparisons do not pass correction. Linguistic accuracy of the primary query glosses was not assessed, so these method differences do not isolate the accuracy of the glosses as their cause. The effects of grammar addition are less uniform: the original primary tests with valid reference translations (Gitksan, Natugu, and Tsez) identify one corrected increase and ten decreases, all in chrF++, while all nine corrected paired text-image differences favour the supplied text packages. Separate supplementary Lezgi tests also find an image-over-text BLEU advantage. These findings describe particular prompts and resources. A separate earlier MTOB study examined grammar-text strategies and saved versus extracted response views under a different protocol. It adds an evaluation lesson: scoring the translation portion instead of the saved response with its explanations can change the ranking of grammar-text strategies without generating new translations (Section 6.2.2). Its prompts and comparators differ from the primary matched study.

For a configuration like those tested, the practical first check is its no-book baseline with the same glossed supports. A grammar package should then be retained only if its matched comparison offers a useful gain for the intended evaluation. The nine corrected primary paired results support the supplied text versions in those comparisons, without establishing a general disadvantage for grammar images. Incomplete Chain-gloss responses were observed, especially for Gemini on Tsez, where generating the gloss could leave no extractable final translation. The completed limited Qwen3 output-budget check does not establish a general benefit from a larger budget.

7.2. Conclusions by Research Question

7.2.1. No-Book Performance of the Adapted Methods (RQ1)

RQ1 compares the three adapted prompting methods without grammar-book material. ModelGloss has the highest observed BLEU and chrF++ scores in every model-language baseline group. This ranking holds across Qwen3, Qwen3.5, and Gemini 2.5 Flash-Lite, including the shared Tsez comparison (Tables 5.2 and 5.3). Explicit Chain-gloss does not consistently outperform Gloss-shot. In these prompts, the method supplied with a query gloss from the dedicated GlossLM system ranks more consistently than the method that generates its own.

The comparison concerns complete prompting methods, rather than gloss quality alone. All three methods receive gold glosses as annotations of the 21 demonstration sentences; ModelGloss alone receives an additional predicted gloss of the new source sentence. A separate supplementary Qwen3 diagnostic substituted a gold query gloss and did not consistently improve automatic translation scores. Completed supplementary method tests support ModelGloss's chrF++ advantages for both Qwen models,

with two Qwen3 Lezgi BLEU exceptions. Gemini and neural method rankings remain descriptive. No primary linguistic gloss-accuracy assessment was performed, so the ranking cannot be attributed causally to better glosses. Section 6.1.1 develops that distinction.

7.2.2. Incremental Effects of Grammar Context (RQ2)

RQ2 compares each grammar addition with the corresponding no-book baseline, holding the model, method, demonstrations, test inputs, and generation policy fixed. Adding prepared grammar material did not produce a consistent improvement in automatic translation scores. In the original primary tests with valid reference translations (Gitksan, Natugu, and Tsez), one contrast passes multiple-testing correction as an increase: Qwen3.5 Tsez Chain-gloss with cheat-sheet text. Ten contrasts pass as decreases, all involving image packages; no BLEU context contrast in these tests passes correction (Table 5.4). The wider matrix contains changes in both directions: chrF++ changes are predominantly negative for Qwen3.5 and Gemini and more often positive for Qwen3 (Appendix F, Table F.4). Thus the study finds a selective positive result, but no basis for recommending grammar addition across the tested configurations.

This comparison measures the addition of a package and its use instruction to a prompt that already contains glossed demonstrations. It does not distinguish rule use from examples or responses to that instruction. Contrasts that did not pass correction do not establish equivalence to the no-book condition. Separate supplementary Qwen tests add a Lezgi context-related decrease; their image-over-text BLEU advantage answers a paired-format comparison, not a test of improvement over the no-book baseline. XCOMET-XXL evaluates source, prediction, and reference, but remains exploratory for these four source languages (Section 4.3.2).

7.2.3. Grammar Representations and Paired Text–Image Packages (RQ3)

RQ3 compares the supplied grammar representations, including paired text and image packages where available. The original primary tests eligible for inference use valid reference translations from Gitksan, Natugu and Tsez. The nine corrected paired findings comprise eight Tsez pairs and one Natugu pair, all chrF++ differences favouring TXT (Table 5.6); Gitksan contributes no significant pair. The supplementary Lezgi image-over-text BLEU advantage shows that this finding does not extend to every tested pair. Choosing a representation therefore requires checking the particular model, method, language, and source document.

The observed scores also provide conditional guidance among package types. For Tsez

Chain-gloss, Qwen3.5 has its highest chrF++ score with cheat-sheet text, followed by summary text and table text; Gemini has its highest chrF++ score with summary text, followed by table text and cheat-sheet text (Table F.3). These orderings are descriptive, rather than tested differences between the three text packages. Only the Qwen3.5 sheet’s increase over its no-book baseline passes the relevant primary correction. They help identify a candidate package for a particular configuration, while providing no universal ranking of summaries, tables, sheets, and selected pages.

The paired labels do not isolate presentation as the cause. The original Lezgi cheat-sheet image omits three mini-gloss rows and four examples present in its text version (Section 3.3). Other pairs still differ in presentation and in the prompt wording that introduces the supplied text or attached images and requests their use. Selected pages also preserve surrounding content that a prepared summary may omit. Whether recognition, content selection, or use of a grammatical cue produced a score difference remains open.

7.2.4. Variation across Models, Languages, and Methods (RQ4)

RQ4 examines how the grammar-addition and representation comparisons vary across models, languages, and prompting methods. Gains and losses occur across the context matrices (Tables F.1–F.3), while the no-book ModelGloss ranking is more consistent. No grammar-context advantage appears across all these configurations. The stored tests assess individual matched condition differences; they do not test whether a material’s effect differs between models or methods and therefore do not establish interactions.

The shared Tsez comparison uses the same first 99 sentences for all models. Some small rankings change on this subset, and no Qwen grammar-context contrast passes its subset-specific correction (Table 5.5). Model decoding policies, grammar-document coverage, and language-specific test sets also vary. The results are most useful for choosing and evaluating a complete configuration against its own baseline, rather than attributing score differences to model architecture or language properties alone.

7.3. Future Work

Content and presentation controls. A direct follow-up could separate the information in a grammar package from its presentation. Text and image versions could be checked for the same examples, forms, and explanations before evaluation, with input length and instructions controlled as far as the model interfaces allow. Separate ablations of lexical cues, parallel examples, and grammatical explanations would then test which supplied content contributes to translation. Rule-relevance and rule-application

checks could further distinguish access to a useful passage from its correct use. Such controls would address the ambiguity in the present paired comparisons; they have not been carried out here.

Evaluation data and expert assessment. Translation assessment should also move beyond the available automatic scores. Source-language experts could evaluate adequacy on audited items, and that judgment could be compared with the automatic metrics for these language pairs. Among the 87 Lezgi test records, three have invalid English references; a separate known overlap with a support example leaves 83 records for primary quality reporting, while generation diagnostics retain all 87. The primary 83-record scores cannot inherit p-values from the original 87-record analysis, but separate valid-reference Qwen tests have already been completed. Future work should extend those tests to new items and other models, correct invalid references, inspect overlap, and specify the evaluation policy beforehand. Expert assessment could also establish how informative the neural evaluator is for these language pairs.

Gloss assessment and alignment. The retained data already contain gold annotations and saved query glosses. Consistency of the retained source sentences and predicted-gloss vectors across ModelGloss conditions has been verified. This does not recover the unavailable original prediction caches or their source-ID mappings, and the primary glosses have not received a linguistic accuracy assessment. A future assessment could compare saved Chain-gloss and ModelGloss glosses with the gold annotations under explicit segmentation, normalization, and tag conventions, accounting for malformed or missing outputs. It should document source-ID alignment for the retained records and distinguish that check from reconstruction of the original cache provenance. Gloss agreement would be a quality diagnostic, rather than proof of how the translator used a gloss.

Demonstrations and completion. Support retrieval offers another testable direction. Ginn et al. (2024a) show variation in glossing outcomes with demonstration selection, but the effect of changing the fixed 21 supports on this translation task remains unknown. A future study could compare the current fixed demonstrations with demonstrations retrieved by similarity to the source sentence, holding other inputs constant. The completed limited Qwen3 budget and predicted-versus-gold query-gloss diagnostics could be extended to Gemini and more grammar-context configurations. For document images, separate checks of recognized text, table relations, and reading order would help distinguish access to a grammar cue from its use (Lee et al., 2023;

Ouyang et al., 2025). Matched-content image controls, including irrelevant or conflicting visual context, could test whether image relevance changes the result (Guerra and Khrapunova, 2026).

Broader replication and reproducibility. Independent evaluation items, additional grammar resources, and other language settings would test the reach of the observed patterns. Repeated generations across the full matrix and other models could extend the completed three-seed check of four selected Qwen3 pairs. Reproducibility requires retaining exact model versions, runtime and decoding settings, and the source-ID–gloss mappings used to supply external predictions. The current saved glosses and verified vector consistency support comparisons of the retained outputs; preserving the original mappings in future studies would also allow their prediction provenance to be reconstructed. Broader replication would help establish whether the ModelGloss ranking persists and where grammar context adds useful information.

A. Prompt Templates

A.1. Primary Translation Prompt Templates

The templates below show the fixed instruction text and block order found in saved prompts. Bracketed fields and the abbreviated support block stand in for the actual source sentence and all 21 support examples, which appeared in a fixed order. Image files were attached separately where specified. The templates do not reproduce complete requests. A no-book prompt has no grammar package or grammar-use instruction, but still includes the gold-glossed supports.

Gloss-shot and ModelGloss use this system message:

You are a direct translation engine. Return only the requested English translation in the requested output format.

Their support entries have the following syntax, repeated 21 times:

[LANGUAGE] sentence: [SUPPORT_SOURCE]
Gloss: [GOLD_SUPPORT_GLOSS]
English translation: [SUPPORT_ENGLISH_TRANSLATION]

The no-book Gloss-shot user body is:

Here are some examples of [LANGUAGE] sentences and their corresponding English translations:

[21 support examples in the format above, in fixed order]

Return exactly one line in this format: FINAL_TRANSLATION: <English translation>

Do not include notes, glosses, reasoning, bullet points, markdown, or extra text.

[LANGUAGE] sentence: [SOURCE_SENTENCE]
FINAL_TRANSLATION:

ModelGloss uses that same body and output instruction, but inserts an externally predicted test-sentence gloss immediately before the final marker:

[LANGUAGE] sentence: [SOURCE_SENTENCE]
Gloss: [EXTERNALLY_PREDICTED_QUERY_GLOSS]
FINAL_TRANSLATION:

The query gloss is externally predicted; it is neither a gold test annotation nor a gloss generated by the translating model. The support glosses remain gold annotations in both direct methods.

Explicit Chain-gloss instead uses this system message and support format:

You are a linguistic translation engine. Produce the requested gloss and translation.

[LANGUAGE] sentence: [SUPPORT_SOURCE]
Gloss: [GOLD_SUPPORT_GLOSS]
FINAL_TRANSLATION: [SUPPORT_ENGLISH_TRANSLATION]

Its no-book user body asks for a gloss before the translation:

Here are some examples of [LANGUAGE] sentences and their corresponding English translations:
An example without a gloss provides translation only; it does not demonstrate an empty gloss.
[21 support examples in the Chain-gloss format above, in fixed order]

[LANGUAGE] sentence: [SOURCE_SENTENCE]
Produce the gloss first and then the English translation.
The gloss must give English lexical meanings and grammatical abbreviations for the source morphemes, in source-word order.
Preserve identifiable morpheme boundaries. Use ? for an uncertain morpheme rather than inventing a meaning.
Do not copy the source sentence as the gloss. Then translate the complete sentence into natural English.
Return exactly two lines:
Gloss: <generated gloss>
FINAL_TRANSLATION: <English translation>
Do not add explanations or analysis.

The sentence about an example without a gloss comes from the prompt builder, although every selected support in the primary study has a nonempty gold gloss.

A.1.1. Grammar-Context Wrappers and Message Order

In text-context conditions, the selected context is inserted after the support block and before the query. The direct-method wrapper is:

Here is a grammar reference summary for [LANGUAGE]. Use it as supporting context:
[GRAMMAR_TEXT]

Use the grammar reference as supporting context.

Chain-gloss instead places “Here is a grammar reference summary for [LANGUAGE]:” before the selected text and replaces its no-book query instruction with “Use the grammar reference to produce the gloss first and then the English translation.” The other Chain-gloss requirements and two output markers remain as displayed above. These fixed strings are accompanied by the actual selected package text in a saved request; [GRAMMAR_TEXT] is not transmitted.

For image-context conditions the text body says “The attached images are pages from a grammar reference for this language.” The direct methods then say “Use the grammar reference as supporting context.” Chain-gloss says “Use the grammar reference to produce the gloss first and then the English translation.” The actual images are separate message items, in configured order. Qwen receives a separate system message and user image items before user text; Gemini receives the separate system string and user text before image entries. Qwen’s chat template disables thinking and requests a generation prompt. These arrangements describe the saved strings and active message construction, not an independently reconstructed visual token stream or a single backend-independent wire request.

A.2. Supplementary Gold-Query Diagnostic

The supplementary Qwen3 ModelGloss diagnostic uses the same no-book system message, 21 support examples and direct-method instructions shown above. Comparison of all 668 paired saved requests found that only the final query’s Gloss: value changes from the retained external prediction to the dataset’s gold annotation. The English query reference is not inserted as an answer. The unchanged final instruction and parameterized query are:

Return exactly one line in this format: FINAL_TRANSLATION: <English translation>
Do not include notes, glosses, reasoning, bullet points, markdown, or extra text.

[LANGUAGE] sentence: [SAME_SOURCE_SENTENCE]
Gloss: [PREDICTED_QUERY_GLOSS or GOLD_QUERY_GLOSS]
FINAL_TRANSLATION:

The bracketed alternative describes the single substitution; it is not a transmitted instruction. In particular, the gold-query condition still says not to output glosses. This supplementary exception does not alter the primary templates.

For Gitksan test index 0, the exact two values of the query field are shown below, in predicted then gold order. The source sentence and final-translation marker are identical in both requests.

Gloss: tell-T-1SG.II COMP LVB-CN COMP-1 first white.man-PL first-CN white.man-PL
go-3.II LOC-CN one.III-PL Kitwancool long.ago
Gloss: PROSP tell-T-TR-1SG.II MANR LVB-CN COMP in toward come.PL-CN PL-white-ATTR
person LOC-CN inhabitants-1PL.II Kitwancool long.ago

A.3. Earlier MTOB Grammar-Text Prompts

The earlier MTOB study uses a different request, with its methods and results documented in Appendix E. Its G_e and G_s strategies retrieve grammar passages by embedding and substring procedures respectively; each selected passage uses the same fixed wrapper, repeated in retained passage order. G_l supplies selected long grammar text. The bracketed context fields below are editorial placeholders; the fixed words agree with saved initial requests for both Qwen models in the earlier MTOB study.

[LANGUAGE] is a language spoken in [LOCATION]. Translate the following sentence from
[LANGUAGE] to English: [SOURCE_SENTENCE]

[GRAMMAR_CONTEXT]

Now write the translation.

[LANGUAGE]: [SOURCE_SENTENCE]

English translation:

For G_e and G_s , each passage within [GRAMMAR_CONTEXT] reads:

To help with the translation, here is a passage retrieved from a [LANGUAGE]-English
grammar book:

[PASSAGE_TEXT]

For G_l , the context has this form:

To help with the translation, here is the full text of a
[LANGUAGE]-English grammar book:

-

[SELECTED_LONG_GRAMMAR_TEXT]

This is the end of the [LANGUAGE]-English grammar book.

-

The phrase “full text” is the earlier MTOB prompt’s wording; the local condition used selected long text, not necessarily a complete book or uniform length. The initial requests have one user message, without a separate system message, the primary 21-support block, a query gloss, or a FINAL_TRANSLATION marker. A best-guess instruction exists as a refusal-retry variant, not in every initial request. The retained attempt history is incomplete, as discussed in Appendix E.

A.4. Grammar-Material Preparation Template Excerpt

The author reports using GPT-5.6 Sol with a shared preparation prompt, adapted by language and paths, to prepare the derived grammar materials. The supplied

historical copy was introduced as a Gitksan example but its literal input block names Tsez. The excerpt below therefore replaces those run-specific values with parameters. It is an abridged template from the author’s account, not a verbatim saved API request or evidence that every requested check succeeded.

You are an expert research coding assistant helping create grammar-context materials for low-resource machine translation experiments.

Goal: Create faithful, compact, model-readable grammar materials from one or more grammar book PDFs for translation from [LANGUAGE] into English.

Inputs: LANGUAGE = [LANGUAGE]; PDF_PATH_OR_FOLDER = [SOURCE_PATH];
OUTPUT_DIR = [OUTPUT_PATH].

Core rule: Use only information that appears in the provided PDF. When information is unclear, incomplete, unreadable, or uncertain, mark it as uncertain instead of guessing. Preserve a mapping from extracted facts to source PDF page numbers.

Create a model-readable summary, a compact cheat sheet, and a structured summary with tables as Markdown/text sources; render the cheat sheet and tables as readable JPG pages. Select and render translation-relevant original PDF pages, recording page numbers and reasons. Prefer paradigms, glossed examples, and grammatical descriptions; avoid pages containing only front matter or references.

Use OCR only when embedded PDF text is missing or unusable. Record extraction, OCR, rendering, or uncertainty issues. Before finishing, verify source faithfulness and correspondence between Markdown and rendered images.

These last instructions describe requested preparation and checks. The retained original Lezgi cheat-sheet JPG omits content present in its TXT, so successful completion of the requested correspondence check cannot be inferred from the prompt.

B. Complete Experiment Matrix

The primary study has 351 conditions: 36 no-book baselines and 315 grammar contexts, with 40,731 persisted sentence-condition records. These counts include failed generations and repeated test sentences. Table B.1 shows which packages have conditions; a dash marks an absent condition. Source names follow Table 3.1.

Language and grammar source	Summary TXT	Tables TXT	Tables JPG	Sheet TXT	Sheet JPG	Pages JPG	Contexts /model
Gitksan – Brown	Yes	Yes	Yes	Yes	Yes	Yes	18
Gitksan – Rigsby	Yes	Yes	Yes	Yes	Yes	Yes	18
Lezgi – Haspelmath (original)	Yes	Yes	Yes	Yes	Yes	Yes	18
Lezgi – Haspelmath (Cyrillic-form)	Yes	Yes	Yes	Yes	Yes	—	15
Natugu – Boerger	Yes	Yes	Yes	Yes	Yes	Yes	18
Tsez – Comrie and Polinsky	Yes	Yes	Yes	Yes	Yes	Yes	18

Table B.1.: Primary grammar-package availability by source and variant. Pages are selected original pages; the other columns are derived packages. Each available package has three methods per model. The Lezgi Cyrillic-form variant has no separate selected-page condition.

B.1. Primary matched conditions

The catalogs below give the exact P key for each saved condition, in Gloss-shot, explicit Chain-gloss and ModelGloss column order. Each row’s n is the generation-record count for each of its three keys, including failures; it is not a count of distinct or successful translations. Lezgi has $n = 87$ generated records, while the primary 83-record quality mask is an evaluation subset. Each model has four no-book language rows, representing twelve method-specific baseline conditions shared across source variants. Printed keys follow the data-row order of `reports/matched_gold_v2/condition_catalog.tsv`; P001 is the first data row.

Table B.2.: Gemini 2.5 Flash Lite: primary matched conditions. Each row’s n applies to each method. The four no-book rows represent twelve baseline conditions shared across source variants.

Grammar package	Gloss-shot	Chain-gloss	ModelGloss	n
No grammar document – shared language baselines				
Gitksan – no grammar document	P003	P001	P002	37
Lezgi – no grammar document	P069	P055	P062	87
Natugu – no grammar document	P090	P076	P083	99
Tsez – no grammar document	P111	P097	P104	99
Gitksan – Brown (original)				
Summary text (TXT)	P021	P009	P015	37
Summary tables (TXT)	P020	P008	P014	37
Summary tables (JPG)	P019	P007	P013	37
Cheat sheet (TXT)	P017	P005	P011	37
Cheat sheet (JPG)	P016	P004	P010	37
Selected original pages (JPG)	P018	P006	P012	37
Gitksan – Rigsby (original)				
Summary text (TXT)	P039	P027	P033	37
Summary tables (TXT)	P038	P026	P032	37
Summary tables (JPG)	P037	P025	P031	37
Cheat sheet (TXT)	P035	P023	P029	37
Cheat sheet (JPG)	P034	P022	P028	37
Selected original pages (JPG)	P036	P024	P030	37

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Table B.2.: Gemini 2.5 Flash Lite primary conditions (continued)

Grammar package	Gloss-shot	Chain-gloss	ModelGloss	<i>n</i>
Lezgi – Haspelmath (original)				
Summary text (TXT)	P075	P061	P068	87
Summary tables (TXT)	P074	P060	P067	87
Summary tables (JPG)	P073	P059	P066	87
Cheat sheet (TXT)	P071	P057	P064	87
Cheat sheet (JPG)	P070	P056	P063	87
Selected original pages (JPG)	P072	P058	P065	87
Lezgi – Haspelmath (Cyrillic-form)				
Summary text (TXT)	P054	P044	P049	87
Summary tables (TXT)	P053	P043	P048	87
Summary tables (JPG)	P052	P042	P047	87
Cheat sheet (TXT)	P051	P041	P046	87
Cheat sheet (JPG)	P050	P040	P045	87
Natugu – Boerger (original)				
Summary text (TXT)	P096	P082	P089	99
Summary tables (TXT)	P095	P081	P088	99
Summary tables (JPG)	P094	P080	P087	99
Cheat sheet (TXT)	P092	P078	P085	99
Cheat sheet (JPG)	P091	P077	P084	99
Selected original pages (JPG)	P093	P079	P086	99
Tsez – Comrie and Polinsky (original)				
Summary text (TXT)	P117	P103	P110	99
Summary tables (TXT)	P116	P102	P109	99
Summary tables (JPG)	P115	P101	P108	99
Cheat sheet (TXT)	P113	P099	P106	99
Cheat sheet (JPG)	P112	P098	P105	99
Selected original pages (JPG)	P114	P100	P107	99

Table B.3.: Qwen3.5-9B: primary matched conditions. Each row’s n applies to each method. The four no-book rows represent twelve baseline conditions shared across source variants.

Grammar package	Gloss-shot	Chain-gloss	ModelGloss	n
No grammar document – shared language baselines				
Gitksan – no grammar document	P120	P118	P119	37
Lezgi – no grammar document	P186	P172	P179	87
Natugu – no grammar document	P207	P193	P200	99
Tsez – no grammar document	P228	P214	P221	445
Gitksan – Brown (original)				
Summary text (TXT)	P138	P126	P132	37
Summary tables (TXT)	P137	P125	P131	37
Summary tables (JPG)	P136	P124	P130	37
Cheat sheet (TXT)	P134	P122	P128	37
Cheat sheet (JPG)	P133	P121	P127	37
Selected original pages (JPG)	P135	P123	P129	37
Gitksan – Rigsby (original)				
Summary text (TXT)	P156	P144	P150	37
Summary tables (TXT)	P155	P143	P149	37
Summary tables (JPG)	P154	P142	P148	37
Cheat sheet (TXT)	P152	P140	P146	37
Cheat sheet (JPG)	P151	P139	P145	37
Selected original pages (JPG)	P153	P141	P147	37
Lezgi – Haspelmath (original)				
Summary text (TXT)	P192	P178	P185	87
Summary tables (TXT)	P191	P177	P184	87
Summary tables (JPG)	P190	P176	P183	87
Cheat sheet (TXT)	P188	P174	P181	87
Cheat sheet (JPG)	P187	P173	P180	87
Selected original pages (JPG)	P189	P175	P182	87
Lezgi – Haspelmath (Cyrillic-form)				
Summary text (TXT)	P171	P161	P166	87
Summary tables (TXT)	P170	P160	P165	87
Summary tables (JPG)	P169	P159	P164	87
Cheat sheet (TXT)	P168	P158	P163	87
Cheat sheet (JPG)	P167	P157	P162	87

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Table B.3.: Qwen3.5-9B primary conditions (continued)

Grammar package	Gloss-shot	Chain-gloss	ModelGloss	<i>n</i>
Natugu – Boerger (original)				
Summary text (TXT)	P213	P199	P206	99
Summary tables (TXT)	P212	P198	P205	99
Summary tables (JPG)	P211	P197	P204	99
Cheat sheet (TXT)	P209	P195	P202	99
Cheat sheet (JPG)	P208	P194	P201	99
Selected original pages (JPG)	P210	P196	P203	99
Tsez – Comrie and Polinsky (original)				
Summary text (TXT)	P234	P220	P227	445
Summary tables (TXT)	P233	P219	P226	445
Summary tables (JPG)	P232	P218	P225	445
Cheat sheet (TXT)	P230	P216	P223	445
Cheat sheet (JPG)	P229	P215	P222	445
Selected original pages (JPG)	P231	P217	P224	445

Table B.4.: Qwen3-VL-8B-Instruct: primary matched conditions. Each row’s n applies to each method. The four no-book rows represent twelve baseline conditions shared across source variants.

Grammar package	Gloss-shot	Chain-gloss	ModelGloss	n
No grammar document – shared language baselines				
Gitksan – no grammar document	P237	P235	P236	37
Lezgi – no grammar document	P303	P289	P296	87
Natugu – no grammar document	P324	P310	P317	99
Tsez – no grammar document	P345	P331	P338	445
Gitksan – Brown (original)				
Summary text (TXT)	P255	P243	P249	37
Summary tables (TXT)	P254	P242	P248	37
Summary tables (JPG)	P253	P241	P247	37
Cheat sheet (TXT)	P251	P239	P245	37
Cheat sheet (JPG)	P250	P238	P244	37
Selected original pages (JPG)	P252	P240	P246	37
Gitksan – Rigsby (original)				
Summary text (TXT)	P273	P261	P267	37
Summary tables (TXT)	P272	P260	P266	37
Summary tables (JPG)	P271	P259	P265	37
Cheat sheet (TXT)	P269	P257	P263	37
Cheat sheet (JPG)	P268	P256	P262	37
Selected original pages (JPG)	P270	P258	P264	37
Lezgi – Haspelmath (original)				
Summary text (TXT)	P309	P295	P302	87
Summary tables (TXT)	P308	P294	P301	87
Summary tables (JPG)	P307	P293	P300	87
Cheat sheet (TXT)	P305	P291	P298	87
Cheat sheet (JPG)	P304	P290	P297	87
Selected original pages (JPG)	P306	P292	P299	87
Lezgi – Haspelmath (Cyrillic-form)				
Summary text (TXT)	P288	P278	P283	87
Summary tables (TXT)	P287	P277	P282	87
Summary tables (JPG)	P286	P276	P281	87
Cheat sheet (TXT)	P285	P275	P280	87
Cheat sheet (JPG)	P284	P274	P279	87

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Table B.4.: Qwen3-VL-8B-Instruct primary conditions (continued)

Grammar package	Gloss-shot	Chain-gloss	ModelGloss	<i>n</i>
Natugu – Boerger (original)				
Summary text (TXT)	P330	P316	P323	99
Summary tables (TXT)	P329	P315	P322	99
Summary tables (JPG)	P328	P314	P321	99
Cheat sheet (TXT)	P326	P312	P319	99
Cheat sheet (JPG)	P325	P311	P318	99
Selected original pages (JPG)	P327	P313	P320	99
Tsez – Comrie and Polinsky (original)				
Summary text (TXT)	P351	P337	P344	445
Summary tables (TXT)	P350	P336	P343	445
Summary tables (JPG)	P349	P335	P342	445
Cheat sheet (TXT)	P347	P333	P340	445
Cheat sheet (JPG)	P346	P332	P339	445
Selected original pages (JPG)	P348	P334	P341	445

B.2. Separate Earlier MTOB Conditions

The earlier MTOB catalog contains 30 conditions and 4,230 persisted records under the distinct prompt and evaluation protocol described in Appendix E. It covers two Qwen models and five grammar-source selections. G_e retrieves passages by embedding, G_s retrieves by substring, and G_l supplies selected long grammar text. Printed M keys follow the data-row order of `experiments/mtob/evaluation_v1/condition_matrix.tsv`; M01 is the first data row. These rows exclude the earlier `matched_v1` baselines used in the separate cross-protocol comparison; matching a model and language name does not make a primary gold-v2 condition directly comparable.

Language and grammar source	Embedding retrieval (G_e)	Substring retrieval (G_s)	Selected long text (G_l)	n
Qwen3-VL-8B-Instruct				
Gitksan – Brown	M01	M02	M03	37
Gitksan – Rigsby	M04	M05	M06	37
Lezgi – Haspelmath	M10	M11	M12	87
Natugu – Boerger	M07	M08	M09	99
Tsez – Comrie and Polinsky	M13	M14	M15	445
Qwen3.5-9B				
Gitksan – Brown	M16	M17	M18	37
Gitksan – Rigsby	M19	M20	M21	37
Lezgi – Haspelmath	M25	M26	M27	87
Natugu – Boerger	M22	M23	M24	99
Tsez – Comrie and Polinsky	M28	M29	M30	445

Table B.5.: All 30 earlier MTOB conditions, grouped into ten source–model rows. Each strategy cell is one condition key; n is the persisted record count for each strategy separately.

B.3. Separate Retained Qwen3 Diagnostic Conditions

Table B.6 lists the 37 retained new conditions once. Its keys identify the same rows in the supplementary outcome tables; every row uses Qwen3, gold-glossed supports and the runtime in Appendix D.4. S, C and MG denote Gloss-shot, Chain-gloss and Model-Gloss. The query-gloss column distinguishes predicted and gold inputs for MG; C generates its gloss, and S receives none. The catalog counts all generation records, includ-

B. Complete Experiment Matrix

ing 87 Lezgi items; quality tables use the main 83-record mask. Every listed empty prediction has a truncation flag. F, G and I identify diagnostic membership, with shared controls printed once. Exact condition identifiers, parent configurations and hashes are retained in `experiments/additional_v1/publication/condition_catalog.tsv` and `publication/retained_manifest.json`.

Table B.6.: Retained supplementary Qwen3 conditions. Cap is the maximum number of new tokens; n and Empty count generation records.

Key	Language	Method	Package	Seed	Cap	Query gloss	n	Empty	Used in
S01	Gitksan	C	No book	20260910	1024	—	37	1	F
S02	Gitksan	C	No book	20260910	512	—	37	1	F
S03	Gitksan	MG	No book	20260910	512	Pred.	37	0	G
S04	Gitksan	MG	No book	20260910	512	Gold	37	0	G
S05	Gitksan	S	No book	20260910	512	—	37	0	I
S06	Gitksan	S	No book	20260925	512	—	37	0	I
S07	Gitksan	S	No book	20261001	512	—	37	0	I
S08	Gitksan	S	Brown sheet TXT	20260910	512	—	37	0	I
S09	Gitksan	S	Brown sheet TXT	20260925	512	—	37	0	I
S10	Gitksan	S	Brown sheet TXT	20261001	512	—	37	0	I
S11	Lezgi	C	No book	20260910	1024	—	87	1	F
S12	Lezgi	C	No book	20260910	512	—	87	1	F/I
S13	Lezgi	C	No book	20260925	512	—	87	2	I
S14	Lezgi	C	No book	20261001	512	—	87	0	I
S15	Lezgi	C	Sheet TXT	20260910	512	—	87	0	I
S16	Lezgi	C	Sheet TXT	20260925	512	—	87	1	I
S17	Lezgi	C	Sheet TXT	20261001	512	—	87	0	I
S18	Lezgi	MG	No book	20260910	512	Pred.	87	0	G
S19	Lezgi	MG	No book	20260910	512	Gold	87	0	G
S20	Natugu	C	No book	20260910	1024	—	99	3	F
S21	Natugu	C	No book	20260910	512	—	99	3	F
S22	Natugu	MG	No book	20260910	512	Pred.	99	0	G/I
S23	Natugu	MG	No book	20260910	512	Gold	99	0	G
S24	Natugu	MG	No book	20260925	512	Pred.	99	0	I
S25	Natugu	MG	No book	20261001	512	Pred.	99	0	I
S26	Natugu	MG	Sheet JPG	20260910	512	Pred.	99	0	I
S27	Natugu	MG	Sheet JPG	20260925	512	Pred.	99	0	I
S28	Natugu	MG	Sheet JPG	20261001	512	Pred.	99	0	I
S29	Tsez	C	No book	20260910	1024	—	445	34	F
S30	Tsez	C	No book	20260910	512	—	445	35	F/I
S31	Tsez	C	No book	20260925	512	—	445	25	I
S32	Tsez	C	No book	20261001	512	—	445	22	I

Continued on next page

B. Complete Experiment Matrix

Key	Language	Method	Package	Seed	Cap	Query gloss	n	Empty	Used in
S33	Tsez	C	Summary TXT	20260910	512	—	445	22	I
S34	Tsez	C	Summary TXT	20260925	512	—	445	23	I
S35	Tsez	C	Summary TXT	20261001	512	—	445	18	I
S36	Tsez	MG	No book	20260910	512	Pred.	445	0	G
S37	Tsez	MG	No book	20260910	512	Gold	445	0	G

C. Grammar-Context Material Examples

These excerpts and crops illustrate the six representation formats described in Section 3.3. They come from retained materials and do not constitute new experimental inputs or a complete catalogue. The spelling *Gitskan* in quoted prepared files is preserved as found; this thesis uses *Gitksan* in its own prose. The source is Brown et al.’s *Gitksan* article (Brown et al., 2016).

C.1. Prepared Text Excerpts

The Brown text summary has headings WORD_ORDER and PHONOLOGY_AND_ORTHOGRAPHY. Its scope note says a general word-order rule and full case, pronoun, and tense–aspect–mood paradigms were not found in the article. One actual WORD_ORDER row reads: *Not found as a general prose rule in this PDF* and points to PDF pages 9–10. An orthographic row links apostrophe conventions to PDF pages 2 and 4. This illustrates how the summary selects topics and retains page pointers rather than reproducing a whole PDF page.

These lines excerpt the Brown *summary tables* TXT, under its original heading PARTICLES_CLITICS_GLOSS. They are quoted artifact content, not a fresh validation of the linguistic analyses.

```
IPFV | imperfective | appears in narrative gloss yukw=t as IPFV=3.I |  
PDF p. 9  
PROSP | prospective | dim is glossed PROSP in narrative examples |  
PDF p. 9  
CN | connective/clause connector label in glosses | appears as =hl or  
gan=hl in narrative glosses | PDF p. 9
```

The *cheat sheet* TXT instead places these shorter rows under HIGH_VALUE_CUES:

```
dim | PROSP | prospective/future-like cue in narrative | [PDF p. 9]  
nee | NEG | negation; appears in 'I can't do it' | [PDF p. 10]  
dii | FOC | focus label in forms like nee=dii | [PDF p. 10]
```

C.2. Prepared and Original-Page Images

Figures C.1 and C.2 crop the frozen Brown table and sheet JPGs. The corresponding conditions used the full image files. Both display literal Markdown table syntax, unlike the drawn tables in the other 17 derived JPGs. The crops do not establish complete TXT/JPG equivalence.

PARTICLES_CLITICS_GLOSS

	Form/label		Function from PDF		Translation cue		Source page	
	---		---		---		---	
	IPFV		imperfective		appears in narrative gloss yukw=t as IPFV=3.I		PDF p. 9	
	PROSP		prospective		dim is glossed PROSP in narrative examples		PDF p. 9	
	CN		connective/clause connector label in glosses		appears as =hl or gan=hl in narrative glosses		PDF p. 9	

Figure C.1.: Display crop of frozen Brown summary-table JPG showing literal Markdown syntax. Source: Brown et al. (2016); the excerpt refers to PDF page 9.

HIGH_VALUE_CUES

	Form/label		Function from PDF		Translation cue		Page	
	---		---		---		---	
	dim		PROSP		prospective/future-like cue in narrative		[PDF p. 9]	
	nee		NEG		negation; appears in 'I can't do it'		[PDF p. 10]	
	dii		FOC		focus label in forms like nee=dii		[PDF p. 10]	

Figure C.2.: Display crop of frozen Brown cheat-sheet JPG showing literal Markdown syntax. Source: Brown et al. (2016); the excerpt refers to PDF pages 9–10.

Orthographic transcription with interlinear English gloss

The top line of each entry is an orthographic representation. The second line indicates where the affix and clitic boundaries are within words.

Yukwt	laseexwhl	bahasxw	ganhl	hlox _s
yukw=t	laseexw=hl	bax-asxw	gan=hl	hlox _s
IPFV=3.I	discuss=CN	run-ANTIP	PH.CNJ=CN	sun
naahl	ky'aa	daxgyadit	as	'nidiit,
naa=hl	ky'aa	daxgyat-it	a=s	'nidiit
who=CN	most	strong-SX	PREP=PN	3PL.III

'The wind and the sun were discussing amongst themselves who was the strongest of them.'

Figure C.3.: Display crop of frozen selected original Brown PDF page 9, preserving the source article's layout. Source: Brown et al. (2016).

C.3. Embedded-PDF Text Extraction

The earlier MTOB study also used embedded PDF text. A short Brown PDF page-2 excerpt reads: *In the notation used here, glottalization is indicated following the segment for plosives and affricates, and preceding the segment for sonorants* (Brown et al., 2016). It is source text extraction, not output from a newly run OCR system. Extraction preserves some words while losing the page's visual organization; the earlier MTOB retrieval and long-context procedures are described in Appendix E.

D. Reproducibility and Evaluation Details

This appendix identifies the retained inputs, settings, outputs, and analysis files needed to inspect the reported comparisons. It supplements the design and evaluation definitions in Chapter 4; the saved files do not constitute a complete portable generation environment. The paths below are relative to the experiment repository named `Leveraging-Grammar-Book-Images-in-Vision-Language-Models-for-Low-Resource-Machine-Translation-main`. In the project workspace, that repository is under `SourceCode/ThesisExperiments/`, separate from the LaTeX source package. Appendix B lists every evaluated condition.

D.1. Frozen Inputs and Condition Identity

The primary condition catalog is `reports/matched_gold_v2/condition_catalog.tsv`. Each of its 351 rows identifies a frozen configuration under `configs/matched_gold_v2/` and a saved response file under `results/matched_gold_v2/`. The configurations retain the selected support examples, test inputs, model policy, applicable predicted query glosses, context-file hashes, and configuration fingerprints. Result and metric provenance link those identities to persisted record counts and prediction hashes. The catalog’s 36 no-book baselines and 315 context conditions contain 40,731 sentence-condition records in total; the same test sentences recur across conditions, and failed outputs remain in their records.

Each language uses 21 fixed, nonempty gold-glossed support examples from its training input, in the same order across models, methods, and packages. The support manifests at `docs/matched_gold_v2/support/` store their training-source fingerprints, selected indices, and support hashes. Gitksan uses the pinned official uncovered annotation revision `190689ac81935359c69a46463c48e25e63e601f7`. Gitksan, Natugu, and Tsez select original training indices 0–20. Lezgi selects 0–4 and 6–21: original training item 5 was replaced because its source exactly overlaps a test source. That replacement is distinct from the three invalid *test references* excluded later for descriptive scoring. No selected support source exactly matches a test source under the recorded exact-string check.

The saved ModelGloss vectors are consistent across their configured conditions. The original loader reads the no-reference GlossLM prediction CSV for each language,

selects unsegmented rows, sorts their record IDs by numerical suffix, normalizes new-lines in each pred value, and pairs the resulting vector with ordered test items by position. It does not compare source strings while pairing. The four originating CSV caches are unavailable in this transfer, so the alignment of their original rows to the retained vectors cannot be independently rechecked from those files. Frozen paths also retain original cluster locations; locally mapped input files were checked against recorded hashes without rewriting configurations. These identities support inspection of the stored study, while missing caches and incomplete runtime pinning limit exact regeneration.

The retained design history does not explain the model selection, Gemini’s first-99 Tsez subset, omission of the ModelGloss warning, the original Lezgi sheet’s missing sections or the absence of the four gloss-prediction CSV caches. It provides no basis for attributing motives or intent to these choices or omissions.

D.2. Models, Generation, and Response Parsing

The primary model IDs are Qwen/Qwen3-VL-8B-Instruct, Qwen/Qwen3.5-9B, and google/gemini-2.5-flash-lite. The Gemini route uses OpenRouter with the Google provider. These names identify configured systems, but do not pin immutable model-weight or provider revisions. Table 4.3 in Chapter 4 is the authoritative generation-settings table. The Qwen jobs used FP32 on one H100 each; the output limit covers both gloss and translation in Chain-gloss. Policies stay fixed within matched pairs and differ between backends. The earlier records under results/diagnostics/tsez_chain_support_decoding_v1/ retain 32 responses per Qwen model: four configurations on eight development examples. Their summaries include gloss_chrf and production_approved=false. The sampled configurations reduced invalid or truncated outputs; docs/sampled_decoding_validation.md and docs/tsez_chain_diagnosis.md document the check. The settings adopted in matched_v1 remained unchanged in matched_gold_v2. This development history concerns complete decoding configurations, not an isolated sampling or penalty test. The run configurations call Gloss-shot shot, explicit Chain-gloss chain_gloss, and ModelGloss modelgloss. These labels identify the three prompt builders described in Chapter 4; the gloss in the second method is generated in the translation response, while the third uses a saved external prediction.

The current renderer specifies white 1900-by-2600-pixel canvases for prepared materials, further pages where needed and JPEG quality 90; its PDF-page renderer defaults to scale factor two and quality 82. These code settings do not establish how every retained image was originally produced. Section 3.3 describes the observed render-

ing differences. In text conditions, demonstrations precede grammar text and the test sentence. In Qwen image conditions, image blocks precede the textual user-message body; in Gemini requests, text precedes image parts. Images follow the fixed configuration order. A hash identifies the supplied file bytes but does not prove equivalent visual recognition.

The active call path is `runners/matched_gold_v2/run.py` through `runners/matched_gold_v2/protocol.py` to `runners/matched/protocol.py`; the Qwen backend calls the generation helper rather than its legacy standalone parser. The primary parser selects the last explicit `FINAL_TRANSLATION` marker. Gloss-shot and ModelGloss also accept a single unmarked answer line subject to format checks; explicit Chain-gloss requires the final-translation label and separately extracts the preceding `Gloss:` line. An absent or multiline gloss is flagged but does not discard an otherwise extractable translation. Detected reasoning output is rejected. Unextractable, content-filtered, empty, or known truncated responses remain empty predictions at their original indices, even when truncation leaves a translation fragment. Format and completion flags describe response handling.

Marker matching is case-insensitive and line-initial, selecting the last `FINAL_TRANSLATION:` marker. Without a marker, only Gloss-shot and ModelGloss may use the fallback: strip each line, discard blank lines and require exactly one nonempty line. A line beginning with `gloss:`, `note:`, `sorry`, `i cannot` or `i can't`, after lowercasing, is rejected. An optional initial English translation:, Translation: or Answer: prefix is removed case-insensitively. The parser records `unmarked_single_line`; the flag does not itself reject the answer, although a line left empty remains an empty prediction.

The active Qwen warning helper lowercases the raw response and divides it into word and punctuation tokens using `\w+|[\^\w\s]`. It detects four consecutive copies of a span of 3–12 tokens and appends a repetition warning without clearing the prediction. The legacy standalone parser’s repetition-rejection rule is not part of this active call path. Gemini has no corresponding detector; its saved zero-warning count cannot establish absence of repetition.

One persisted semantic attempt is retained per item, without a corrective prompt after a malformed answer. An interrupted condition resumes only unsaved test indices and preserves previously saved failures. API transport errors permit at most five identical-request attempts in total (the initial attempt plus up to four resends), without requesting a revised answer. A response lost before persistence cannot be recovered from the stored record. Appendix A shows the parameterized prompt text.

D.3. Evaluation Cohorts, Metrics, and Saved Analyses

The native cohort follows each configured test sequence: 37 Gitksan, 87 Lezgi, and 99 Natugu records per condition; Tsez has 445 for each Qwen model and 99 for Gemini. The common99 cohort restricts Qwen Tsez to the same fixed first 99 items used by Gemini; the other languages are unchanged. It reuses native-cohort outputs and is neither random nor an independent replication.

Lezgi reference-mask provenance. Lezgi’s 87 generated records include three invalid English references at zero-based indices 37, 62 and 81, and test index 11 has a reference embedded in a selected support translation. Every primary quality condition and metric uses the same retained indices, giving $87 - 3 - 1 = 83$ records; failed predictions remain included. The 84-record view after only invalid-reference exclusion is an intermediate historical reporting view, not a second primary analysis. Existing `valid_reference_no_overlap83` rows in `reports/matched_gold_v2/lezgi_sensitivity_scores.tsv` provide scores for all 108 Lezgi conditions in each cohort. No 83-record significance tests were stored in the original primary analysis; the original 87-record tests remain in their original correction families and are not attached to these primary scores.

Table D.1 records software and metric identities found in saved provenance. BLEU and chrF++ are on a 0–100 scale. XCOMET-XXL remains on its native scale and is exploratory because the evaluator’s listed source-language coverage excludes all four study languages. XCOMET-XL is retained separately as supplemental evidence, not substituted for xCOMET-XXL. Versions absent from the saved records, including an immutable Qwen weight revision or a complete PyTorch/Transformers/CUDA environment, are not inferred from current installations.

The native and common99 lexical analyses are under `docs/matched_gold_v2/analysis_native/` and `docs/matched_gold_v2/analysis_common99/`; xCOMET-XXL analyses are under `docs/matched_gold_v2/analysis_xcomet_xxl/`. Individual metric files are under `metrics/matched_gold_v2/`, while `reports/matched_gold_v2/thesis_scores.tsv` gives the selected reporting rows with cohort and reference-policy fields. The original lexical Holm families contain 630 context-baseline and 216 paired TXT/JPG tests *per cohort*, pooling BLEU and chrF++ within each family. The xCOMET-XXL families contain 630 and 216 tests respectively while pooling the two cohorts for one metric. Excluding Lezgi’s contaminated inference from reporting does not recompute those original correction families. Chapter 4 specifies the bootstrap procedures and their limits; no new test is introduced here.

Component	Recorded identity or setting
Primary lexical	SacreBLEU 2.6.0; one reference; BLEU mixed case, 13a tokenization, exponential smoothing, no effective-order adjustment. chrF++ uses character order 6, word order 2, $\beta = 2$, and no space inclusion.
Neural evaluator	Unbabel/xCOMET-XXL, COMET 2.2.7; retained checkpoint SHA256 e760e1f568af69b7a1bf7aeb46d8f3be 21f01be7cbda480f8225ee81eb0af27a. Supplementary xCOMET-XL has a separate checkpoint.
Stored lexical analysis	100,000 paired bootstrap samples; seed 20260915.
Stored xCOMET-XXL analysis	NumPy 2.2.6; SciPy 1.15.3; 100,000 samples; base seed 20260919.

Table D.1.: Recorded primary evaluation identities. A listed version is supported by retained provenance, not a complete environment lock.

Lexical paired bootstrap. SacreBLEU 2.6.0 recomputes both systems’ corpus scores on the same 100,000 index samples, using seed 20260915. For the saved p-value, the implementation forms the absolute score difference for every sample and centres those differences by subtracting their mean. If c centred absolute differences strictly exceed the observed absolute difference, it reports $p = (c + 1)/(100000 + 1)$. The stored individual-system bootstrap confidence half-width is distinct from a paired-difference interval. The implementation is `_paired_bs_test` and `_compute_p_value` in SacreBLEU 2.6.0.

Neural paired bootstrap. The xCOMET-XXL analysis draws 100,000 paired item-score-difference samples and computes their means. For a two-sided null-centred test, it subtracts the observed mean difference from each sample mean and counts absolute centred values at least as large as the observed absolute difference, allowing tolerance 10^{-12} . The p-value uses a plus-one correction; an observed absolute difference at or below the tolerance receives $p = 1$. Comparison seeds are deterministically derived from base seed 20260919 and a hash of the comparison identity. The 95% percentile interval concerns the paired mean difference and has no multiple-testing adjustment.

Holm adjustments. For ordered raw p-values $p_{(j)}$ in a family of m tests, the adjusted value at rank i is $\min\{1, \max_{j \leq i} (m - j + 1)p_{(j)}\}$. The stored family definitions, including all original Lezgi 87-record tests, are retained when interpreting filtered descriptive views; the original tests are not recomputed for 84 or 83 Lezgi items. The

separate supplementary source-cluster tests use those masks.

D.4. Retained Supplementary Provenance and Procedures

The source revision is 7411b8aa64fcce178c806d06daa8638e6f171a1b. The supplement is under `experiments/additional_v1/`; its `publication/retained_manifest.json` and `publication/condition_catalog.tsv` identify 37 distinct Qwen3 conditions, 6,049 generation records and 192 empty/truncated predictions. Shared controls appear once in the catalog and may belong to more than one diagnostic. The 74-condition proposal is not a completed matrix. Completed A/B/C analyses also use historical Qwen3.5 predictions, although no new Qwen3.5 output survived. The original primary matrix, score files and statistical families remain unchanged.

Every retained new runtime record identifies Qwen/Qwen3-VL-8B-Instruct revision 0c351dd01ed87e9c1b53cbc748cba10e6187ff3b, Torch 2.7.1+cu118 and Transformers 4.57.6, FP32 on one GPU, thinking disabled and no retries. The supports remain gold-glossed; `test_gold_gloss` is true only for G. The output cap is 512 except F's 1,024-token variant. Base seeds are 20260910, 20260925 and 20261001, with item seed $s + 2i$ for original index i ; F/G use the first base seed. These date-like identifiers do not establish execution dates. The Qwen sampling settings in Table 4.3 otherwise remain unchanged, with no documented precision, image-resolution, prompt or extraction fallback.

A saved-response identity check aligns 13 corresponding Qwen3 conditions at base seed 20260910 and the 512-token cap. All 2,041 predictions and raw responses are identical to the primary records. This check supports same-seed agreement for these Qwen3 conditions; complete-environment reproducibility and Gemini are outside its scope.

A/B source-cluster bootstrap. Distinct source strings are sampled with replacement, keeping all records of each sampled string on both sides. The stored lexical procedure uses 100,000 draws and deterministic cohort-specific seeds derived from base 20260925 and cohort identity. If D is the observed signed corpus-score difference and D_b is a resampled difference, the two-sided null-centred plus-one p-value counts $|D_b - D| \geq |D|$ (with the implementation's numerical tolerance). The 95% percentile interval is pointwise for the paired difference. Holm correction pools all declared views, both models and both metrics into separate families: `A_methods`, 72; `B_context`, 264; `B_format`, 96. Reporting only 83-record Lezgi or first99 Tsez does not shrink these families. A/B scores retain failures under each mask; main Lezgi has 83 records and 81 distinct-source clusters. Its 84-record view is sensitivity evidence. In the saved A/B

reports, sensitivity83 maps to the thesis’s main 83-record cohort, while valid84 maps to the overlapping 84-record appendix sensitivity.

The saved A/B reports are reports/A_comparisons.tsv and reports/B_comparisons.tsv relative to the supplement. F/G comparisons are in metrics/generation_comparisons.tsv; their planned 24-test families remain incomplete, with blank adjusted p-values. Neither their raw p-values nor pointwise intervals establish adjusted significance. I has no significance test, and supplementary xCOMET-XL/xCOMET-XXL scores in reports/xcomet_xl_scores.tsv and reports/xcomet_xxl_scores.tsv are descriptive. Result, configuration, runtime and scoring hashes in the retained manifests connect each contrast to its saved inputs. Table B.6 provides readable catalog keys used by the outcome tables.

Input sizes. C’s reports/C_input_sizes_qwen3.tsv and reports/C_input_sizes_qwen35.tsv describe 117 existing conditions and 15,999 logged records per Qwen model, including all 87 Lezgi generation items. Logged lengths are attempt.input_tokens, the numbers of token IDs in the actual processor/model input. First-prompt text counts reconstruct saved system text, a newline and user text with the pinned tokenizer; they include textual context and differ from chat-template processing. Grammar-only counts encode grammar_text without special tokens. Reconstructed visual counts use the stored images and processor image_grid_thw, dividing its grid product by merge_size². These quantities are model-specific. They cannot be added to obtain the logged totals and do not reconstruct the complete original vision stream. Table F.10 reports ranges of condition means separately from per-item extrema; it does not demonstrate that input length caused a quality difference.

E. Earlier MTOB Methods and Results

This appendix documents the separate earlier MTOB-inspired supplementary study. Its 30 conditions and 4,230 retained records use grammar text without the primary study’s 21 gold-glossed supports. Its cross-protocol comparator is the earlier `matched_v1` family, not `matched_gold_v2`. The prompt template is in Appendix A; the 30 condition rows are in Appendix B. The differences in inputs and generation policy prevent this suite from serving as another matched test of the primary grammar-package effect.

E.1. Context Construction and Execution

E.1.1. Text Extraction and Context Construction

The supplementary study adapts MTOB’s embedding retrieval (G_e), substring retrieval (G_s), and long grammar context (G_l). The original work also uses a word list and translation examples (Tanzer et al., 2024). Here, the prompt supplies grammar text but no demonstrations or predicted query gloss. Word-list retrieval and the medium-context G_m condition are not used.

PyMuPDF extracts embedded PDF text, normalizes whitespace, and removes repeated marginal lines. Character spans retain the source-page links. This is text extraction, not OCR. MTOB instead converts grammar LaTeX source to plaintext and includes manual correction (Tanzer et al., 2024). The present extraction retains noise, particularly in the Rigsby resource, and does not preserve the original page layout.

Retrieval uses 512-token GPT-2 chunks with 256-token overlap, following MTOB’s chunking scheme (Tanzer et al., 2024). Final chunks can be shorter. Both retrieval methods supply two passages, using only the source sentence as the query.

For G_e , `all-mpnet-base-v2` embeds the query and passages, following MTOB’s model choice (Tanzer et al., 2024). This checkpoint produces sentence and paragraph embeddings for similarity search (Sentence Transformers, n.d.). The retriever applies attention-mask mean pooling and L2 normalization, then ranks passages by cosine similarity. Chunk indices break ties. Embedding input is capped at 384 model tokens, but the translator receives the whole selected GPT-2 chunk. A passage may therefore be ranked from a representation of only its beginning.

For G_s , the retrieval score is the longest common character substring between query and passage. The two selected passages are not explicitly sorted by descending score; the prompt uses their fixed order in the saved retrieval data. The retained passages have been checked against their source chunks. That check confirms the passage content, but does not independently establish relevance or reconstruct embedding rankings.

The G_l condition joins fixed PDF-page ranges: Brown 2–10, Rigsby 1–7, Natugu 2–68, Lezgi 19–35 and 48–59, and Tsez 1–36 and 40. These selections contain approximately 9,181, 13,310, 90,307, 18,935, and 25,903 GPT-2 tokens, respectively. They cover selected portions of the documents and differ greatly in length. Natugu’s range includes PDF pages 26 and 47, whose examples correspond to test indices 20 and 50. The saved prompts for both Qwen models contain index 20’s English reference in G_s and G_l , and index 50’s source example in G_e , G_s , and G_l . For index 20, G_e retrieves other pages and lacks the known reference. This establishes prompt overlap for these two earlier items; its effect on scores and the extent of overlap elsewhere remain unknown.

E.1.2. Prompting and Model Execution

The text-only prompt follows MTOB’s Appendix C: it identifies the language, introduces the source sentence, supplies grammar text, and repeats the sentence before an English-translation prefix (Tanzer et al., 2024). The two Qwen checkpoints, five sources, and three context methods give 30 conditions and 4,230 sentence–condition records. The completed evaluation uses the existing outputs without generating new responses.

These runs sample at temperature 0.05 and allow 256 new tokens. MTOB reports the same temperature (Tanzer et al., 2024), although the models and other settings differ. The implementation uses FP32 on one device and disables thinking. The initial seed is 2024 plus the example index, with additional offsets for retries.

The earlier generation policy allows up to three attempts to meet the no-reasoning requirement. A refusal may trigger up to three further best-guess requests, each with reasoning retries. This retry policy differs from the primary suite’s one-attempt policy. The retained responses do not provide a complete attempt history. Four Qwen3.5 records have a recorded rerun round, but their earlier responses were overwritten. The stored answers can therefore be evaluated, but not every original attempt can be reconstructed. Usable finish metadata are also absent, so truncation is unknown for all MTOB records.

The final evaluation retains all expected items. Two Qwen3.5 Natugu G_e records and one G_s record have failed statuses and receive empty predictions. Both conditions retain their full 99-item denominators. The earlier successful-only scores are not used

as the final evaluation. Response extraction and comparison with the earlier matched baselines are defined below.

E.2. Response Views, Metrics and Comparisons

MTOB evaluation has two response views. The raw view uses the final saved response after basic removal of a leading English-translation prefix. A retry response saved as text takes precedence over the initial response and prediction fields.

The extracted view identifies explicit translation sections and removes following labelled explanations, breakdowns, or notes. When an explanation comes first, extraction requires a unique translation section. If multiple labels or unclear boundaries make extraction ambiguous, the procedure retains the raw answer and adds an ambiguity flag. It never selects text by consulting the reference or metric score. Both views replace known failed responses with empty predictions.

Comparing the raw and extracted views provides a post-hoc sensitivity analysis, not an externally preregistered experiment. The extraction rules remove some explicitly marked commentary. They neither select the first sentence or paragraph by default nor reclassify ordinary explanations as hidden reasoning. Extraction does not establish linguistic accuracy or recover missing text when truncation is unknown.

MTOB cleaning removes the literal substring `=` and ASCII punctuation from predictions and references. Its BLEU uses 13a tokenization, four n-gram orders, no smoothing, and no effective-order adjustment. Its chrF uses character order six, word order zero, and $\beta = 2$, on a 0–100 scale. This is plain chrF (Popović, 2015), the metric family used by MTOB (Tanzer et al., 2024), rather than chrF++.

Additional MTOB metrics are descriptive. The local ROUGE implementation calculates sentence-level F1 from case-folded ASCII alphanumeric tokens, without stemming, and then averages those scores. ROUGE-Lsum equals ROUGE-L, and the scores use a 0–1 scale. CharacTER uses word shifts and character edits (Wang et al., 2016). The cap and empty-hypothesis convention in the following sentence are local evaluation choices. It normalizes by hypothesis length and caps the score at one; an empty hypothesis receives one. This definition differs from ordinary reference-normalized character error rate. No MTOB XCOMET scores are included.

MTOB and the primary study use different preprocessing and metric definitions. Table E.1 states which records enter each evaluation and which exclusions were applied. An *evaluated record* is a configured test item included in a score or test, even when its prediction is empty after a known failure. A reference exclusion removes an item because its English reference is invalid or overlaps a prompt; this differs from retaining a failed prediction as empty. The retained common-evaluator comparison applies to ear-

lier matched baselines only; it does not rescore or compare the primary gold-support outputs against MTOB.

Evaluation	Metric conventions	Evaluated records and exclusions
Primary gold-support	BLEU and chrF++, 0–100; XCOMET-XXL on its native scale	Native/common99 cohorts; primary Lezgi 83-reference scores, but saved inference uses 87; failures retained empty
Earlier MTOB	BLEU, plain chrF, ROUGE F1 and capped CharacTER; raw and extracted views	Complete configured sequences; failures retained empty; Lezgi still has all 87 references, including three invalid ones
Earlier cross-protocol	MTOB and earlier <code>matched_v1</code> baselines under one MTOB BLEU/chrF evaluator	Aligned complete sequences; no success filtering or Lezgi reference repair; no current gold-support outputs

Table E.1.: Metric and record policies of three distinct evaluations. MTOB’s plain chrF has no word n-grams, unlike primary chrF++. MTOB BLEU is stored on a 0–1 scale in condition metrics and displayed on a 0–100 scale in statistical comparisons.

The earlier cross-protocol evaluation compares 30 MTOB systems with 24 unique Qwen no-book baseline systems from the earlier `matched_v1` family. Those earlier baselines use empty support-gloss fields. Each MTOB condition is paired with earlier Shot, Chain-gloss, and ModelGloss for the same model and language, giving 90 comparisons. Both sets of systems are scored with the MTOB evaluator in its raw and extracted views. These are not comparisons with the current gold-support baselines. Before a matched baseline is scored, an adapter removes the required final-translation framing and the generated Chain gloss. If the Chain translation section is missing, the prediction remains empty. The adapter leaves the stored baseline predictions unchanged, and the raw and extracted views of each baseline are identical.

Common scoring makes these outputs comparable under one metric definition. It does not make the generation protocols identical. They differ in demonstrations, query-gloss information, prompts, temperature, output limits, seeds, and retry policy. The contrast therefore evaluates complete MTOB and matched-baseline configurations, not an isolated grammar-addition effect.

E.2.1. Earlier MTOB Paired Tests

MTOB uses 100,000 resamples and seed 20260917. Within each model and source, the three context-method pairs are compared in both response views using BLEU and chrF. This gives $2 \times 5 \times 3 \times 2 \times 2 = 120$ tests in one Holm family. The 90 cross-protocol comparisons form a separate $90 \times 2 \times 2 = 360$ -test family. Both analyses are complete. In these two supplementary analyses, an absolute observed difference at or below 10^{-12} receives $p = 1$. The package’s original value is retained separately. The raw and extracted views remain correlated analyses of the same answers.

E.3. Earlier MTOB Results

E.3.1. Raw and Extracted Responses

The earlier MTOB study contains 30 conditions and 4,230 sentence–condition records, divided equally between Qwen3 and Qwen3.5. Each source uses the native language cohort, including all 445 Tsez records. Two Qwen3.5 Natugu G_e records and one G_s record failed and remain empty in both response views. Both affected Natugu conditions retain a denominator of 99. The two Gitksan sources evaluate the same 37 sentences, so the record total does not count distinct source sentences.

Tables E.2 and E.3 give every condition’s BLEU and chrF scores before and after response extraction. G_e uses embedding retrieval, G_s uses substring retrieval, and G_l uses the longer selected grammar text. Both displayed metrics use 0–100. The MTOB evaluator uses plain chrF and BLEU without smoothing, with the cleaning rules specified above; its scores cannot be subtracted directly from primary-study scores. Two Natugu grammar examples corresponding to test indices 20 and 50 appear in earlier context. In both Qwen models’ saved prompts, the known index-20 English reference is present for G_s and G_l ; the index-50 source example is present for all three methods. This confirmed prompt overlap limits interpretation of the earlier Natugu scores and method rankings, though its influence on the scores is unknown.

Extraction changes 801 responses: 458 for Qwen3 and 343 for Qwen3.5. A separate diagnostic marks 212 responses as ambiguous for extraction: 50 from Qwen3 and 162 from Qwen3.5. Ambiguous cases keep the raw answer. Since both views use the same generated responses, score differences measure sensitivity to response handling rather than the effect of generating a new translation.

Across the 30 conditions, extraction increases BLEU in 21 and leaves it unchanged in nine. It increases chrF in 28 and leaves it unchanged in two. The largest BLEU increase is Qwen3.5 Natugu G_l , from 1.15 to 2.38. The largest chrF increase is Qwen3.5 Lezgi G_l , from 14.75 to 20.04. Qwen3 Tsez G_s has the largest number of changed responses,

Source	Method	BLEU		chrF	
		Raw	Extracted	Raw	Extracted
Gitksan: Brown	G_e	0.00	0.00	17.39	18.59
	G_s	0.00	0.00	17.20	19.65
	G_l	0.00	0.00	19.84	20.10
Gitksan: Rigsby	G_e	0.00	0.00	15.80	17.62
	G_s	0.28	0.36	14.69	16.63
	G_l	0.00	0.00	20.23	20.23
Lezgi	G_e	0.18	0.25	11.84	13.76
	G_s	0.19	0.26	11.94	13.83
	G_l	0.53	0.56	16.98	17.22
Natugu	G_e	0.47	0.92	14.35	17.40
	G_s	1.07	1.94	15.62	19.55
	G_l	1.21	1.31	15.08	15.79
Tsez	G_e	0.09	0.10	14.70	15.46
	G_s	0.00	0.00	13.38	16.67
	G_l	0.00	0.00	17.89	17.89

Table E.2.: Earlier MTOB-inspired results for Qwen3. Both metrics use 0–100. chrF is the character-only variant. Every source uses its full native cohort: Gitksan 37, Lezgi 87, Natugu 99 and Tsez 445. The two views reuse the same responses and retain failed items. Lezgi retains three invalid references and has not received the primary study’s reference exclusions.

Source	Method	BLEU		chrF	
		Raw	Extracted	Raw	Extracted
Gitksan: Brown	G_e	0.70	0.78	21.73	22.74
	G_s	0.58	0.83	19.57	22.59
	G_l	0.58	0.88	20.00	23.54
Gitksan: Rigsby	G_e	0.61	0.77	18.58	20.55
	G_s	0.39	0.48	17.62	19.59
	G_l	0.28	0.36	16.73	18.41
Lezgi	G_e	1.24	1.39	22.19	22.96
	G_s	1.91	2.08	23.08	23.63
	G_l	0.42	0.84	14.75	20.04
Natugu	G_e	0.61	0.78	15.45	16.83
	G_s	1.86	2.07	18.57	19.41
	G_l	1.15	2.38	15.72	19.84
Tsez	G_e	0.00	0.00	18.17	18.43
	G_s	0.00	0.00	17.93	18.83
	G_l	0.11	0.17	14.95	17.41

Table E.3.: Earlier MTOB-inspired results for Qwen3.5. Both metrics use 0–100. chrF is the character-only variant. Every source uses its full native cohort: Gitksan 37, Lezgi 87, Natugu 99 and Tsez 445. The two views reuse the same responses and retain failed items. Lezgi retains three invalid references and has not received the primary study’s reference exclusions.

259 of 445. These are descriptive comparisons of the two views; the saved analysis does not test extraction itself for statistical significance. Removing commentary can improve reference overlap without correcting the retained translation, as illustrated at the end of this appendix.

E.3.2. Comparison of Grammar-Context Methods

For Qwen3, G_l has the highest observed chrF on both Gitksan sources, Lezgi, and Tsez in both response views; G_s leads on Natugu. Qwen3.5 shows a less stable ranking. In the raw view, G_e leads on both Gitksan resources and Tsez, while G_s leads on Natugu and Lezgi. After extraction, G_l leads on Brown and Natugu, G_e on Rigsby, and G_s on Lezgi and Tsez. A method ranking can therefore change when explicitly marked commentary is removed.

Lezgi retains all 87 earlier references, including the three invalid entries discussed in Section 3.2.2. Its method comparisons are therefore reference-contaminated and cannot establish an effect on valid references alone. The 120-test within-MTOB family contains twelve significant comparisons, all in chrF: nine in the raw view and three after extraction (Table E.4). Only Lezgi and Tsez have significant method contrasts. No BLEU method comparison meets the corrected threshold. The table includes significant effects in both directions.

Qwen3 Tsez gives a direct example of sensitivity to the response view. Raw chrF favours G_e over G_s by 1.31 points ($p_{\text{Holm}} = 0.00120$). Extracted chrF instead favours G_s over G_e by 1.21 points ($p_{\text{Holm}} = 0.02507$). Both tests compare methods within one view; neither is a test of the raw-to-extracted change. The underlying responses are unchanged.

The long-context comparisons also differ by model. In raw Lezgi scores, Qwen3 G_l exceeds G_e and G_s by 5.14 and 5.04 chrF points. For Qwen3.5, it falls below them by 7.43 and 8.33 points. Those four earlier differences pass the stored correction, but the invalid references limit their interpretation. They neither establish a reference-clean Lezgi effect nor test an interaction between models.

E.3.3. Comparison with Earlier Matched Baselines

The separate earlier cross-protocol analysis pairs the 30 MTOB configurations with 24 distinct Qwen no-book baselines from `matched_v1`. Those earlier baselines had empty support-gloss fields; they are not the gold-support baselines reported in the preceding sections. Pairing each MTOB condition with Shot, Chain, and ModelGloss gives 90 configuration pairs. Each pair contributes raw and extracted BLEU and chrF tests, for a separate family of 360. Both configurations are evaluated with the MTOB evaluator

Model	Language	Contrast	View	ΔchrF	p	p_{Holm}
Qwen3	Lezgi	$G_l - G_e$	Raw	+5.14	2.00×10^{-5}	0.00224
Qwen3	Lezgi	$G_l - G_s$	Raw	+5.04	5.00×10^{-5}	0.00550
Qwen3	Tsez	$G_s - G_e$	Raw	-1.31	1.00×10^{-5}	0.00120
Qwen3	Tsez	$G_s - G_e$	Extracted	+1.21	2.30×10^{-4}	0.02507
Qwen3	Tsez	$G_l - G_e$	Raw	+3.19	1.00×10^{-5}	0.00120
Qwen3	Tsez	$G_l - G_e$	Extracted	+2.43	1.00×10^{-5}	0.00120
Qwen3	Tsez	$G_l - G_s$	Raw	+4.50	1.00×10^{-5}	0.00120
Qwen3.5	Lezgi	$G_l - G_e$	Raw	-7.43	1.00×10^{-5}	0.00120
Qwen3.5	Lezgi	$G_l - G_s$	Raw	-8.33	1.00×10^{-5}	0.00120
Qwen3.5	Tsez	$G_l - G_e$	Raw	-3.22	1.00×10^{-5}	0.00120
Qwen3.5	Tsez	$G_l - G_s$	Raw	-2.98	1.00×10^{-5}	0.00120
Qwen3.5	Tsez	$G_l - G_s$	Extracted	-1.42	2.00×10^{-5}	0.00224

Table E.4.: All twelve significant method tests from the earlier MTOB study. Lezgi rows retain invalid references and are not reference-clean inference. Differences follow the displayed subtraction. All use chrF; none of the BLEU comparisons survives correction. The 120-test Holm family jointly includes both response views and both metrics.

on identical ordered test records. Failed predictions remain in both denominators. The raw and extracted baseline predictions are identical after removal of the required response framing.

Every one of the 360 observed metric-view differences favours the earlier matched baseline. Of these, 217 pass the saved Holm correction: 105 for Qwen3 and 112 for Qwen3.5. Table E.5 separates the counts by baseline method, metric, and response view. The raw view has 116 significant differences and the extracted view 101. These counts refer to correlated tests of 90 configuration pairs, not 217 independent experiments.

Model	Baseline method	BLEU		chrF	
		Raw	Extracted	Raw	Extracted
Qwen3	Shot	3	3	10	4
	Chain	3	3	14	11
	ModelGloss	12	12	15	15
Qwen3.5	Shot	3	3	11	8
	Chain	3	3	12	9
	ModelGloss	15	15	15	15
Total		39	39	77	62

Table E.5.: Significant advantages of earlier empty-support-gloss matched baselines over MTOB configurations under the common MTOB evaluator. Each model-method cell contains 15 tests (five grammar sources by three MTOB methods). The total is 217 of 360 correlated metric-view tests, using a separate 360-test Holm family. Lezgi retains invalid references. No current gold-support output enters these comparisons.

The size of the advantage depends on the baseline and metric. Table E.6 illustrates this with Qwen3 Natugu G_l in the extracted view. Shot exceeds G_l by 0.76 BLEU and 4.36 chrF points, without passing correction. ModelGloss exceeds the same G_l output by 9.30 BLEU and 22.09 chrF points, both significant. These values use the common evaluator and earlier predictions. They must not be read as scores or differences for the current gold-support baselines.

Under the common evaluator, the earlier comparison favours the earlier matched configurations. Its Lezgi comparisons remain reference-contaminated, and none of these tests compares MTOB with the current gold-support study. The result does not isolate a harmful effect of grammar: the protocols differ in demonstrations, query-gloss information, prompts, decoding, output limits, and retry policy. There is no separately

Baseline	Metric	Baseline score	G_l score	Difference	p_{Holm}
Shot	BLEU	2.07	1.31	+0.76	1.00000
Shot	chrF	20.15	15.79	+4.36	0.34710
ModelGloss	BLEU	10.61	1.31	+9.30	0.00360
ModelGloss	chrF	37.88	15.79	+22.09	0.00360

Table E.6.: Illustrative earlier cross-protocol comparisons for Qwen3 on the 99 Natugu records, using extracted responses. The difference is the earlier empty-support-gloss matched baseline minus MTOB G_l . All values come from the common evaluator, not the primary BLEU or chrF++ tables.

generated grammar-free MTOB condition that holds those choices fixed. Chapter 5 supplies the primary matched grammar-addition contrasts.

E.4. Provenance and Interpretation Limits

E.4.1. Retained MTOB Artifacts

Section E.1 specifies the earlier generation settings and incomplete attempt history. The following paths identify the retained evaluation artifacts.

The retained response-view evaluation is under `experiments/mtob/evaluation_v1/`; its provenance records SacreBLEU 2.6.0, NumPy 2.2.6, tiktoken 0.11.0, evaluator `mtob-evaluation-1.0.0`, and parser `mtob-response-views-1.0.0`. Raw and extracted views use the same saved responses and keep all expected indices, with known failures represented by empty predictions. This evaluator removes specified punctuation and reports unsmoothed BLEU with 13a tokenization and plain chrF with word order zero, unlike the primary chrF++. Its per-condition BLEU JSON uses a 0–1 scale, whereas the statistical and thesis displays use 0–100. The earlier 100,000-resample analyses use seed 20260917, a 120-test within-suite Holm family and a separate 360-test cross-protocol family.

The cross-protocol files under `experiments/mtob/grammamt_comparison_v1/` compare these outputs with earlier `matched_v1` baselines, whose support-gloss fields are empty. They do not compare with the current gold-support baselines or isolate grammar addition. The earlier MTOB evaluation of Lezgi retains the invalid references, and known Natugu source/reference examples occur in some saved prompts. The retained outputs allow the two response views to be compared. Missing earlier attempts, unequal inputs, and the absence of a matched earlier no-book ablation limit conclusions from this supplement.

E.4.2. A Response-Extraction Example

A separate earlier MTOB example illustrates response extraction. For Qwen3.5 with Brown’s Gitksan resource and G_l , test index 19 has the reference “They mobbed the surveyors.” The raw response starts “They are surveyors.” and continues with a labelled breakdown. Extraction retains exactly “They are surveyors.” Removing the explanation changes the evaluated text, but does not restore the action expressed by *mobbed*.

F. Primary and Supplementary Results

F.1. Complete Native chrF++ Matrices

The following three matrices retain all 315 native-cohort chrF++ differences, calculated as context minus the matched no-book baseline. Each cell is a condition contrast, not an independent observation. Qwen Tsez uses 445 records and Gemini Tsez uses 99; the common99 analysis reuses the first 99 Qwen records. Lezgi uses the uniform primary 83-record mask; the original primary tests do not match this mask, and separate supplementary Qwen tests appear in Section F.3. The same differences appear visually in Figures 5.2 and 5.3. Stars in the non-Lezgi tables mark the original native context-family Holm-adjusted $p < 0.05$ decisions. S means Gloss-shot, C explicit Chain-gloss, MG ModelGloss, Sheet cheat sheet and Pages selected original pages. Rounded zero retains the unrounded sign.

	Qwen3			Qwen3.5			Gemini		
Context	S	C	MG	S	C	MG	S	C	MG
Gitksan: Brown									
Summary TXT	+0.55	+3.00	+0.42	-0.03	+0.00	-3.06	-0.28	+2.50	-1.05
Table TXT	+2.59	+2.76	+0.47	+0.61	+1.85	-2.29	+1.40	-1.13	-0.73
Table JPG	+2.15	+2.24	+3.12	-1.69	-4.04	-3.82	-1.00	-2.80	-2.57
Sheet TXT	+2.30	+2.98	+1.13	+0.08	-0.77	-2.49	-0.06	+0.07	-1.77
Sheet JPG	+1.31	+0.75	+0.13	-1.17	-0.57	-4.06	-2.10	-0.14	+0.10
Pages JPG	-0.77	+1.90	-2.44	-1.07	-3.28	-3.39	-3.17	-2.27	-5.21
Gitksan: Rigsby									
Summary TXT	+1.68	+1.71	+0.81	+0.49	-0.53	+0.24	+1.39	-2.20	-0.76
Table TXT	+1.24	+2.01	+0.31	+0.56	+1.09	-0.95	+0.74	+1.42	-1.78
Table JPG	+1.38	+0.77	+0.82	-1.29	-2.75	-1.88	-1.63	-3.82	-3.41
Sheet TXT	+0.76	+1.82	+0.62	+0.25	+0.49	-1.33	-0.41	-1.11	-1.62
Sheet JPG	+1.54	-1.90	+0.52	-1.95	-1.12	-3.94	-0.60	-1.11	-1.53
Pages JPG	+1.62	+0.09	+0.19	-1.10	-3.99	-2.65	-4.49	-1.81	-6.49

Table F.1.: Native Gitksan chrF++ changes, 37 records. S: Gloss-shot; C: explicit Chain-gloss; MG: ModelGloss; Sheet: cheat sheet; Pages: selected original pages. Each chrF++ difference is context minus its matched no-book baseline; positive/negative values mean higher/lower scores. Rounded zero retains the unrounded sign. A star marks saved native context-family Holm-adjusted $p < 0.05$.

	Qwen3			Qwen3.5			Gemini		
Context	S	C	MG	S	C	MG	S	C	MG
Lezgi: original									
Summary TXT	-0.06	+1.56	+1.39	-0.77	-2.80	-2.04	-0.64	-3.43	-1.81
Table TXT	-0.05	+0.94	+0.91	+0.99	-1.97	-0.99	-1.04	-2.10	-0.85
Table JPG	-1.11	-1.48	-1.05	-1.56	-1.96	-2.58	-0.40	-0.01	+0.52
Sheet TXT	-0.34	+0.98	+0.91	+0.29	-1.80	-0.65	-0.89	-1.83	-1.38
Sheet JPG	-0.15	-1.56	-0.01	-1.19	-1.34	-3.19	-0.86	+0.34	+0.50
Pages JPG	-2.07	-1.66	+0.01	-1.53	-3.11	-0.95	+0.31	+1.31	-0.64
Lezgi: Cyrillic-oriented									
Summary TXT	+0.35	+0.76	+1.57	-0.76	-2.15	-0.81	-1.25	-3.12	-1.15
Table TXT	+0.09	+0.35	+1.43	+0.56	-1.88	-0.31	-1.35	-1.59	-1.04
Table JPG	-1.16	-1.93	-0.27	-1.34	-1.14	-2.54	-0.76	+0.37	-0.11
Sheet TXT	-0.12	+0.91	+0.88	+0.35	-1.68	-1.36	-0.66	-1.30	-1.03
Sheet JPG	-1.76	-1.61	-0.75	-0.39	-3.40	-2.65	-0.36	+1.27	-0.03

Table F2.: Native Lezgi chrF++ changes, uniformly 83 retained-reference records per cell. S: Gloss-shot; C: explicit Chain-gloss; MG: ModelGloss; Sheet: cheat sheet; Pages: selected original pages. Each chrF++ difference is context minus its matched no-book baseline; positive/negative values mean higher/lower scores. Rounded zero retains the unrounded sign. No primary significance stars apply; the Cyrillic bundle has no Pages condition.

	Qwen3			Qwen3.5			Gemini		
Context	S	C	MG	S	C	MG	S	C	MG
Natugu									
Summary TXT	-0.11	+1.19	+0.31	+1.26	+0.27	-1.27	+0.43	+0.28	+1.53
Table TXT	+0.72	+0.84	+1.07	-0.09	-0.54	-0.40	-0.60	-0.05	+1.19
Table JPG	-0.71	-1.19	-1.07	-1.25	-1.25	-2.07	-2.98*	-1.51	-0.40
Sheet TXT	+0.03	+1.32	-0.34	+1.09	+0.22	-0.54	-1.04	-0.56	+1.16
Sheet JPG	-0.30	-0.14	-0.83	-1.74	-1.41	-2.08	-2.41	-1.34	+0.50
Pages JPG	-1.33	-1.34	-1.02	-0.78	-1.20	-1.13	-3.99*	-1.45	-2.05
Tsez									
Summary TXT	+0.03	+0.58	+0.61	+0.48	+0.93	-0.11	+0.61	+5.35	-0.23
Table TXT	+0.55	+0.96	-0.03	+0.31	+0.83	+0.18	-0.03	+5.04	-1.18
Table JPG	-0.05	-1.60*	-0.07	-0.99	-1.18*	-1.97*	-1.31	+3.78	-1.87
Sheet TXT	+0.19	+0.96	-0.06	+0.35	+1.38*	-0.36	+0.07	+4.93	-0.32
Sheet JPG	-0.27	-2.57*	-0.60	-1.02*	-0.75	-2.07*	-1.34	+2.06	-1.71
Pages JPG	+0.10	-0.59	-1.26	-0.81	-0.66	-2.17*	-2.97*	+0.98	-1.47

Table F.3.: Native chrF++ changes: Natugu, 99 records; Tsez, 445 for Qwen and 99 for Gemini. S: Gloss-shot; C: explicit Chain-gloss; MG: ModelGloss; Sheet: cheat sheet; Pages: selected original pages. Each chrF++ difference is context minus its matched no-book baseline; positive/negative values mean higher/lower scores. Rounded zero retains the unrounded sign. A star marks saved native context-family Holm-adjusted $p < 0.05$.

F.2. Direction Counts and Saved Test Detail

Table F.4 counts the signs of individual contrasts in the complete native matrix. Raw p-values for the corrected rows shown in Chapter 5 follow in Tables F.5, F.6 and F.7. No additional tests or correction families are introduced.

Model	Metric	$\Delta > 0$	$\Delta < 0$	Ties	Sig. +	Sig. −
Qwen3	BLEU	42	63	0	0	0
Qwen3	chrF++	63	42	0	0	2
Qwen3	xCOMET	33	72	0	0	0
Qwen3.5	BLEU	24	81	0	0	0
Qwen3.5	chrF++	25	80	0	1	5
Qwen3.5	xCOMET	28	77	0	0	1
Gemini	BLEU	29	76	0	0	0
Gemini	chrF++	28	77	0	0	3
Gemini	xCOMET	19	86	0	0	0

Table F.4.: Each row counts signs of 105 context-minus-own-baseline differences across languages, resources and methods. These contrasts share sentences and baselines and are not independent observations. Lezgi contributes its 83-record descriptive signs. Significant columns count only eligible non-Lezgi saved Holm-adjusted findings. xCOMET uses XXL and its pooled-cohort family; BLEU/chrF++ use the native lexical family.

Cohort	Model	Language	Method	Context	Raw p	Holm p
Native	Gemini	Natugu	S	Pages JPG	9.9999×10^{-6}	0.00630
Native	Gemini	Natugu	S	Table JPG	3×10^{-5}	0.01869
Native	Gemini	Tsez	S	Pages JPG	9.9999×10^{-6}	0.00630
Native	Qwen3.5	Tsez	C	Sheet TXT	9.9999×10^{-6}	0.00630
Native	Qwen3.5	Tsez	C	Table JPG	3×10^{-5}	0.01869
Native	Qwen3.5	Tsez	MG	Sheet JPG	9.9999×10^{-6}	0.00630
Native	Qwen3.5	Tsez	MG	Pages JPG	9.9999×10^{-6}	0.00630
Native	Qwen3.5	Tsez	MG	Table JPG	9.9999×10^{-6}	0.00630
Native	Qwen3.5	Tsez	S	Sheet JPG	6.9999×10^{-5}	0.04340
Native	Qwen3	Tsez	C	Sheet JPG	9.9999×10^{-6}	0.00630
Native	Qwen3	Tsez	C	Table JPG	5.9999×10^{-5}	0.03726
Shared99	Gemini	Natugu	S	Pages JPG	9.9999×10^{-6}	0.00630
Shared99	Gemini	Natugu	S	Table JPG	3×10^{-5}	0.01884
Shared99	Gemini	Tsez	S	Pages JPG	9.9999×10^{-6}	0.00630

Table F.5.: Saved p-values for all chrF++ rows displayed in the main context–baseline tables. Native and Shared99 denote the original native and common99 lexical families, respectively. Their sizes are 630 tests per cohort, including BLEU and chrF++. All materials are original; method and package codes follow Chapter 5.

Cohort	Model	Language	Method	Context	Raw p	Holm p
Native	Qwen3.5	Natugu	S	Sheet JPG	2×10^{-5}	0.00418
Native	Qwen3.5	Tsez	C	Sheet JPG	9.9999×10^{-6}	0.00216
Native	Qwen3.5	Tsez	C	Table JPG	9.9999×10^{-6}	0.00216
Native	Qwen3.5	Tsez	MG	Sheet JPG	3×10^{-5}	0.00624
Native	Qwen3.5	Tsez	MG	Table JPG	9.9999×10^{-6}	0.00216
Native	Qwen3.5	Tsez	S	Sheet JPG	9.9999×10^{-6}	0.00216
Native	Qwen3.5	Tsez	S	Table JPG	9.9999×10^{-6}	0.00216
Native	Qwen3	Tsez	C	Sheet JPG	9.9999×10^{-6}	0.00216
Native	Qwen3	Tsez	C	Table JPG	9.9999×10^{-6}	0.00216
Shared99	Qwen3.5	Natugu	S	Sheet JPG	2×10^{-5}	0.00430
Shared99	Qwen3.5	Tsez	C	Sheet JPG	0.00017	0.03604
Shared99	Qwen3.5	Tsez	C	Table JPG	2×10^{-5}	0.00430
Shared99	Qwen3	Tsez	C	Sheet JPG	9.9999×10^{-6}	0.00216

Table F.6.: Saved p -values for all chrF++ rows displayed in the main TXT/JPG tables. Native and Shared99 denote the original native and common99 lexical families, respectively. Their sizes are 216 tests per cohort, including BLEU and chrF++. All materials are original; method and package codes follow Chapter 5.

Comparison	Raw p	Holm p	95% paired interval
Context – baseline	5.9999×10^{-5}	0.03780	$[-0.02126, -0.00729]$
JPG – TXT	0.00013	0.02808	$[-0.02249, -0.00702]$

Table F.7.: Saved detail for Table 5.8, Qwen3.5 Tsez ModelGloss table JPG, native $n = 445$. The first row uses the 630-test xCOMET context family, the second the separate 216-test presentation family; each pools both cohorts. Intervals are unadjusted paired-difference intervals. Significance follows Holm p , rather than interval exclusion of zero.

F.3. Separate Supplementary Results

The following tables report the retained supplement separately from the primary matrices above. A/B inferential decisions use their complete source-cluster Holm families, while F/G/I and all supplementary neural scores are descriptive. Table F.8 gives every adjusted-significant A metric row, including the overlapping Lezgi 84 sensitivity. Q3 and Q3.5 denote Qwen3 and Qwen3.5. Method codes are S (Gloss-shot), C (Chain-gloss) and MG (ModelGloss); scores run from the right comparator to the left method, and differences are left minus right. The two Chain-gloss/Gloss-shot decreases occur only on native Tsez chrF++. The complete 72-test report, including nonsignificant rows and source identities, is `experiments/additional_v1/reports/A_comparisons.tsv`.

Table F.8.: Adjusted-significant supplementary no-book method contrasts. Parentheses give records/source clusters. Intervals are pointwise 95%; all Holm values retain the 72-test family.

Model, language, view, pair (<i>n/c</i>)	Metric	Right→left	Δ	Interval	<i>p</i>	<i>p</i> _{Holm}
Q3 Gitksan native MG-S (37/37)	chrF++	22.93→46.10	+23.17	[17.83, 28.21]	0.00001	0.00072
Q3 Gitksan native MG-S (37/37)	BLEU	4.86→14.98	+10.12	[4.88, 14.73]	0.00019	0.00513
Q3 Gitksan native MG-C (37/37)	chrF++	18.11→46.10	+27.99	[22.02, 33.37]	0.00001	0.00072
Q3 Gitksan native MG-C (37/37)	BLEU	3.80→14.98	+11.18	[6.40, 15.51]	0.00001	0.00072
Q3 Lezgi 84 MG-S (84/82)	chrF++	19.89→34.78	+14.89	[11.57, 18.29]	0.00001	0.00072
Q3 Lezgi 84 MG-C (84/82)	chrF++	19.26→34.78	+15.52	[12.26, 18.92]	0.00001	0.00072
Q3 Lezgi 83 MG-S (83/81)	chrF++	19.40→34.51	+15.11	[11.80, 18.55]	0.00001	0.00072
Q3 Lezgi 83 MG-C (83/81)	chrF++	18.78→34.51	+15.73	[12.52, 19.18]	0.00001	0.00072
Q3 Natugu native MG-S (99/99)	chrF++	19.54→39.14	+19.61	[17.18, 22.17]	0.00001	0.00072
Q3 Natugu native MG-S (99/99)	BLEU	2.34→14.09	+11.75	[8.66, 15.10]	0.00001	0.00072
Q3 Natugu native MG-C (99/99)	chrF++	18.70→39.14	+20.45	[17.65, 23.37]	0.00001	0.00072
Q3 Natugu native MG-C (99/99)	BLEU	3.19→14.09	+10.89	[7.60, 14.42]	0.00001	0.00072

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F. Primary and Supplementary Results

Model, language, view, pair (<i>n/c</i>)	Metric	Right→left	Δ	Interval	<i>p</i>	<i>p</i> _{Holm}
Q3 Tsez native MG-S (445/445)	chrF++	18.14→38.95	+20.81	[19.78, 21.85]	0.00001	0.00072
Q3 Tsez native MG-S (445/445)	BLEU	2.17→12.08	+9.90	[8.75, 11.12]	0.00001	0.00072
Q3 Tsez native C-S (445/445)	chrF++	18.14→15.52	-2.61	[-3.48, -1.76]	0.00001	0.00072
Q3 Tsez native MG-C (445/445)	chrF++	15.52→38.95	+23.42	[22.23, 24.63]	0.00001	0.00072
Q3 Tsez native MG-C (445/445)	BLEU	1.22→12.08	+10.86	[9.66, 12.12]	0.00001	0.00072
Q3 Tsez first99 MG-S (99/99)	chrF++	17.33→38.18	+20.84	[18.61, 23.16]	0.00001	0.00072
Q3 Tsez first99 MG-S (99/99)	BLEU	1.25→11.84	+10.58	[8.07, 13.10]	0.00001	0.00072
Q3 Tsez first99 MG-C (99/99)	chrF++	15.64→38.18	+22.53	[19.91, 25.13]	0.00001	0.00072
Q3 Tsez first99 MG-C (99/99)	BLEU	1.26→11.84	+10.58	[7.99, 13.18]	0.00001	0.00072
Q3.5 Gitksan native MG-S (37/37)	chrF++	23.61→49.14	+25.53	[20.60, 30.18]	0.00001	0.00072
Q3.5 Gitksan native MG-S (37/37)	BLEU	4.01→21.06	+17.05	[10.93, 23.73]	0.00001	0.00072
Q3.5 Gitksan native MG-C (37/37)	chrF++	23.46→49.14	+25.68	[20.15, 30.87]	0.00001	0.00072
Q3.5 Gitksan native MG-C (37/37)	BLEU	4.14→21.06	+16.92	[10.63, 23.70]	0.00001	0.00072
Q3.5 Lezgi 84 MG-S (84/82)	chrF++	25.38→39.80	+14.42	[10.84, 18.01]	0.00001	0.00072
Q3.5 Lezgi 84 MG-S (84/82)	BLEU	6.72→13.86	+7.13	[3.97, 10.42]	0.00001	0.00072
Q3.5 Lezgi 84 MG-C (84/82)	chrF++	26.26→39.80	+13.54	[10.16, 16.89]	0.00001	0.00072
Q3.5 Lezgi 84 MG-C (84/82)	BLEU	6.37→13.86	+7.49	[4.28, 10.76]	0.00003	0.00090
Q3.5 Lezgi 83 MG-S (83/81)	chrF++	25.00→39.52	+14.52	[10.90, 18.08]	0.00001	0.00072
Q3.5 Lezgi 83 MG-S (83/81)	BLEU	6.35→13.57	+7.21	[4.01, 10.55]	0.00006	0.00168
Q3.5 Lezgi 83 MG-C (83/81)	chrF++	25.82→39.52	+13.70	[10.30, 17.04]	0.00001	0.00072
Q3.5 Lezgi 83 MG-C (83/81)	BLEU	5.80→13.57	+7.76	[4.51, 11.10]	0.00001	0.00072

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F. Primary and Supplementary Results

Model, language, view, pair (<i>n/c</i>)	Metric	Right→left	Δ	Interval	<i>p</i>	<i>p</i> _{Holm}
Q3.5 Natugu native MG-S (99/99)	chrF++	18.82→41.59	+22.77	[20.39, 25.28]	0.00001	0.00072
Q3.5 Natugu native MG-S (99/99)	BLEU	3.54→16.55	+13.02	[9.80, 16.44]	0.00001	0.00072
Q3.5 Natugu native MG-C (99/99)	chrF++	19.64→41.59	+21.95	[19.39, 24.62]	0.00001	0.00072
Q3.5 Natugu native MG-C (99/99)	BLEU	3.60→16.55	+12.95	[9.65, 16.52]	0.00001	0.00072
Q3.5 Tsez native MG-S (445/445)	chrF++	18.44→39.95	+21.52	[20.47, 22.58]	0.00001	0.00072
Q3.5 Tsez native MG-S (445/445)	BLEU	1.82→13.72	+11.90	[10.60, 13.23]	0.00001	0.00072
Q3.5 Tsez native C-S (445/445)	chrF++	18.44→17.20	-1.24	[-1.82, -0.66]	0.00005	0.00145
Q3.5 Tsez native MG-C (445/445)	chrF++	17.20→39.95	+22.75	[21.70, 23.84]	0.00001	0.00072
Q3.5 Tsez native MG-C (445/445)	BLEU	1.00→13.72	+12.72	[11.49, 14.02]	0.00001	0.00072
Q3.5 Tsez first99 MG-S (99/99)	chrF++	17.28→39.72	+22.44	[20.20, 24.74]	0.00001	0.00072
Q3.5 Tsez first99 MG-S (99/99)	BLEU	1.24→12.96	+11.72	[9.37, 14.07]	0.00001	0.00072
Q3.5 Tsez first99 MG-C (99/99)	chrF++	16.64→39.72	+23.09	[20.66, 25.61]	0.00001	0.00072
Q3.5 Tsez first99 MG-C (99/99)	BLEU	0.85→12.96	+12.12	[9.70, 14.58]	0.00001	0.00072

Table F.9 gives the two adjusted-significant Lezgi comparisons under the main 83-record mask and their overlapping 84-record sensitivities. The context row compares Cyrillic-form sheet JPG with no book; the presentation row compares Cyrillic-form table JPG with TXT. These are different contrasts and correction families. All other saved B tests do not pass their adjustment; the full reports and exact identities remain in `experiments/additional_v1/reports/B_comparisons.tsv`.

Table F.9.: Supplementary Qwen3.5 Lezgi Chain-gloss findings: main 83 and sensitivity 84 records. Intervals are pointwise 95%. Context uses the 264-test family; JPG-TXT uses the 96-test family.

Contrast/view	Metric	Right→left	Δ	Interval	p	p_{Holm}
Context 83 ($c = 81$)	chrF++	25.82→22.42	-3.40	[-5.14, -1.69]	0.00006	0.01584
JPG-TXT 83 ($c = 81$)	BLEU	4.15→6.77	+2.62	[1.42, 3.98]	0.00011	0.01045
Context 84 ($c = 82$)	chrF++	26.26→22.82	-3.44	[-5.18, -1.74]	0.00015	0.03945
JPG-TXT 84 ($c = 82$)	BLEU	4.59→7.36	+2.78	[1.58, 4.11]	0.00005	0.00480

Table F.10 separates ranges of logged condition means from per-item extrema and reconstructed first-prompt components. Both models include all generation items. Text reconstruction includes textual context; visual ranges cover image conditions only. Grammar-only counts are available by condition in the saved C reports defined in Appendix D.4. None of the reconstructed columns can be added to the logged total.

Table F.10.: Input-token ranges for existing Qwen conditions. Logged means range over conditions; logged extrema range over individual records. Reconstructed text and visual counts refer to first prompts.

Model	Language	Conditions	Logged means	Logged items	Text	Visual
Q3	Gitksan	39	1697.51–14103.51	1669–14202	1694–4175	4779–12276
Q3	Lezgi	36	2093.62–12185.62	2071–12240	2077–6406	4779–9963
Q3	Natugu	21	2244.78–20023.78	2227–20104	2224–6487	9558–17649
Q3	Tsez	21	2408.96–20187.96	2378–20348	2398–6404	9558–17649
Q3.5	Gitksan	39	1735.95–14148.95	1708–14247	1728–4284	4779–12276
Q3.5	Lezgi	36	2062.87–12162.87	2041–12212	2039–6432	4779–9963
Q3.5	Natugu	21	2300.16–20086.16	2282–20162	2276–6679	9558–17649
Q3.5	Tsez	21	2464.01–20250.01	2433–20404	2451–6587	9558–17649

Table F.11 gives every retained condition’s main/native scores and the overlapping Lezgi 84 and common99 Tsez sensitivities. Keys refer to Table B.6; that table supplies generation failure counts, seeds, caps and the changed query-gloss field. Failed predictions remain included at retained indices. For F, scores in Gitksan, the 83-record Lezgi view and Natugu are identical across caps, while native Tsez changes slightly and shared99 lexical scores are unchanged. For G, gold-query lexical scores decline in three languages and rise in Tsez; native Tsez xCOMET-XXL instead declines. No adjusted inference is supplied for these diagnostics.

Table F.11.: Retained Qwen3 diagnostic scores. Lezgi 83 is the main quality view; 84 and first99 Tsez are overlapping sensitivities. xCOMET-XL and xCOMET-XXL are exploratory native-scale scores without supplementary tests.

Key	Language	View	<i>n</i>	BLEU	chrF++	xCOMET-XL	xCOMET-XXL
S01	Gitksan	native	37	3.80	18.11	0.290	0.184
S02	Gitksan	native	37	3.80	18.11	0.290	0.184
S03	Gitksan	native	37	14.98	46.10	0.540	0.432
S04	Gitksan	native	37	11.45	38.68	0.500	0.380
S05	Gitksan	native	37	4.86	22.93	0.353	0.220
S06	Gitksan	native	37	6.09	25.90	0.362	0.225
S07	Gitksan	native	37	4.41	23.83	0.342	0.218
S08	Gitksan	native	37	5.14	25.23	0.352	0.224
S09	Gitksan	native	37	4.58	26.12	0.371	0.226
S10	Gitksan	native	37	4.47	24.28	0.353	0.228
S11	Lezgi	83 main	83	3.63	18.78	0.336	0.151
S11	Lezgi	84 sens.	84	4.26	19.26	0.342	0.155
S12	Lezgi	83 main	83	3.63	18.78	0.336	0.151
S12	Lezgi	84 sens.	84	4.26	19.26	0.342	0.155
S13	Lezgi	83 main	83	4.10	17.70	0.319	0.138
S13	Lezgi	84 sens.	84	4.69	18.20	0.325	0.143
S14	Lezgi	83 main	83	4.53	19.14	0.330	0.163
S14	Lezgi	84 sens.	84	5.08	19.63	0.336	0.168
S15	Lezgi	83 main	83	3.09	19.76	0.318	0.153
S15	Lezgi	84 sens.	84	3.85	20.25	0.324	0.157
S16	Lezgi	83 main	83	3.40	18.57	0.312	0.144
S16	Lezgi	84 sens.	84	4.08	19.08	0.318	0.148
S17	Lezgi	83 main	83	3.74	19.52	0.313	0.145
S17	Lezgi	84 sens.	84	4.40	20.02	0.319	0.149
S18	Lezgi	83 main	83	8.46	34.51	0.496	0.333
S18	Lezgi	84 sens.	84	8.81	34.78	0.501	0.335
S19	Lezgi	83 main	83	7.27	33.63	0.494	0.314
S19	Lezgi	84 sens.	84	7.80	34.00	0.498	0.317
S20	Natugu	native	99	3.19	18.70	0.296	0.176
S21	Natugu	native	99	3.19	18.70	0.296	0.176
S22	Natugu	native	99	14.09	39.14	0.512	0.397
S23	Natugu	native	99	11.98	36.21	0.495	0.393
S24	Natugu	native	99	14.14	38.73	0.515	0.392
S25	Natugu	native	99	13.85	38.69	0.521	0.395
S26	Natugu	native	99	13.43	38.32	0.524	0.389
S27	Natugu	native	99	13.49	37.81	0.529	0.388

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F. Primary and Supplementary Results

Key	Language	View	n	BLEU	chrF++	xCOMET-XL	xCOMET-XXL
S28	Natugu	native	99	14.45	38.43	0.521	0.396
S29	Tsez	first99	99	1.26	15.64	0.234	0.130
S29	Tsez	native	445	1.22	15.55	0.237	0.130
S30	Tsez	first99	99	1.26	15.64	0.234	0.130
S30	Tsez	native	445	1.22	15.52	0.237	0.131
S31	Tsez	first99	99	0.97	14.52	0.240	0.136
S31	Tsez	native	445	1.13	15.53	0.237	0.131
S32	Tsez	first99	99	1.09	16.06	0.233	0.133
S32	Tsez	native	445	1.29	16.00	0.236	0.132
S33	Tsez	first99	99	1.35	16.22	0.239	0.135
S33	Tsez	native	445	1.57	16.11	0.233	0.133
S34	Tsez	first99	99	1.38	16.29	0.230	0.130
S34	Tsez	native	445	1.47	16.03	0.230	0.131
S35	Tsez	first99	99	1.51	16.81	0.224	0.127
S35	Tsez	native	445	1.80	16.58	0.229	0.131
S36	Tsez	first99	99	11.84	38.18	0.482	0.352
S36	Tsez	native	445	12.08	38.95	0.486	0.362
S37	Tsez	first99	99	13.04	39.10	0.484	0.344
S37	Tsez	native	445	12.29	39.34	0.493	0.361

Table F.12 makes the I comparisons explicit across all reported metrics. The selected packages are Gitksan Gloss-shot/Brown sheet TXT, Lezgi Chain-gloss/sheet TXT, Natugu ModelGloss/sheet JPG and Tsez Chain-gloss/summary TXT. Differences are context minus the same-seed no-book score; the condition scores are in Table F.11. Positive chrF++ signs persist for the first, second and fourth pairs, and negative signs for Natugu, but the other metrics do not show the same stability in every pair.

In seed order 20260910, 20260925, 20261001, the main/native BLEU signs are Gitksan (+, -, +), Lezgi (-, -, -), Natugu (-, -, +) and Tsez (+, +, +). xCOMET-XL signs are respectively (-, +, +), (-, -, -), (+, +, -) and (-, -, -); xCOMET-XXL signs are (+, +, +), (+, +, -), (-, -, +) and (+, +, -). These signs use unrounded stored scores, so a change can be hidden by rounded equality in the table.

Table F.12.: Selected-pair seed diagnostics. Lexical columns are saved context-minus-baseline differences; neural columns show stored baseline-to-context scores. All outcomes are descriptive.

Language	View	Seed	Δ BLEU	Δ chrF++	xCOMET-XL baseline→context	xCOMET-XXL baseline→context
Gitksan	native	20260910	+0.29	+2.30	0.353→0.352	0.220→0.224
Gitksan	native	20260925	-1.51	+0.22	0.362→0.371	0.225→0.226
Gitksan	native	20261001	+0.07	+0.45	0.342→0.353	0.218→0.228
Lezgi	83 main	20260910	-0.54	+0.98	0.336→0.318	0.151→0.153
Lezgi	83 main	20260925	-0.70	+0.87	0.319→0.312	0.138→0.144
Lezgi	83 main	20261001	-0.78	+0.38	0.330→0.313	0.163→0.145
Lezgi	84 sens.	20260910	-0.41	+0.99	0.342→0.324	0.155→0.157
Lezgi	84 sens.	20260925	-0.61	+0.87	0.325→0.318	0.143→0.148
Lezgi	84 sens.	20261001	-0.68	+0.39	0.336→0.319	0.168→0.149
Natugu	native	20260910	-0.66	-0.83	0.512→0.524	0.397→0.389
Natugu	native	20260925	-0.65	-0.93	0.515→0.529	0.392→0.388
Natugu	native	20261001	+0.60	-0.26	0.521→0.521	0.395→0.396
Tsez	first99	20260910	+0.09	+0.58	0.234→0.239	0.130→0.135
Tsez	first99	20260925	+0.41	+1.77	0.240→0.230	0.136→0.130
Tsez	first99	20261001	+0.42	+0.75	0.233→0.224	0.133→0.127
Tsez	native	20260910	+0.35	+0.58	0.237→0.233	0.131→0.133
Tsez	native	20260925	+0.35	+0.51	0.237→0.230	0.131→0.131
Tsez	native	20261001	+0.50	+0.58	0.236→0.229	0.132→0.131

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